Informatica® Axon Data Governance
7.3.8

Data Governance Administrator Guide

Informatica Axon Data Governance Data Governance Administrator Guide
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Preface

Use the *Axon Data Governance Administrator Guide* to configure Axon after it is installed in your organization. In addition to the configuration steps, the guide also includes tasks to customize the Axon interface and steps to monitor Axon on a regular basis.

This guide is written for the Axon system administrator. This guide assumes that you have installed Axon successfully. For details on installing Axon, see the *Axon Data Governance Installation Guide* in the Informatica Documentation Portal.

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To find online support resources on the Informatica Network, visit https://network.informatica.com and select the Support option.

CHAPTER 1

Introduction

This chapter includes the following topics:

- Introduction to Axon Data Governance, 11
- Audience, 12
- Axon Architecture, 12
- Users, 15

Introduction to Axon Data Governance

Axon Data Governance is a knowledge repository tool that you can use to document the data objects that require governance. Data owners, data stewards, subject matter experts, and other responsible stakeholders collaborate across the organization to progressively chart the business reality of data, its lineage, and usage across processes, policies, projects, and regulation.

Use Axon to inventorize the objects in your organization. You can connect an object to another object within the inventory, and view the details on dashboards, maps, and search results. You can control the access to objects by creating roles and responsibilities.

You can decide whether you want to make the data available to everyone or restrict access to a particular set of users in Axon. You can create segments based on rules, regulations, or geography to authorize data access to users.

After you govern the data, you can publish the data to the Axon Data Marketplace. In the Data Marketplace, you can search for data that other users in your organization have published. Users from various departments in the organization can find data that can be used to make decisions that impact the business. Users can request access to the data and place the order from the Data Marketplace on a self-service basis. To administer the Axon Data Marketplace, see the *Axon Data Governance Data Marketplace Administrator Guide* on the Informatica Documentation Portal.

Audience

As an administrator, you can perform various administrative tasks, such as managing users, configuring roles, assigning permissions, adding users, customizing the Axon interface, and troubleshooting issues. You can do some administrative tasks in the Linux environment and some tasks in the Axon web interface.

Axon Administrator

A non-root user performs some of the administrative tasks from the Linux environment. The root user must create a non-root user to manage Axon. This user is called the **Axon Administrator** in this guide.

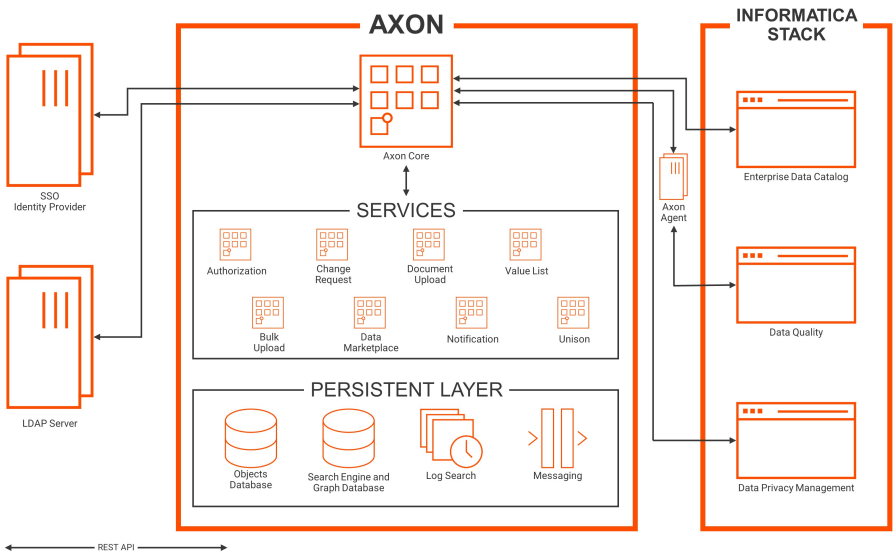
Admin or SuperAdmin Profile

If you have the Admin or SuperAdmin profile, you can perform many of the administrative tasks in the Axon Admin Panel. To access the Admin Panel, perform the following steps:

1. Log in to Axon with an Admin or SuperAdmin profile.
2. From the Axon toolbar, click the **Admin Panel** menu item under your user name.

Axon Architecture

The following image shows the architecture components of Axon:



The following sections describe the architecture components:

Axon

The Axon application consists of the core Axon component, several application services and microservices, and databases. These components exchange information with each other using REST APIs.

Component	Description
Axon Core	The service that allows Axon users to implement data governance in their organization through a Web interface. Axon uses the Apache HTTPD service for the Web interface.
Axon Agent	The service that reads data quality rules in Axon and retrieves the scores from Informatica Data Quality according to the schedules defined in Axon.

Services

Axon uses several application services and microservices to provide specific functionality that enable you to implement data governance. These services are independent modular services that work together to enhance Axon capabilities.

Component	Description
Authorization	Provides access to Axon objects based on the segments that are assigned to users.
Bulk Upload	Uploads a large number of Axon objects and saves them to the PostgreSQL database.
Change Request	Provides a controlled and audited process to make changes to Axon objects. Axon uses the Camunda service for managing and executing workflows within a change request.
Data Marketplace	Enables you to use Data Marketplace. The Data Marketplace allows users to search and access data collections that are created by Axon users. Data collections can be used for business use.
Document Upload	Attaches and maintains reference documents to an Axon object.
Notification	Notifies Axon users about changes to user roles, status of workflows and change requests, and status of bulk upload tasks.
Unison	Allows you to perform the Unison search. When you run a Unison search, Axon searches for the objects and the linked objects in all the available facets.
Value List	Creates or updates a large number of values for data set attributes and saves the values to the PostgreSQL database.

Persistent Layer

The components in the persistent layer store data that can be retrieved by the application services and microservices.

Component	Description
Objects Database	Axon stores all user-created objects in a PostgreSQL database. This is the primary data store.
Search Engine and Graph Database	Axon stores object relationships in an OrientDB database. The objects in this database are generated from the objects database. Axon uses this database for maps and Unison Search. The graph database also enables Quick Search. When you run a Quick Search, Axon searches the database and displays all results that match the search terms that you enter.
Log Search	The service that creates and stores logs for admin activities and user errors.
Messaging	The messaging queues that Axon requests and processes based on the rules defined by the administrator. Axon uses the RabbitMQ service as the message broker.

Informatica Stack

Axon connects to other Informatica products to retrieve and display metadata so that you can make informed business decisions.

Component	Description
Enterprise Data Catalog	Enterprise Data Catalog brings together all data assets in an enterprise and presents a comprehensive view of the data assets and their relationships. Axon integrates with Enterprise Data Catalog to discover and retrieve enterprise metadata.
Data Quality	Data Quality is a comprehensive set of tools that measure and verify if your enterprise data complies with quality standards. Axon integrates with Data Quality to retrieve data quality scores and display them in Axon dashboards.
Data Privacy Management	Data Privacy Management is a data security product that identifies sensitive data across your organization. Axon integrates with Data Privacy Management to display the security level for your enterprise data and metadata.

External Components

Axon connects to external components to securely retrieve user credentials so that users can seamlessly log in to Axon.

Component	Description
SSO Identity Provider	Third-party Identity Providers (IDPs) that integrate with Axon to provide Single Sign-on (SSO) for Axon users in your organization.
LDAP Server	Axon retrieves users and user groups from your organization's LDAP server and creates People objects in Axon.

Users

Users in Axon have privileges that allow them to interact with the Axon application in different ways. To install and manage Axon effectively, you must have users with the correct privileges in the Linux environment and Axon interface.

Users in the Linux Environment

The installer file installs Axon in a home directory that is designated for the installer binaries and application files. This directory must be managed by a non-root user.

To install and manage Axon, you must have the following users in the Linux environment:

Root User

The Linux root user creates the Axon Administrator in Linux and provides a home directory to install Axon. The root user is also required to upgrade Axon from version 5.2 or earlier.

Axon Administrator

This user is a non-root user who installs and manages Axon. The Axon Administrator monitors the Axon application in the Linux environment, troubleshoots Axon services that might have stopped, and configures Axon in Linux to connect with other applications.

The Axon Administrator must have read and write permissions on the Axon installation directory. The Axon Administrator does not require the root or sudo privilege.

User Profiles in Axon Data Governance

Every Axon Data Governance user has a user profile. The type of profile decides the tasks that a user can perform in Axon. An administrator associates user profiles to user accounts.

Users can have one of the following profiles associated with their account:

WebUser

A WebUser is the most basic user profile in Axon. WebUsers have permission to view content.

An administrator can provide advanced permissions to a WebUser on a facet by assigning a role. SuperAdmin users can adjust the permissions applicable to a role. Any user with the Edit privileges for an object can assign roles to WebUsers. The WebUser then becomes a stakeholder for the facet and can perform other tasks to create or manage content for the facet. Stakeholder roles can provide New permissions for a facet and Edit permissions for individual objects. For example, a WebUser with the relevant role and permission can bulk upload data to Axon. A WebUser with the New permission on a Glossary facet can create Glossary objects.

If a SuperAdmin user hides non-public objects from Unison search, the non-public objects do not appear in Unison search results for a WebUser that has no stakeholder role on any object of the facet type.

Admin

Users can perform administrative tasks on facets if they are assigned the Admin profile. Users with the admin profile have access to the Admin Panel.

Users with the Admin profile can perform the following tasks from the Admin Panel:

- Automate data quality rules
- Bulk update objects from the Unison grid
- Bulk upload objects

- Configure default values for mandatory change requests
- Configure saved searches
- Create custom fields for facet objects
- Download logs
- Link Enterprise Data Catalog resources to systems and fields to attributes
- Manage locks
- Send Axon diagnostic information to Informatica Global Customer Support
- View licensed users
- View non-public and deleted objects
- View roles and permissions
- Import migrated data
- Use the static page editor

SuperAdmin

Users with the SuperAdmin profile can perform all administrative tasks in Axon. In addition to the tasks that a user with the Admin profile can perform, a user with a SuperAdmin profile can perform the following tasks:

- Assign roles and permissions
- Create segments and assign users to the segments
- Change the organization logo in the toolbar and the colors of the Axon interface
- Configure access control for users
- Configure audit history period
- Configure automatic glossary assignment
- Configure Axon access to other systems
- Configure change request options
- Configure default values for mandatory change requests
- Configure default facets
- Configure dropdown values
- Configure email delivery options
- Configure regional settings
- Configure threshold for map objects
- Create default workflows for facet objects
- Create or edit roles and permissions
- Delete Axon objects
- Edit the glossary rules engine
- Enable fuzzy search
- Hide non-public objects from Unison search
- Enable data migration
- Enable or disable the dashboard
- Hide fields with empty values

- Import data
- Specify prefix for object reference numbers
- Specify support email address
- Use the static page editor
- View content of log files
- Transfer the ownership of dashboards and shared searches

CHAPTER 2

Integrating Axon with other Informatica Products

This chapter includes the following topics:

- Integration Overview, 18
- Configure Access to Enterprise Data Catalog, 18
- Configure Access to Informatica Data Quality, 30
- Configure Access to Data Privacy Management, 36

Integration Overview

As the administrator, you can configure Axon to connect with other Informatica products. You typically do these configurations right after you install Axon.

Configure Access to Enterprise Data Catalog

In Axon, you can configure Enterprise Data Catalog parameters from the Admin Panel. You must have the Super Admin profile to perform this task. After you configure the parameters, you can test the connectivity to Enterprise Data Catalog to ensure the entered parameters are correct.

After you install Axon, you can configure Enterprise Data Catalog parameters from the Admin Panel in the following scenarios:

- You did not select advanced configuration during Axon installation.
- You want to update the connection settings for Enterprise Data Catalog after the installation.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Enterprise Data Catalog**.

4. Click **Edit**, and configure the following properties:

Property	Description
Enterprise Data Catalog Server Host	URL of the Enterprise Data Catalog service in the following format: `http(s)://<host_name>` or `http(s)://<IP_address>`
Enterprise Data Catalog Server Port	Port number of the Enterprise Data Catalog service.
Enterprise Data Catalog Server Login User Name	User name to log in to Enterprise Data Catalog server.
Enterprise Data Catalog Server Login Password	Password to log in to Enterprise Data Catalog server.
Enterprise Data Catalog Server Login Namespace	User security domain of Enterprise Data Catalog server
Enterprise Data Catalog Resource Name	Name of the resource created in Enterprise Data Catalog that connects to your Axon application.
Axon Super Admin Email	Required. The email address of the SuperAdmin user responsible for onboarding discovered objects.
Automatically Accept Lineage Recommendations	Create attribute links automatically based on the lineage recommendations from Enterprise Data Catalog. If you do not select this option, you can manually create links later from the lineage recommendation notifications.
Enable Lineage Recommendation Notifications	Send notifications for the recommended attribute relationships based on the lineage in Enterprise Data Catalog.
Default Glossary	The default glossary name that you want to associate with the objects onboarded from Enterprise Data Catalog. Axon uses this default glossary if you choose to not onboard objects with associated glossaries to a System object.
Automatically Delete Attribute Relationships	Automatically deletes attribute relationships that are no longer valid if a linked physical field is deleted from Enterprise Data Catalog. This deletion applies to attribute relationships that are automatically created based on the lineage recommendations from Enterprise Data Catalog.
View Custom Attributes for Physical Fields	View custom attributes for physical fields from Enterprise Data Catalog.
Restrict Object Access Based on Enterprise Data Catalog	Restrict access to resources in Axon based on the access permissions to the assets in Enterprise Data Catalog. If you do not select this option, you can access all the assets in Axon irrespective of the access permissions to the assets in Enterprise Data Catalog. The default value is "False." **Note:** The access permission for a user from Enterprise Data Catalog is retrieved based on the email address. Ensure that the email filtering option is enabled in Enterprise Data Catalog. For more information, refer to *Informatica 10.4.1 Enterprise Data Catalog User Guide* .
Configure Timeout Period for Enterprise Data Catalog API	The maximum time period in seconds that Axon waits for the Enterprise Data Catalog API to respond.

Property	Description
Enterprise Data Catalog Server Proxy Host	The proxy host name for the Enterprise Data Catalog server. For more information on configuring the proxy parameters, contact Informatica Global Customer Support.
Enable Onboarder Failure Email Notification	Activate an email alert for the SuperAdmin when a data onboarding job fails. Select the checkbox to enable notification to SuperAdmin for onboarding process failures.
Enterprise Data Catalog Server Proxy Port Number	The proxy port number for the Enterprise Data Catalog server. For more information on configuring the proxy parameters, contact Informatica Global Customer Support.
Resume Failed Onboarding Job	Resume the previously failed onboarding job from the point of failure instead of restarting it. Select this check box to resume the previously failed onboarding job.

5. To test the connectivity to Enterprise Data Catalog, click **Test Connection**.

   If all the parameters are correct, you can view that the connection is successful. If any parameter is incorrect, the connection fails and an error message appears.

   Note: You cannot test the connectivity for the Enterprise Data Catalog proxy parameters.

6. Click **Save**.

7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Catalog Notifications

If the relationships between attributes are created or deleted, Axon sends notifications to stakeholder based on the configuration of Enterprise Data Catalog properties in the Admin Panel.

Created Link Notifications

Axon sends email and interface notifications to the stakeholders based on configuration of the **Automatically Accept Lineage Recommendations** and **Enable Lineage Recommendation Notifications** options on the Admin Panel.

The following table describes the different combination of the values based on which Axon sends lineage notifications:

Automatically Accept Lineage Recommendations	Enable Lineage Recommendation Notifications	Notifications
True	True	Data Steward receives email and interface notifications about the automatically created relationships.
True	False	Data Steward receives email and interface notifications about the automatically created relationships.
False	True	Data Steward receives email and interface notifications about the lineage recommendations.
False	False	Axon does not send any notification.

Deleted Link Notifications

Axon sends email and interface notifications about the deleted links to the stakeholders based on configuration of the **Automatically Delete Attribute Relationships** and **Enable Lineage Recommendation Notifications** options on the Admin Panel.

The following table describes the different combination of the values based on which Axon sends link removal notifications:

Automatically Delete Attribute Relationships	Enable Lineage Recommendation Notifications	Notifications
True	True	Links are automatically deleted and the Data Set stakeholder receives email and interface notifications about the deleted links.
True	False	Links are automatically deleted and the Data Set stakeholder receives email and interface notifications about the deleted links.
False	True	Links are not deleted and the Data Set stakeholder receives email and interface notifications about the relationships that are not valid.
False	False	Links are not deleted and Axon does not send any notification.

Configure Custom Resource Types

In Axon, you can view custom resource types from Enterprise Data Catalog. For more information on custom resources, refer to *Informatica® Catalog Administrator Guide*.

To view a custom resource type, perform the following steps:

1. Create a file with the name `custom_edc_fields.yml` in the following location:
   `<INSTALLATION_DIR>/axonhome/axon/src/Diaku/EIC/EICBundle/Resources/config/queries/custom`
2. Use the following syntax to configure the parameters in the `custom_edc_fields.yml` file:

```
EDC_FIELDS :
```

```
<FIELD_UNIQUE_NAME> :
   classType : "<Class Type for the field>"
   label : "<Unique label for the field>"
   parentClassType : "<Parent Class Type for the field>"
   parentIconFileName : "<Generic icon for the parent file name>"
```

To start configuring the parameters for a field in a custom resource type in the `custom_edc_fields.yml` file, ensure that you start with the `EDC_FIELDS :` parameter. In `FIELD_UNIQUE_NAME`, enter a unique name for the field in Enterprise Data Catalog resource that you want to display in Axon. Within `FIELD_UNIQUE_NAME`, configure the following custom resource type parameters :

Parameter	Description
`classType`	Required. The class type of the Enterprise Data Catalog field.
`label`	Required. A unique label that you want to see in the Axon interface for this field.
`parentClassType`	Required. The parent class type of the Enterprise Data Catalog field.
`parentIconFileName`	Optional. The generic icon that you use for the parent file name.

To configure multiple custom resource types, create a separate section for each field.

3. Before you deploy the configuration on the Axon server, validate the file using a YML validator tool.
4. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Whenever you modify the `custom_edc_fields.yml` file, make sure that all Axon users clear their browser cache before they use Axon.

Example

The following sample file contains a field with parameters configured for a custom resource type:

```
EDC_FIELDS :
    RELATIONAL_COLUMN :
       classType : "com.ldm.custom.relational.column"
       label : "RELATIONAL_COLUMN"
       parentClassType : "com.ldm.custom.relational.table"
       parentIconFileName : "dbcolumn"
```

In the example, the file starts with the `EDC_FIELDS :` parameter. The `RELATIONAL_COLUMN` field shows the configured parameters.

Configure Data Onboarding Rules

You can configure data onboarding rules to include or exclude objects that are onboarded from Enterprise Data Catalog to Axon. You must be an Admin or SuperAdmin user to configure data onboarding rules.

When you create, edit, and delete a data onboarding rule from the Admin Panel, Axon records the activity in Admin logs.

Creating a Data Onboarding Rule

Configure data onboarding rule parameters to apply filters while data is onboarded from Enterprise Data Catalog to Axon.

1. From the Axon toolbar, click your user name and click **Admin Panel**.
2. From the navigation pane, click **Operational Management** > **Data Onboarding Rules**.

   You can view the list of existing rules.
3. To create a data onboarding rule, click **Create**.
4. In the **Definition** section, enter a name and description for the data onboarding rule.

   The **Name** and **Description** fields are required.
5. In the **Rules for Onboarding** section, configure the following properties:

   Context

   Specify the context based on which you want the objects to be onboarded from Enterprise Data Catalog to Axon. You can select the parent, type, or path. Parent refers to the parent of a physical field. Type refers to the class type of a physical field. Path refers to the location of a physical field.

   Regular Expression

   Enter a regular expression to specify a pattern based on which you want the objects to be onboarded from Enterprise Data Catalog to Axon. The regular expression that you enter is case-sensitive.
   To know more about the regular expressions that you can use, see "Supported Characters in Regular Expressions" on page 28.

   Note: If you want to enter the slash (/) in a regular expression, ensure that you first enter the backslash (\) followed by the slash (/). For example, if you select Path, enter `CMAXON\/AXON_CUSTDATA` in the regular expression field.

   Action

   If you want to add more rows, click **+**. If you want to delete rows, click **-**.

   If you add multiple rows in the **Rules for Onboarding** section, the data is onboarded based on the combination of all the conditions that you specified in the rows.
6. In the **Onboarding Rule Configuration** section, select the following options:

   Requires an Associated Glossary

   If you select this option, the objects are onboarded to an Axon system only if a physical field or data domain is linked to a glossary. If you do not select this option and if a physical field is not linked to a glossary or data domain in Enterprise Data Catalog, the default glossary that is configured in the Admin Panel is used for automated onboarding of objects.

   Marked as a Key Data Element

   If you select this option, the objects are onboarded to an Axon system only if a physical field or data domain is connected to a glossary that is a key data element. If you select this option and if the default glossary configured in the Admin Panel is used for onboarding, the default glossary has to be a key data element.

   Marked as a Certified Object

   If you select this option, the objects are onboarded only if a physical field is marked as a certified object in Enterprise Data Catalog.

Confidence Score (50 - 100)

Enter the confidence score threshold value based on which you want to onboard the object from Enterprise Data Catalog. You can enter a value from 50 to 100. The default confidence score value is 95%.

If you want to onboard the values of the **Accepted** field, you can set the confidence score as 100 while configuring the onboarding rule.

If you want to onboard the values of the **Inferred** field with the data match of 100% on Enterprise Data Catalog and **Accepted** field, set the confidence score as 95 while configuring the onboarding rule.

The confidence score value is applicable if the **Requires an Associated Glossary** option is selected. If the **Requires an Associated Glossary** option is not selected, the onboarded attribute has 100% confidence score.

7. In the **Onboarding** section, configure the following properties:

   Name

   Select the name that you want to use for the onboarded attribute. You can choose to use the glossary or physical field name. The default value is "Use Glossary Name."

   Description

   Select the description that you want to use for the onboarded attribute. You can choose to use the glossary or physical field description. The default value is "Use Glossary Description."

8. In the **Roles Inherited From System**, configure the following properties:

   System Role

   The role that the onboarding job uses to get the stakeholders. While onboarding, the Dataset create job gets the stakeholders for this role from the system and assigns it to the mapped dataset role. To add or remove stakeholders, see "Configure Roles" on page 40.

   Dataset Role

   The child role that inherits stakeholders from the mapped system role. For example, Data Steward (dataset role) is mapped to the System Steward (system role). You must map a unique system role with a unique dataset role; duplicating role mappings is not allowed.

   To map dataset roles to system roles, click the plus icon () and add a dataset role and a relevant system role.

   Note: Mapping dataset roles with system roles is optional.

9. Click **Save and Close**.

   You can view the onboarding rule that you created.

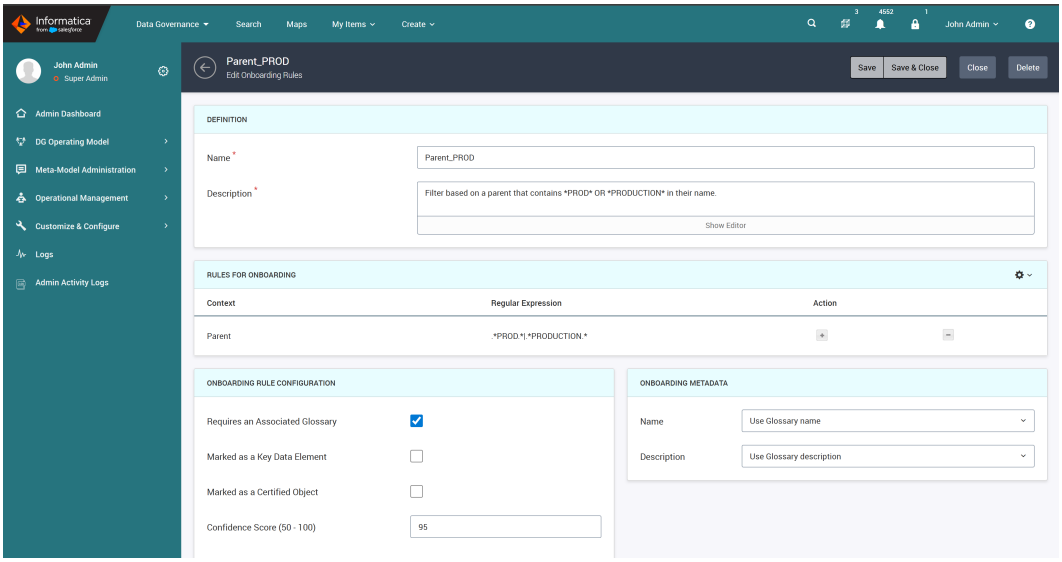
Data Onboarding Rule Examples

You can configure data onboarding rule parameters, such as regular expressions, metadata, and glossary requirements. Based on the rule configuration, the data sets and attributes are onboarded from Enterprise Data Catalog to Axon systems.

Consider the following examples for onboarding rules:

Specify a Parent Name

The following image shows a sample onboarding rule with a regular expression for filtering objects based on a specific parent name:



Consider that you want to filter onboarding of objects based on the parent that contains "PROD" or "PRODUCTION" in the name. In this example, the onboarding rule has the following filter criteria:

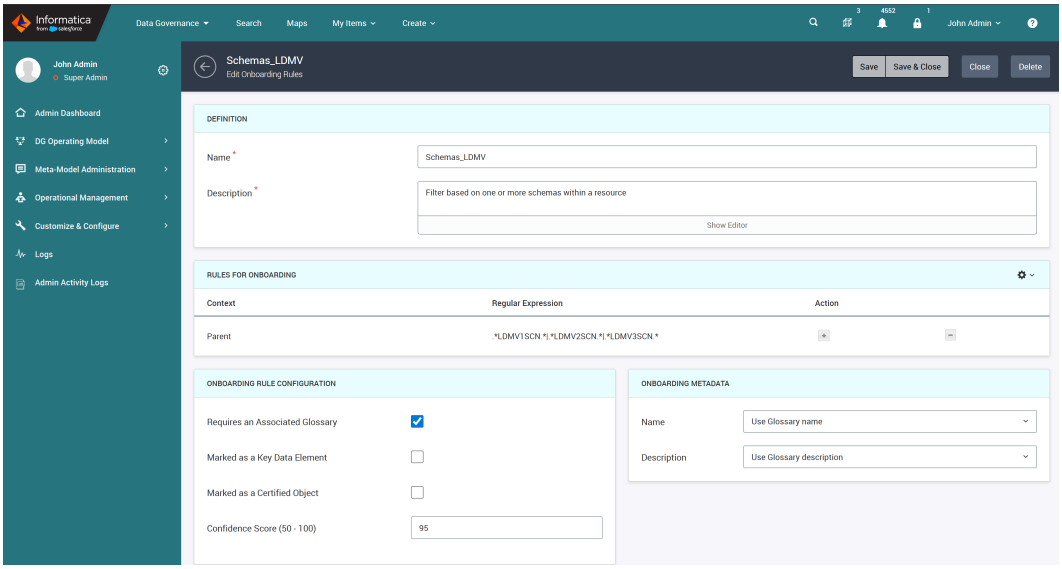
- Context is "Parent" to find the parent with the name `.*PROD.*|.*PRODUCTION.*` specified in the regular expression.
- A glossary association is required for a physical field in Enterprise Data Catalog.
- Confidence score threshold value must be 95%.
- Onboarded attributes must have the name same as the glossary name and the description same as the glossary description.

Based on the rule configuration, the filter is applied on physical fields with the parent that has "PROD" or "PRODUCTION" in the name. During onboarding process, the job checks whether the physical fields are associated to a glossary and then attribute is onboarded with the glossary name and description.

To know more about the regular expressions that you can use, see "Supported Characters in Regular Expressions" on page 28.

Specify One or More Schemas Within a Resource

The following image shows a sample onboarding rule with a regular expression for filtering objects based on one or more schemas within a resource:



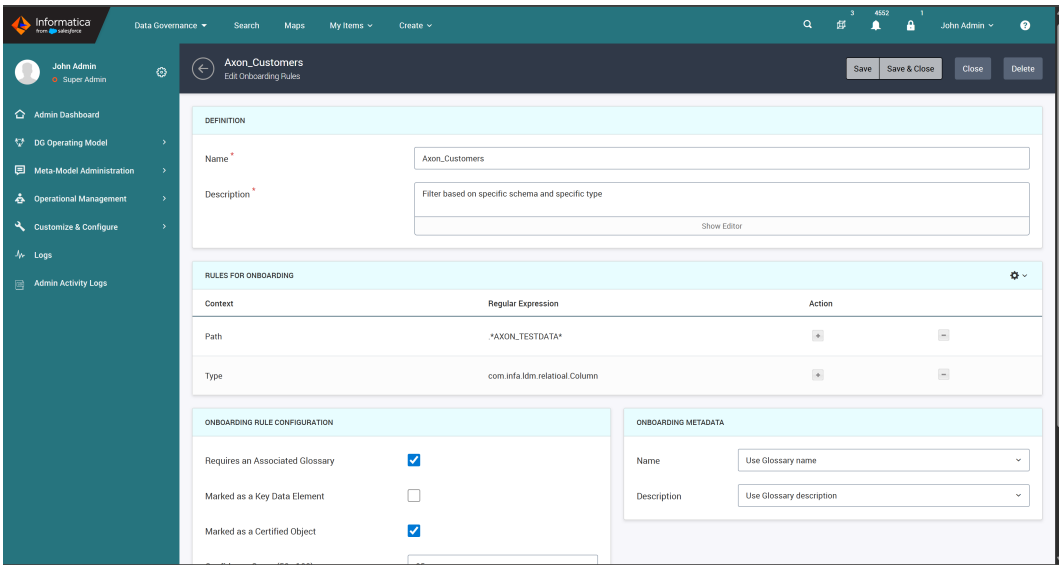
Consider that you want to filter onboarding of objects based on one or more schemas, such as LDMV1SCN, LDMV2SCN, or LDMV3SCN, within a resource. In this example, the onboarding rule has the following filter criteria:

- Context is "Path" to locate the schemas with names `.*LDMV1SCN.*|.*LDMV2SCN.*|*.LDMV2SCN.*` specified in the regular expression.
- A glossary association is required for a physical field in Enterprise Data Catalog.
- Confidence score threshold value must be 95%.
- Onboarded attributes must have the name same as the glossary name and the description same as the glossary description.

Based on the rule configuration, the filter is applied on resources that contain schema names, such as LDMV1SCN, LDMV2SCN, or LDMV3SCN. During onboarding process, the job checks whether the physical fields are associated to a glossary and then data is onboarded with the glossary name and description.

Specify a Schema and Type

The following image shows a sample onboarding rule with a regular expression for filtering objects based on a specific schema and type:



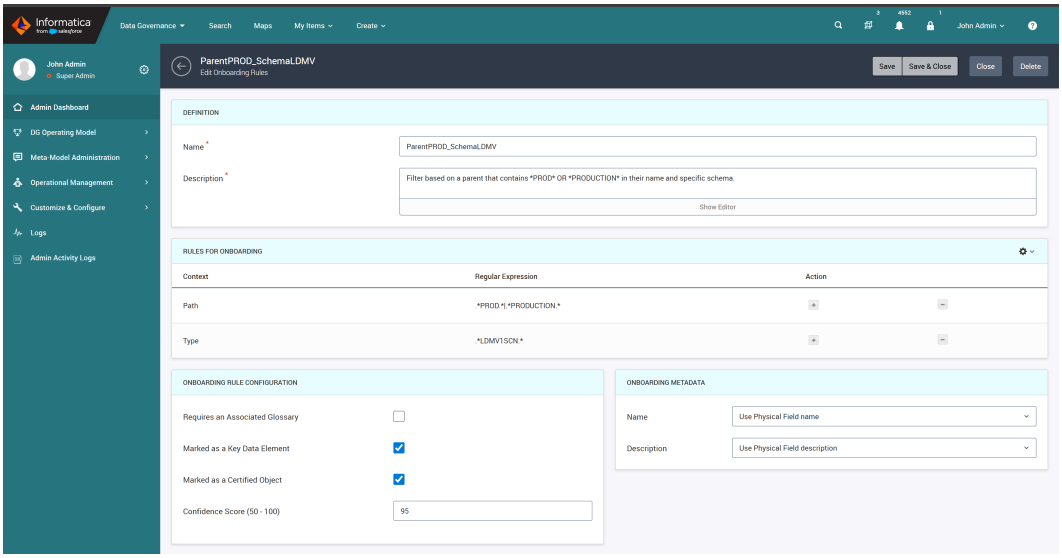
Consider that you want to filter onboarding of objects based on the schema name "AXON_TESTDATA" and physical field of Column type. In this example, the onboarding rule has the following filter criteria:

- Context is "Path" to locate the schema with name `.*AXON_TESTDATA.*` specified in the regular expression.
- Context is "Type" to search for physical field of type `com.infa.ldm.relational.Column` specified in the regular expression.
- A glossary association is required for a physical field in Enterprise Data Catalog.
- Physical fields must be certified objects in Enterprise Data Catalog.
- Confidence score threshold value must be 95%.
- Onboarded attributes must have the name same as the glossary name and the description same as the glossary description.

Based on the rule configuration, the filter is applied on resources that contain the schema name "AXON_TESTDATA" and the physical field of type "com.infa.ldm.relational.Column". During onboarding process, the job checks whether the physical fields are marked as certified objects and associated to a glossary in Enterprise Data Catalog. Then, data is onboarded with the glossary name and description.

Specify a Parent Name and Schema

The following image shows a sample onboarding rule with a regular expression for filtering objects based on a specific parent name and schema:



Consider that you want to filter onboarding of objects based on the parent that contains "PROD" OR "PRODUCTION" in the name and schema name "LDMV1SCN." In this example, the onboarding rule has the following filter criteria:

- Context is "Parent" to find the parent with the name `.*PROD.*|.*PRODUCTION.*` specified in the regular expression.
- Context is "Path" to locate the schema with name `.*LDMV1SCN.*` specified in the regular expression.
- Physical fields must be certified objects in Enterprise Data Catalog.
- The glossary that is used for onboarding process must be a key data element.
- Confidence score threshold value must be 95%.
- Onboarded attributes must have the name same as the physical field name and the description same as the physical field description.

Based on the rule configuration, the filter is applied on resources that contains schema name LDMV1SCN and the physical fields with the parent that has "PROD" or "PRODUCTION" in the name. During onboarding process, the job checks whether the physical fields are marked as certified objects in Enterprise Data Catalog. Because the physical field to glossary association is not required for onboarding process, the default glossary specified in the Admin Panel is used for onboarding objects if there are no glossaries associated to physical fields. If the default glossary is a key data element, then data is onboarded with the physical field name and description.

Supported Characters in Regular Expressions

You can use the following characters in the regular expression that you define in a data onboarding rule:

Character	Function
$	Matches the end of the input.
(?:x)	Matches the value "x" but does not remember the match.
(x)	Matches the value "x" and remembers the match.

Character	Function
*	Matches zero or any number of occurrences of the preceding element. For example, xy*z matches "z", "xz", "xyz", and "xyyz".
+	Matches 1 or any number of occurrences of the preceding element. For example, xy +z matches "xyz", "xyyz", and "xyyyz".
.	Matches any single character except for newline characters.
?	Matches zero or one occurrence of the preceding element. For example, flavo?r matches both "flavour" and "flavor" but it does not match "flavoour".
[\b]	Matches a backspace.
[^abc]	Matches anything that is not enclosed in the brackets.
[abc]	Matches any of the enclosed characters. You can create range of characters using the hyphen (-) character, such as A-Z (A to Z). In character sets, special characters, such as ., *, and +, do not have any special meaning.
\	- Indicates that the next character must not be literally interpreted. For example, the character "w" is interpreted as match the character "w". If you use "\w", it matches an alphanumeric character including the underscore (_). - Indicates that a metacharacter must be literally interpreted. For example, the "." metacharacter indicates to match any single character but with a new line. If you use "\.", it matches the dot (.) character.
\0	Matches a null character.
\b	Matches a word boundary. Boundaries are determined when a word character is neither followed nor preceeded by another word character.
\B	Matches a non-word boundary. Boundaries are determined when two adjacent characters are word characters or non-word characters.
\cX	Matches a control character. The character "X" must be included within A to Z.
\d	Matches a digit character. For example, you can enter [0-9] or [0123456789].
\D	Matches a non-digit character. For example, you can enter [^0-9] or [^0123456789].
\s	Matches a single white space character. This includes a space, tab, form feed, and line feed.
\S	Matches anything other than a single white space character. It matches anything other than a space, tab, form feed, and line feed.
\t	Matches a tab.
\w	Matches any alphanumeric character including the underscore (_). For example, it matches [A-Za-z0-9_].
\W	Matches anything other than an alphanumeric character including the underscore (_). For example, it matches [^A-Za-z0-9_].

Character	Function
^	Matches the beginning of the input.
x(?!y)	Matches the character "x" only if "x" is not followed by character "y".
x(?=y)	Matches the character "x" only if "x" is followed by character "y".
x|y	Matches either the character "x" or "y".
{n,m}	Matches the preceding character at least "n" times and at the most "m" times. If the values of "n" and "m" are zero, they are omitted.
{n}	Matches the preceding character exactly n times.

Connect to an SSL-Enabled Enterprise Data Catalog Instance

To connect Axon to an SSL-enabled instance of Enterprise Data Catalog, you must upgrade cURL and the related libraries after you install Axon. If you do not upgrade, objects might not appear in Axon facets..

You must have the following library versions running on your Linux system:

- cURL version 7.29.0 or higher
- Network Security Services (NSS) version 3.21 or higher

To check the existing versions of cURL and NSS libraries, run the following command:

```
curl --version
```

To upgrade the cURL and NSS libraries to the latest versions, run the following commands:

```
yum upgrade curl
yum upgrade nss
```

Configure Access to Informatica Data Quality

If you have installed the Axon Agent, you must specify the details of the Axon Agent installation in Axon. Axon uses the parameters you specify to connect to Informatica Data Quality. You can open profiles or scorecards in Informatica Analyst tool from Axon local data quality rules.

You must have the SuperAdmin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Data Quality**.
4. Click **Edit**, and configure the following properties:

Property	Description
Axon Agent Server Host	Host name of the system where you installed the Axon Agent.

Property	Description
Axon Agent Server Port	Port number on which the Axon Agent runs.
IDQ Instance URL	URL of the Analyst Tool. To create a clickable link that associates data quality rule with profiles and scorecards, enter the Analyst Tool URL followed by the slash (/) character. For example, enter the URL in the following format: `http(s)://<hostname>:<port>/analyst/`

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Working with the Axon Agent

The Axon Agent is the service that exchanges information between Axon and Informatica Data Quality.

The Axon Agent reads the data quality rules in Axon that you create for profiles and scorecards. The Axon Agent also reads the scheduling information for the data quality rules.

At the scheduled time, the Axon Agent retrieves data quality scores from Informatica Data Quality. The Axon Agent then updates the corresponding data quality rules in Axon with the scores.

The following sections provide information on setting up and configuring the Axon Agent if it does not work as expected.

Start or Stop the Axon Agent

If the Axon Agent stops, you can restart the Axon Agent. For example, if you want to modify the connection details of the Axon Agent, you must stop and restart the Axon Agent.

Run the following command to start the Axon Agent:

```
<Axon_Agent_Home>/sh startup.sh
```

Run the following command to stop the Axon Agent:

```
<Axon_Agent_Home>/sh shutdown.sh
```

When the Axon Agent starts, it verifies that the Informatica Domain, Model Repository Service, and the Data Integration Service are running. The Axon Agent does not start if any of these components is not running.

Monitor the Axon Agent

When the Axon Agent is running, you can monitor its status by viewing the log file.

Open the following file to monitor the Axon Agent:

```
<Axon_Agent_Home>/logs/dq-axon/dqagent.log
```

Modify Connection Details

You must configure the Axon Agent to connect to other components that are required by Axon. For example, if you update the LDAP namespace of the Informatica domain or update the Axon application URL, you must update the connection details in the Axon Agent to point to the new location.

Connect to LDAP-Configured Informatica Domain

To point the Axon Agent to the LDAP-configured Informatica domain, edit the `dis.securityDomain` and `mrs.securityDomain` properties in the following file:

```
<Axon_Agent_Home>/conf/application-mrs-dis.properties
```

Connect to Informatica Domain, Model Repository Service, or Data Integration Service

To point the Axon Agent to a new location for the Informatica Domain, Model Repository Service, or the Data Integration Service, edit the properties in the following file:

```
<Axon_Agent_Home>/conf/application-mrs-dis.properties
```

Connect to Axon

To point the Axon Agent to a new location for the Axon application, edit the properties in the following file:

```
<Axon_Agent_Home>/conf/application-axon.properties
```

Change Password

To change the password for Axon, Model Repository Service, or Data Integration Service, run the following command during startup:

```
<Axon_Agent_Home>/sh startup.sh --mrs.userName=Administrator --
mrs.password=<MRS_password> --dis.userName=Administrator --dis.password=<DIS_password> --
axon.userName=admin@informatica.com --axon.password=<Axon_password>
```

Note: Do not update the password by modifying the `application-mrs-dis.properties` and `application-axon.properties` files.

Configure Synchronization Delay and Synchronization Interval

You can update the synchronization delay and synchronization interval between Axon, the Axon Agent, and Informatica Data Quality.

Synchronization Delay Between Axon and the Axon Agent

After the Axon Agent starts, it synchronizes with Axon. Synchronization delay is the time after which Axon and the Axon Agent start synchronization after the Axon Agent is started. For example, if you set the synchronization delay to 5 minutes, Axon and the Axon Agent start the synchronization task 5 minutes after the Axon Agent starts.

1. To configure the synchronization delay, modify the following file:
   `<Axon_Agent_Home>/conf/application-app.properties`
2. Modify the value for the `app.syncDelay` property. Enter the synchronization delay in minutes. The default value is 1 minute.

Synchronization Interval Between Axon and the Axon Agent

You can configure the interval frequency at which Axon and the Axon Agent synchronize with each other. For example, if you set the synchronization interval to 3 minutes, Axon and the Axon Agent synchronize with each other every 3 minutes.

1. To configure the synchronization interval, modify the following file:
   `<Axon_Agent_Home>/conf/application-axon.properties`
2. Modify the value for the `axon.delayInterval` property. Enter the synchronization interval in minutes. The default value is 1 minute.

Note: The synchronization interval you configure determines the frequency of login logs in Axon. Each synchronization interval generates two login log entries.

Synchronization Interval Between the Axon Agent and Informatica Data Quality

You can configure the interval frequency at which the Axon Agent and Informatica Data Quality synchronize with each other. For example, if you set the synchronization interval to 60 minutes, the Axon Agent and Informatica Data Quality synchronize with each other every 60 minutes.

1. To configure the synchronization interval, modify the following file:
   `<Axon_Agent_Home>/conf/application-axon.properties`
2. Modify the value for the `axon.ruleDelayInterval` property. Enter the synchronization interval in minutes. The default value is 120 minutes.

Configure Axon with Operating System Profiles Enabled User for Data Integration Service

After you install or upgrade Axon, you can configure the OS profile enabled user with Data Integration Service. After you configure the profile, you can use the operating system profile in the Axon Agent to run Data Quality rules.

To update the operating system profile user name, choose any of the following options:

Start DQAgent with your operating system profile user credentials

1. In the command prompt, navigate to the following Axon Agent home location:
   `DQAGENT_HOME` directory
2. To start the Axon Agent with the operating system profile user name, run the following command:
   `sh startup.sh --dis.ospUsername=<OSP Username>`

Configure DQAgent with the updated `application-mrs-dis properties` file

1. Navigate to the following Axon Agent home location:
   `DQAGENT_HOME/conf/directory`
2. Edit the `application-mrs-dis.properties` file.
3. Update the `dis.ospUsername` parameter with the user name enabled with the Data Integration Service.
4. To start the Axon Agent, run the following command from the command prompt:
   `sh startup.sh`

Configure Data Quality Rule Recommendation

You can modify the default Axon settings to recommend and automatically create rules in Informatica Data Quality.

1. In the Linux system, open the `<Axon_Agent_Home>/conf/application-app.properties` file.

2. To configure the supported date format for technical descriptions, modify the `app.generateRulesDateFormat` parameter.

   The default format is MM/DD/YYYY. To see the supported date values that you can enter, see the *TO_DATE and IS_DATE Format Strings* topic in the *Informatica Data Quality Transformation Language Reference* on the Informatica Documentation Portal.
3. To configure the project in Informatica Data Quality where Axon saves the data quality rules, modify the `app.generatedRulesProject` parameter.
4. Save the file.

Configure Data Quality Rule Automation

Using data quality rule automation, you can automatically run data quality rules on fields scanned by Enterprise Data Catalog and display the scores in Axon. Axon acts as an interface between Enterprise Data Catalog and Informatica Data Quality to run the rules. Configure the properties for rule automation.

You must have the Admin or Super Admin profile to perform this task.

Make sure that the PowerExchange for JDBC license option is present on the license that is used by the Data Integration Service in Informatica Data Quality.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **Application Settings**.
3. In the **Application Settings** page, click **Data Quality**.
4. Configure the following properties for data quality rule automation:

Property	Description
Auto-generate local rules for standard rules	Select to automatically run data quality rules.
Recommend Data Quality Rules	Select to recommend data quality rules using Informatica CLAIRE® and automatically create rules in Informatica Data Quality.

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settingsn** pages. Additionally, perform this action during a maintenance period when very few users use Axon.

Specify Output Values for Automated Rules

If you choose to automate data quality rules, the output values of the rules in Informatica Data Quality must conform to the output criteria. You can modify or remove the default values, or add your own values. For example, you can specify `OK` as an output value.

When you install Axon, the following output values are defined by default:

- Valid
- True
- Yes
- 1

The output values can be in uppercase or lowercase. If the output values do not conform to the value criteria, Axon displays the data quality scores as 0% and marks all row counts as exceptions.

1. Run the following command to stop the Axon Agent:

   ```
   <Axon_Agent_Home>/sh shutdown.sh
   ```

2. Open the `<Axon_Agent_Home>/conf/application-app.properties` file.
3. Add, remove, or modify the output values in the `app.edgeRuleValidValues[index]` parameters.

   Make sure that you follow these guidelines when you add, remove, or modify output values:

   - The index values in the `app.edgeRuleValidValues[index]` variable must be positive integer values. Do not enter duplicate index values.
   - The `application-app.properties` file supports only Latin-1 (ISO-8859-1) encoded characters. To use a character that is not encoded in Latin-1, enter the character in Unicode escape notation.
   - Do not modify any other property in the file.
4. Save the file.
5. Run the following command to start the Axon Agent:

   ```
   <Axon_Agent_Home>/sh startup.sh
   ```

Example: Add OK and Delta Symbol as Output Values

You want to add `OK` and the Delta symbol as output values for data quality rules.

The following image shows the default values in the `application-app.properties` file:

```
# Application level properties
app.jobStatusMonitorQueue = 50
app.jobResultProcessorQueue = 50
app.resultMonitorThreadPoolSize = 1
app.resultProcessorThreadPoolSize = 3
#syncDelay should be minute
app.syncDelay = 1
aap.ruleSyncDelay = 1

# List of Valid Values
app.edgeRuleValidValues[0] = 1
app.edgeRuleValidValues[1] = Yes
app.edgeRuleValidValues[2] = True
app.edgeRuleValidValues[3] = Valid
app.edgeConnectionName = PWH
```

The Delta symbol is not encoded in Latin-1 scheme. The Unicode escape notation for the Delta symbol is `\u225C`. The following image shows the new values in the in the `application-app.properties` file:

```
# Application level properties
app.jobStatusMonitorQueue = 50
app.jobResultProcessorQueue = 50
app.resultMonitorThreadPoolSize = 1
app.resultProcessorThreadPoolSize = 3
#syncDelay should be minute
app.syncDelay = 1
aap.ruleSyncDelay = 1

# List of Valid Values
app.edgeRuleValidValues[0] = 1
app.edgeRuleValidValues[1] = Yes
app.edgeRuleValidValues[2] = True
app.edgeRuleValidValues[3] = Valid
app.edgeRuleValidValues[4] = OK
app.edgeRuleValidValues[5] = \u225C
app.edgeConnectionName = PWH
```

Configure Access to Data Privacy Management

If you selected the advanced configuration option during Axon installation, you would have entered the configuration settings for Data Privacy Management. If you did not select advanced configuration during Axon installation or if you want to update the connection settings for Data Privacy Management after the installation, go to the Axon Admin Panel.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Data Privacy Management**.
4. Click **Edit**, and configure the following properties:

Property	Description
Data Privacy Management Server Host	URL of the Data Privacy Management service in the following format: `http(s)://<host_name>` or `http(s)://<IP_address>`
Data Privacy Management Server Port	Port number of the Data Privacy Management service
Data Privacy Management Server Login User Name	User name to log in to the Data Privacy Management server
Data Privacy Management Server Login Password	Password to log in to the Data Privacy Management server
Data Privacy Management Server Login Namespace	User security domain of Data Privacy Management

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

CHAPTER 3

User Administration

This chapter includes the following topics:

- User Administration Overview, 38
- User Roles, 38
- Segmentation, 40
- Retrieve Users from LDAP Directory to Axon, 46
- Configure Single Sign-On, 53
- Configure Role Assignments, 57
- Configure Access Control, 58
- Change Default Administrator Password, 59
- Unlock Locked Users, 59
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User Administration Overview

As the administrator, you can view the privileges of Axon users, and manage user permissions so that they can log on to Axon successfully and work with the Axon objects. You can also configure Axon to connect with your organization's LDAP directory or Single Sign-On (SSO) Identity Provider (IDP).

User Roles

All objects that need governance must have an assigned stakeholder. When you describe the assets of an organization, you need to ensure that you assign the assets to responsible people in the organization. You can assign roles to an object stakeholder in Axon to reflect various aspects of responsibilities, such as ownership, stewardship, and subject matter expertise.

You can classify roles based on different role types. You can view dashboards and scorecards for an object based on these role types. From the Admin Panel, you can specify a role type for each role in a facet based on the expertise of the role. You cannot edit the role types.

You can assign different roles to an object stakeholder based on the expertise of user. Select the following role types:

Authority Role

An important stakeholder with no direct responsibility on the object. For example, a project owner, a person who approves funding from another department, or a data protection officer who is responsible for ensuring that they are compliant with GDPR.

Knowledge Role

A stakeholder who has expertise on the object. For example, a Subject Matter Expert can advise the owner and steward to ensure that the object information is accurate and complete.

Ownership Role

A stakeholder who is directly responsible for an object, but might not involve in daily activities for the object except for workflow approval. An owner is responsible to define and manage the workflow of an object.

SMR Role

The Senior Management Regime (SMR) role that is used for UK-based banking regulations.

Stewardship Role

A stakeholder who is responsible to manage the content daily, handle enquiries, and perform most of the workflow activities. A steward manages relationships across facets and coordinates with other stakeholders to ensure data quality.

An Admin profile user can assign stakeholder roles to users. A stakeholder with Edit permissions on an object can also assign stakeholder roles to other users for the object. You can view the stakeholders for an object from the **Stakeholder** tab of the object. You can also perform a Unison search based on the stakeholder roles.

A SuperAdmin profile user configures and manages roles at the facet level. Axon has two default roles per facet, an object Owner and object Steward. For each facet, you can configure role names and number of roles from the Admin Panel.

You can assign stakeholder roles to a user with any profile type. Most Axon users have the WebUser profile, and the role permissions determine which objects they can create or change. If an Admin user is also a stakeholder of an object, the permissions granted to the Admin user profile always overrides their assigned roles. For example, if a System SME role has only View permissions for a System object, users with that role cannot edit the System objects. However, an Admin user with the System SME role can create and edit System objects.

You can assign different permissions to each stakeholder role. Permission control how a user can interact with Axon objects. A stakeholder role can have the following permissions:

- New. Create an object in a facet.
- Edit. Edit an object if the user has a stakeholder role in the object.
- View. View objects. It is recommended that all stakeholder roles have this permission because you can limit the users who can view the objects in Axon.

Any permission for a user lasts till the user has an assigned role on one or more objects. If the stakeholder role is removed from an object, the user does not have any permission on the object.

Configure Roles

Axon provides a number of default roles. You can configure new roles based on the needs of your organization. You can configure different roles for the stakeholders of an object.

You must have the Admin profile to view roles and responsibilities. You must have the Super Admin profile to edit roles and responsibilities.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, click **Roles and Responsibilities** under **DG Operating Model**.
3. Click **Edit**.
4. In the **Roles and Responsibilities** page, click the **+** icon to add a new role.
5. Configure the following properties:

Option	Description
PrimaryName	Provide a name for the role.
Description	Provide a description for the role.
Default	Select the checkbox to make this a default role when you save an object. All objects need at least one default role. If you select this option, you must assign the Edit permission to the role.
Module	From the list, select a module. In the object page, you can filter the grid by module to quickly find the role.
Role Type	From the list, select a role type. The role type affects dashboards and scorecards.

6. In the **Roles and Responsibilities** page, click the **-** icon to delete the role.

   Note: You cannot delete or update roles when the role is used in the following configurations:

   - Workflows
   - Object Stakeholder
   - Data Onboarding Rule
   - Default Change Request
   - Role Assignment

   You can edit the Description, role type, and facet for the object role.

   You can delete the role when the role permission is present.
7. Click **Save and Close**.

Segmentation

Segmentation is the capability to restrict information access to a limited set of users based on their location, business area, and function within an organization. Create and use segments to control access to the content that might be sensitive or requires to be granted limited access due to regulatory needs. Create segments

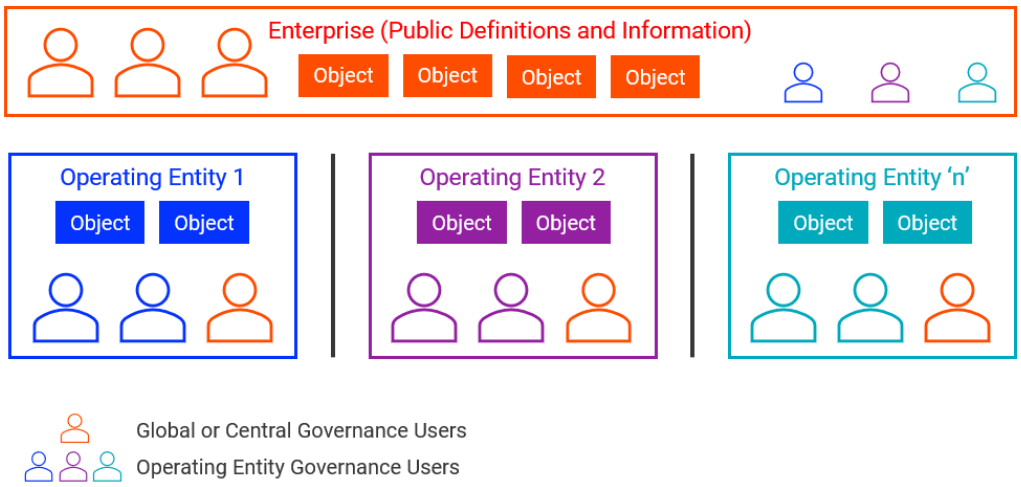
based on your organization requirements such as geography, business area, business function, and legal entity.

In an organization, you can make information available to users for simplicity, security, or regulatory reasons. Some Axon objects are made available to everyone, while some objects are restricted based on the work profile or location of an individual user. For example, a bank in Singapore with offices in Malaysia wants a common platform for data governance, but the regulations might not allow the information from Singapore to be shared with the employees of Malaysia. Similarly, two business areas might want to maintain their own content without sharing it to others for security reasons.

You can restrict access to sensitive information based on rules and regulations. You can update access control capabilities to allow information access based on the role type, profile, and organization structure. You can restrict the data access for the some of the following reasons:

- To prevent access to sensitive information.
- To ensure compliance with country-specific regulations.
- To provide a way of driving multitenancy.

The following image describes different segments and users that are part of the segments:



You can assign an object to the Enterprise segment or a specific restricted segment that you create. An Enterprise segment is a public segment that all users can access. In the example, you can see that the operating entities are defined as segments. Each segment has assigned users and contains Axon objects. Though the operating entity governance users belong to different segments, they can access the Enterprise segment.

You can create multiple segments as per your requirements. You can assign a user to a single segment or multiple segments. You can associate an object to a segment. An object and its parent object must belong to the same segment. Only the users with access to a segment can view the objects that belong to the segment.

If an object belongs to a private segment, the related objects can be part of the same private segment or Enterprise segment. For example, if a Process object belongs to a private segment and a Policy object belongs to Enterprise segment, you can create Policy X Process relationships.

If an object belongs to one private segment, the related objects cannot be part of another private segment. For example, if two Process objects belong to separate private segments, you cannot create the Process X Process relationships.

For a parent-child relationship, the entire hierarchy must be part of the same segment. For example, glossary G1 has parent glossary G2, and glossary G2 has parent glossary G3. If glossary G3 belongs to segment HR, the entire hierarchy of glossaries G3 > G2 > G1 must belong to the same segment HR.

If a system belongs to a Private segment, the data set within the system must belong to the same private segment. If a system belongs to the Enterprise segment, the data set within in the system can belong to the Enterprise segment or any private segment.

A SuperAdmin user can create a segment and assign a Segment Admin user to the segment. You can assign only Admin users as a Segment Admin. Segment Admin users are responsible to manage access to segments. A segment can have one or more Segment Admin users.

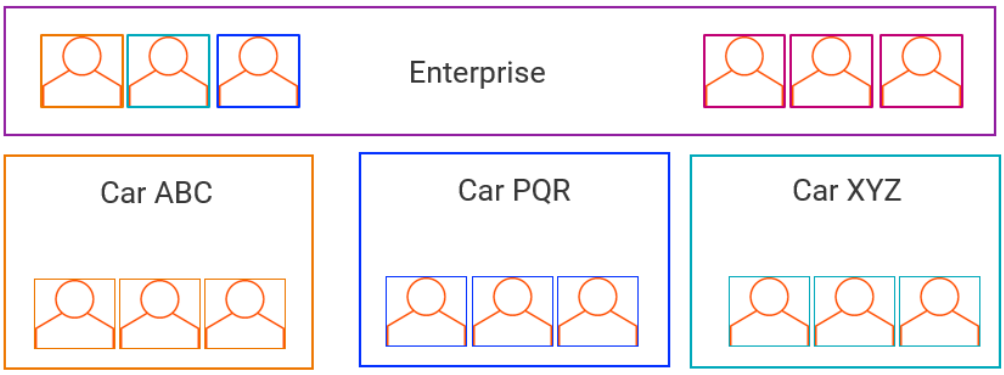
A SuperAdmin or Segment Admin user can associate users to the segments in the following ways:

- Assign individual users to a segment.
- Assign an org unit to a segment. All members of the org unit can access the segment.
- Assign users to a segment via Single Sign-on (SSO).

Segmentation Example

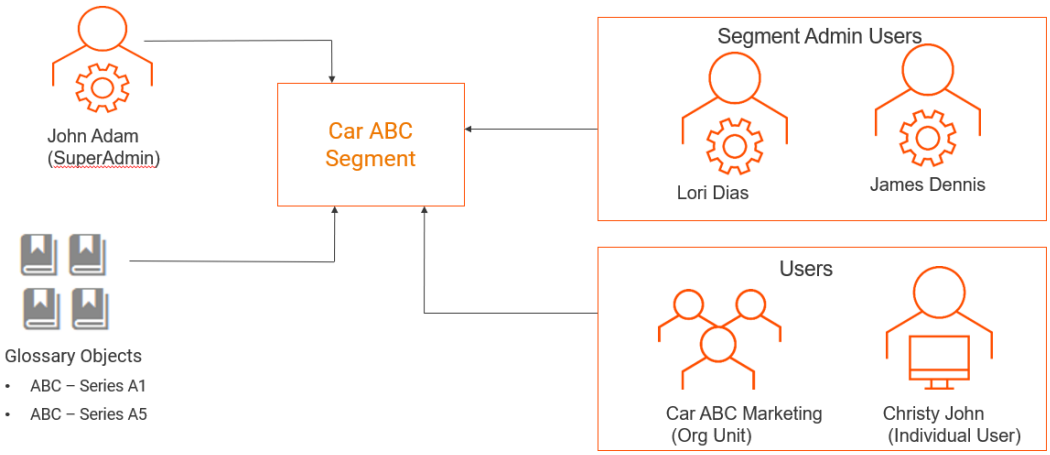
Consider a car manufacturing company that has several subsidiaries, such as Car ABC, Car PQR, and Car XYZ. Each unit manages their own finance data. To restrict information access, you can create a segment for each car subsidiary.

The following image shows an example of different segments for a car company:



All users of each segment can access the Enterprise segment. Assign Segment Admin users and other users for each segment.

The following image shows a segment that is associated to different users and objects:



In the example, a SuperAdmin user creates a segment for the subsidiary Car ABC and adds Segment Admin users. You can add an org unit and individual users to the segment. Create glossaries for the Car ABC

subsidiary and add the glossaries to the Car ABC segment. When you access the ABC - Series A1 and ABC - Series A5 glossaries from the Unison search, you can see that the glossaries belong the Car ABC segment.

Create a Segment

Create and manage segments from the Admin Panel. A SuperAdmin user assigns Segment Admin users to a segment.

1. From the Axon toolbar, click your user name and click **Admin Panel**.
2. From the navigation pane, click **Meta-Model Administration** > **Segments**.

   You can view a list of the existing segments.
3. To create a segment, click **Create**.

   The **Summary** tab appears.
4. In the **Definition** section, enter a name and description for the segment.
5. In the **Segment Admin Users** section, click **Assigned Manually** to assign a Segment Admin user to the segment.

   The page displays the list of Segment Admin users that are manually assigned.
6. Click **Add** to add a Segment Admin user.

   You must assign at least one Segment Admin user to a segment. The Segment Admin user that you add must be a user with an Admin profile.
7. Configure the following properties to assign users to the segment manually:

Property	Description
Name	Name of the Axon user.
Org Unit	Organizational unit name of the Axon user.
Profile	User profile that Axon assigns to all the users.
Function	Function of the Axon user in the organization

8. You can click the name and org unit to modify existing segment users.
9. In the **Segment Admin Users** section, click **Assigned via SSO** to assign a Segment Admin user to the segment through SSO.

   The page displays the list of the SSO properties assigned to the segment for Segment Admin users.
10. Click **Add** to configure the SSO properties to assign users to the segment.
11. Configure the following SSO properties:

Property	Description
Property	SSO properties to retrieve users according to values you specify. For example, `email`. To retrieve users that match the email address that you specify, enter `email`.
Value	Value associated with the property. For example, enter `John@yourorganization.com` to assign user with the email address "John@yourorganization.com".

12. You can click the assigned SSO property to modify the configured property and value.

13. Click **Save and Close**.

    You can view the segment that you created.

Assign Users to a Segment

The SuperAdmin or Segment Admin user can assign Axon users to a segment.

Before you assign Axon users to a segment from an LDAP server or Single Sign-On (SSO) identity provider, you must configure Axon to retrieve users from an LDAP directory or SSO identity provider.

To configure Axon to access the LDAP directory, refer to "Retrieve Users from LDAP Directory to Axon" on page 46. To configure Axon to use SSO to authenticate users based on the Identity Provider (IDP) credentials, refer to "Configure Single Sign-On" on page 53.

1. From the Axon toolbar, click your user name and click **Admin Panel**.
2. From the navigation pane, click **Meta-Model Administration** > **Segments**.

   The page displays the existing segments configured in your axon instance.
3. Click the segment in which you want to add users.

   The **Summary** tab appears.
4. Click the **Assigned Users** tab.
5. To assign users based on organizational units, go to the **Assigned by Org Units** sub-tab.
   a. Click **Add** to add organizational units to the segment.

      The **Select Org Units** dialog box appears.
   b. Select an organizational unit from the list, and click **Select**.

      All the users in the selected org unit is part of the segment.
6. To assign users individually, go to the **Assigned Manually** sub-tab.
   a. Click **Add** to add a new user.

      The **Add Manually dialog** box appears.
   b. Enter the name of the user.
   c. Select an org unit from the list.
   d. Select a profile from the list.
   e. Enter the function of the user.
   f. Click **Add**.
7. To assign users from the Single Sign-On (SSO) identity provider, go to the **Assigned via SSO** sub-tab.
   a. To assign users from the SSO identity provider, click **Add**.

      The **Add SSO Property** dialog box appears.
   b. In the **Property** field, select the SSO property for which you want to retrieve users from the identity provider.
   c. In the **Value** field, add the value of the SSO property for which you want Axon to retrieve SSO users.

      You can also specify a regular expression as the value of the SSO property. The supported regular expression characters are `?` and `*`.
   d. Click **Add**.

      The SSO property is configured. Axon retrieves SSO users based on the configuration.

e. To modify the assigned SSO property, click the SSO property.

   The **Edit SSO Property** dialog box appears.

f. You can change the property and value associated with the selected SSO property.

g. Click **Save**.

8. Click **Save and Close**.

Delete a Segment

When you delete a segment, you need to move all the objects from the segment to the Enterprise segment or another segment. You need to be a SuperAdmin user to delete a segment.

1. From the Axon toolbar, click your user name and click **Admin Panel**.
2. From the navigation pane, click **Meta-Model Administration** > **Segments**.

   You can view a list of existing segments.
3. Click the segment that you want to delete.
4. Click **Delete**.

   The **Delete Segment** dialog box appears.
5. Choose one of the following options:
   - Move all objects from this segment to the default Enterprise segment that all users can access.
   - Move all objects from this segment to another segment. Select a target segment in which you want to move the objects. Ensure that stakeholders of all the objects from the segment that you want to delete have access to the target segment.
6. Click **OK**.

   The segment is deleted after the objects are moved to an existing target segment.

Configure Default Segments

You can configure a default segment for a user who logs in to Axon.

1. From the Axon toolbar, click your user name and click **Admin Panel**.
2. From the navigation pane, click **Customize & Configure** > **System Settings**.
3. In the **Group** list, select **Default Segment**.
4. Click **Edit**.
5. Choose the following options:

Option	Description
Enterprise Segment	Enable to set Enterprise Segment as the default segment for a user to view Axon content.
Assigned Segments	Enable to set Assigned Segments as the default segment for a user to view Axon content.

   Note: If you choose either the Enterprise or assigned segment as the default segment for a user, the segments list on the Unison search page shows the default segment when the user logs in to Axon for the first time. If the user updates the segments selection list, the default segment that you configured does not persist when the user logs in again.
6. Click **Save**.

7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users use Axon.

Disable or Re-enable Segmentation

After you install or upgrade Axon, segmentation is automatically enabled, and you can restrict Axon objects by various segments. To make all Axon objects available to all Axon users, disable segmentation.

You must have the Super Admin profile to perform this task.

If you want to disable segmentation after the segments are created, delete all segments, and then disable segmentation.

1. To disable segmentation, go to the `<INSTALLATION_DIR>/bin` directory in the Linux environment, and run the following command:

   ```
   disable_authorization.sh
   ```

2. To enable segmentation again, go to the `<INSTALLATION_DIR>/bin` directory in the Linux environment, and run the following command:

   ```
   enable_authorization.sh
   ```

Retrieve Users from LDAP Directory to Axon

Axon can connect to the multiple LDAP directories in your organization, retrieve users, assign them user profiles, and display them in the Axon interface. You do not need to manually create users. Axon automatically creates the users after it connects to the LDAP server.

You can configure Axon to retrieve users from a specific domain in the LDAP directory. You can also search for users that match certain criteria within the domain that you specify, and assign them specific user profiles. After the users from the LDAP directory are created, users can log in to Axon using their email addresses and passwords that are configured in the LDAP directory.

Note:

- If you configure Axon to access the LDAP server and, in addition, use SSO to authenticate the users, the SSO authentication takes precedence. When Axon users attempt to log in, the browser displays the login window of the IDP. To configure Axon to use SSO authentication, refer to "Configure Single Sign-On" on page 53.
- If a user forgets the LDAP password in the Axon login page, Axon does not display the **Forgot password** link. The user must contact the system administrator to retrieve or reset the password.

Example

You want to retrieve users from the LDAP directory and assign them user profiles according to the following search critera:

- You want to assign the WebUser profile to all users that match the following search criteria:

  ```
  (&(ou=allUserGroup)(sAMAccountName=*)(objectclass=user))
  ```

  To assign the WebUser profile to LDAP users, enter the search criteria in the **WebUser Search Filter** field in the **LDAP Server** configuration page of the Admin Panel.

- You want to assign the Admin profile to all users that match the following search criteria:

  ```
  (&(ou=adminUserGroup)(sAMAccountName=*)(objectclass=user))
  ```

  To assign the Admin profile to LDAP users, enter the search criteria in the **Admin Search Filter** field in the **LDAP Server** configuration page of the Admin Panel.

- You want to assign the SuperAdmin profile to all users that match the following search criteria:

  ```
  (&(ou=superAdminUserGroup)(sAMAccountName=*)(objectclass=user))
  ```

  To assign the SuperAdmin profile to LDAP users, enter the search criteria in the **SuperAdmin Search Filter** field in the **LDAP Server** configuration page of the Admin Panel.

To specify the search criteria for LDAP users, refer to "Configure Access to the LDAP Server" on page 49.

LDAP Users and Axon Objects

Axon creates new users or modifies the existing users when it retrieves users from the LDAP directory.

Axon creates or modifies users based on the following criteria:

- If new users are retrieved from the LDAP directory, Axon creates **People** objects and assigns to them the user profile that you configure in the Admin Panel.
- If users belong to organizational units in the LDAP directory, Axon assigns them to the matching **Org Unit** objects.
- If users are deleted from the LDAP directory, the **Axon Status** field of the users are marked as `Deleted` in the Axon interface.
- If the **Axon Status** value of a user is `Deleted`, and the user is retrieved from the LDAP directory, the **Axon Status** value of the user changes from `Deleted` to `Active` in the Axon interface.
- To retrieve users and create **People** objects in Axon, the users in the LDAP directory must have valid entries for the following fields:

  - First Name
  - Last Name
  - Distinguished Name
  - Email

  If the entry for any of these fields is invalid or blank, Axon does not create the users.

- If users in the LDAP directory are modified, these users are updated in Axon. Axon updates the following fields:

  - User First Name
  - User Last Name
  - Function Name
  - Function Description

- Org Unit Name
- Office Location
- Internal Mail Code
- Office Telephone
- Mobile Phone

- The following fields in Axon are read-only for users retrieved from the LDAP directory:
  - Email
  - Password
  - LAN ID
- If the search retrieves LDAP users that require multiple user profiles, the profile with the greater privileges apply to the user. For example, if the search in the LDAP directory retrieves a user that must be assigned the WebUser and Admin profiles, Axon assigns the Admin profile to the user.

Configure the Linux System to Connect to an SSL-Enabled LDAP Server

If the LDAP server in your organization requires SSL authentication, you must create a truststore in your Linux system and import the SSL certificate to the truststore before you configure the LDAP server. Axon uses the SSL certificate from the keystore to access the LDAP server.

1. Run the following command to create a keystore in the Linux system:

   ```
   <INSTALLATION_DIR>/axonhome/java/jre/bin/keytool -keystore clientkeystore -genkey -
   alias ldapClient
   ```

2. Enter the password to access the keystore. Enter the password again to confirm.
3. Enter the following details about your user profile to create the keystore:
   - First and last name
   - Organizational unit
   - Organization name
   - City or locality
   - State or province
   - 2-letter country code

   When the console prompts you to confirm your user profile details, type `Yes`.
4. Enter the password to access the LDAP client. Type the password that you entered in Step 2.
5. Run the following command to import the SSL certificate of the LDAP server to the keystore:

   ```
   <INSTALLATION_DIR>/axonhome/java/jre/bin/keytool -importkeystore -srckeystore /
   <full_path_to_SSL_certificate_file> -srcstoretype pkcs12 -destkeystore /
   <location_of_SSL_certificate>/clientkeystore -deststoretype JKS
   ```

   Note: The SSL certificate file must be in the PFX format.
6. Enter the password of the destination keystore. This is the keystore that you created in Step 1. Type the password that you entered in Step 2.
7. Enter the password of the source keystore. This is the password of the SSL certificate.

The page displays the status of the SSL file import.

Configure Access to the LDAP Server

Configure Axon to connect to the one or more LDAP servers in your organization, and retrieve users based on the criteria that you specify.

You must have the SuperAdmin profile to perform the task. You must enter the following types of parameters:

- Details of the LDAP server.
- Fields in the LDAP directory that represent fields in Axon for the **People** objects.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **LDAP Server**.

   The page displays the list of LDAP servers configured for the Axon **People** objects.
4. Click **Add** to configure Axon to connect to a new LDAP server

   The page displays the list of properties that you must configure to connect to the new LDAP server.
5. Configure the following properties:

Property	Description
Name	Required. Name of the LDAP server.
Description	Required. Description of the LDAP server.
Host	Required. LDAP host name or IP address. If the LDAP server is configured with SSL authentication, enter the LDAP host name.
Port	Required. Port number of the LDAP server.
Principal Username	Required. Distinguished Name of the LDAP administrator. For example, enter the Distinguished Name (dn) value as `cn=Administrator, cn=users`.
Password	Required. Password to log in to the LDAP server.
SSL Enabled	Optional. Select this option if the LDAP server requires SSL authentication.
Search Base	Required. Point in the hierarchy of root objects from where Axon starts the user search. For example, enter the search base as `cn=users, dc=PAADS, dc=com`.
User Search Filter	Optional. Search criteria to retrieve users. Axon retrieves the users and assigns that profile that you specify in the **Profile** field. For example, enter the search filter `(&(sAMAccountName=*)(objectclass=user))` to search for all entries that match the object class `user`.
User First Name	Required. Attribute of an LDAP user that represents the first name of the Axon user.
User Last Name	Required. Attribute of an LDAP user that represents the last name of the Axon user.
Distinguished Name (dn)	Required. Attribute that represents the Distinguished Name (dn) of an LDAP user.

Property	Description
User Email	Required. Attribute of an LDAP user that represents the email address of the Axon user.
Function Name	Optional. Attribute of an LDAP user that represents the function of the Axon user.
Function Description	Optional. Attribute of an LDAP user that represents the function description of the Axon user.
Office Location	Optional. Attribute of an LDAP user that represents the office location of the Axon user.
Internal Mail Code	Optional. Attribute of an LDAP user that represents the internal mail code of the Axon user.
Office Telephone	Optional. Attribute of an LDAP user that represents the office telephone number of the Axon user.
Mobile Phone	Optional. Attribute of an LDAP user that represents the mobile phone number of the Axon user.
Org Unit Name	Optional. Attribute of an LDAP user that represents the organizational unit name of the Axon user.
Default Org Unit Ref	Optional. Reference number of the default organization unit that Axon assigns to all users.
Org Unit Status	Required. Default status that Axon assigns to all the organization units retrieved from the LDAP directory.
Profile	Optional. Default user profile that Axon assigns to all the users retrieved from the search criteria that you specify in the **User Search Filter** field.
Employment Type	Required. Default employment type that Axon assigns to all the users retrieved from the LDAP directory.
Lifecycle	Required. Default lifecycle that Axon assigns to all the users retrieved from the LDAP directory.
Axon Super Admin Email	Required. Email address of the Axon SuperAdmin that administers the users retrieved from the LDAP directory. **Note:** If you have not assigned the SuperAdmin role to a different user, use the default SuperAdmin email address `admin@informatica.com`. If you have assigned the SuperAdmin role to another user then you can either use the default `admin@informatica.com` value or replace the default value with the relevant SuperAdmin email address.
LDAP SSL Certificate File	Path to the SSL certificate in Linux that authenticates the LDAP server. Enter this value if the LDAP server requires SSL authentication. **Note:** The SSL certificate file must be in the PEM format.
Trust Store for LDAP Synchronization	Path to the truststore in Linux that contains the SSL certificate. Enter this value if the LDAP server requires SSL authentication.

Property	Description
Trust Store Password for LDAP Synchronization	Password of the truststore in Linux that contains the SSL certificate. Enter this value if the LDAP server requires SSL authentication.
WebUser Search Filter	Search criteria to retrieve users that need to have the WebUser profile in Axon. For example, enter the search filter `(&(sAMAccountName=*)(objectclass=user))` to search for all entries that match the object class `user`.
Admin Search Filter	Optional. Search criteria to retrieve users that need to have the Admin profile in Axon.
SuperAdmin Search Filter	Optional. Search criteria to retrieve users that need to have the SuperAdmin profile in Axon.
Update Axon User Profiles with LDAP	Optional. Select this option if you want to automatically update the existing user profiles in Axon to match the user privileges in the LDAP directory. The update happens when you manually run a synchronization job from the Admin Panel to retrieve users or when Axon uses a scheduled job to synchronize the users.

Note:

- You cannot enter values in the **Default Org Unit Ref** and **Org Unit Name** fields simultaneously. If you enter a value in one of the fields, you must leave the other field blank.
- Use the following guidelines to assign profiles to the users retrieved from the LDAP directory:
  - When you use the **User Search Filter** and **Profile** fields, you can assign only one profile to the users.
  - Effective from Axon version 6.2, you can retrieve users from the LDAP directory and automatically assign them the WebUser, Admin, and SuperAdmin profiles.
  - If you are running Axon version 6.2, you can ignore the **User Search Filter** and **Profile** fields. Instead, go to the **WebUser Search Filter**, **Admin Search Filter**, and **SuperAdmin Search Filter** fields to specify the user profiles that Axon must assign to the users retrieved from the LDAP directory.
  - If you choose to ignore the **User Search Filter** and **Profile** fields, you must assign at least one user profile in the **WebUser Search Filter**, **Admin Search Filter**, or **SuperAdmin Search Filter** field.
  - If you assign different user profiles for the same user, the **WebUser Search Filter**, **Admin Search Filter** and **SuperAdmin Search Filter** fields take precedence over the **User Search Filter** and **Profile** fields. For example, if you specify the WebUser profile for a user in the **Profile** field and the search criteria for the same user in the **Admin Search Filter** field, Axon assigns the Admin profile to the user.

6. Click **Save**.

   The page displays the list of connected LDAP servers.
7. You can click the LDAP server name to view the configured properties.
8. Click **Edit** to modify the configured properties.
9. You can select one or more servers, and click **Delete** to delete the selected servers configurations.
10. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

    ```
    sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
    ```

    When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save

your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Retrieve Users from the LDAP Directory

After you configure Axon to connect to the LDAP server, you must initiate the process to retrieve the users from the LDAP directory and create People objects in Axon.

You can manually retrieve the users from the LDAP directory, or you can create a job that automatically retrieves the users based on the schedule that you specify.

Manually retrieve users from the LDAP directory

You can run a synchronization job from the Admin Panel to retrieve users from the LDAP directory.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left of the page, under the **Operational Management** category, click **Administrator's Panel**.
3. Select **Synchronize With LDAP Server** on the left panel, and then click the **Run** button.

The page displays the status of the synchronization job.

Schedule a job to retrieve users from the LDAP directory

You can create a time-based job scheduler to retrieve users from the LDAP directory. To retrieve users according to schedule, Axon uses the following files:

- The script file `sync_ldap.sh` in the `<INSTALLATION_DIR>/axonhome/axon_ldap_synchronizer/scripts` directory executes the job to retrieve users according to the schedule you define.
- The `axon.jobs` file in the `<INSTALLATION_DIR>/axonhome/axon_scheduler` directory specifies the schedule to run the `sync_ldap.sh` script file.

At the time scheduled in the `axon.jobs` file, Axon runs the `sync_ldap.sh` script and retrieves the users from the LDAP directory.

Note: If the LDAP bulk upload job fails due to invalid records, the `sync_ldap.sh` script removes them and attempts to run the job again. The script saves the invalid records into a .CSV file in the `<INSTALLATION_DIR>/axonhome/axon_ldap_synchronizer/tmp` directory.

Perform the following tasks to define the job schedule:

1. Go to the `<INSTALLATION_DIR>/axonhome/axon_scheduler` directory and open the `axon.jobs` file.
2. Configure the following properties to schedule the job:

Property	Description
name	Job name
cmd	Enter the following path to the script file: `<INSTALLATION_DIR>/axonhome/axon_ldap_synchronizer/scripts/sync_ldap.sh`

Property	Description
time	Specify the daily, weekly, or monthly email schedule. The six asterisks `* * * * * *` represent `<second> <minute> <hour> <date> <month> <day_of_the_week>`. For example, to run the job at 2.00 p.m. everyday, enter the following value: `0 0 14 * * *`
onError	Enter `Stop`.

3. Save your changes.
4. If you make changes to the `axon.jobs` file, you must reload the Jobber schedule settings. Run the following command to reload the settings:

```
<INSTALLATION_DIR>/axonhome/third-party-app/scripts/jobber reload
```

Note: You must have the write permission in the `<INSTALLATION_DIR>` directory for the Jobber utility to run the script file.

Check Status of Users Retrieved

Axon uses the bulk upload service to create or update the user profiles that it retrieves from the LDAP directory.

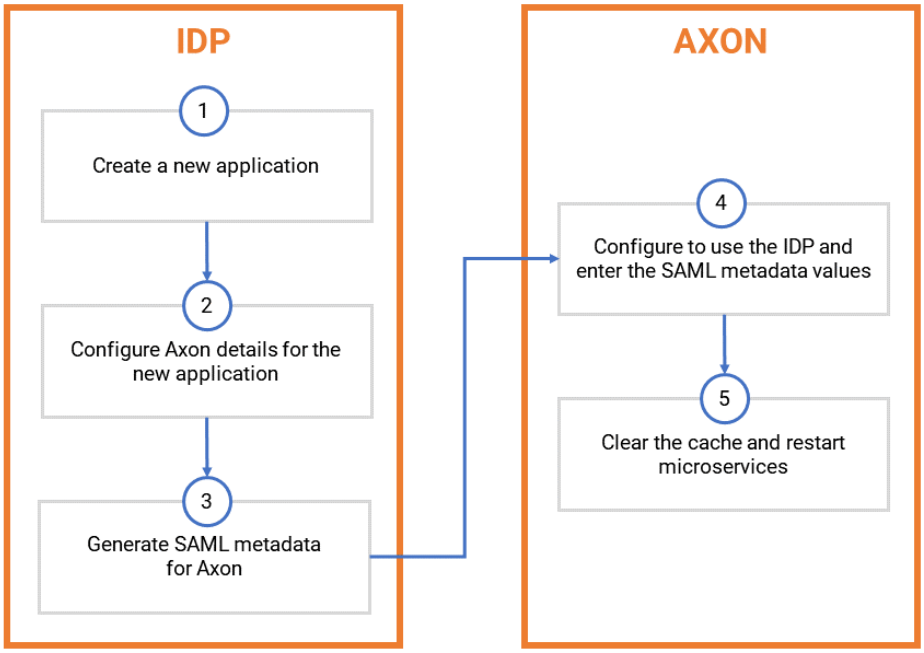
To see the status of the users that Axon creates or modifies, click the **My Account** menu item under your user name, and select **My Jobs**. If you are the Super Admin recorded in the **Axon Super Admin Email** field of the Admin Panel, you can view the status in the **My Jobs** section.

Configure Single Sign-On

When you enable Single Sign-On (SSO), users can log in to Axon using login credentials that are common to other applications in your organization. Axon uses SSO to authenticate users based on the Identity Provider (IDP) credentials.

To enable SSO, you must configure Axon as an application in the IDP tool. Axon supports any IDP that uses SAML 2.0 authentication.

The steps to configure SSO for Axon are shown in the following diagram:



Note: If you configure Axon to access the LDAP server and in addition use SSO to authenticate the users, the SSO authentication takes precedence. When Axon users attempt to log in, the browser displays the login window of the IDP. To configure Axon to access the LDAP directory, refer to "Retrieve Users from LDAP Directory to Axon" on page 46.

Create a New Application in the IDP

You must first add Axon as an application in the IDP.

1. Log in to the IDP configuration page and choose to add a new SAML 2.0 application.
2. Enter the new application name as `Axon`.

Configure Axon Details for the New Application in the IDP

To configure Axon in the IDP, perform the following tasks:

1. Set the name ID format and user name format of users to `Email`. This step makes sure that the IDP expects the Axon user's name ID and user name in the format of an email address
2. Set the SSO domain URL to `http(s)://<Axon_host>:<9999_or_9443_or_custom_port_number>/saml/acs`. This URL is the endpoint for SSO login to Axon.
3. Set the audience URL to `http(s)://<Axon_host>:<9999_or_9443_or_custom_port_number>/saml/metadata`. This URL is the service provider metadata that is obtained from Axon.
4. Set the following mandatory properties for the SAML attributes:

Property	Value
firstName	First name of the Axon user

Property	Value
lastName	Last name of the Axon user
email	Email address of the Axon user

Note: Make sure that the SAML attribute properties are configured for every Axon user in your organization. If a property is not configured, you must add the property to the IDP and enter valid values of the property for each Axon user.

5. You can set the following optional properties for the SAML attributes:

Property	Value
orgunit	Organization unit of the Axon user
orgunitTitle	orgunitTitle
orgunitDescription	Description of the Axon user's organization. **Note:** If you do not select the **orgunitDescription** property, or if you select the **orgunitDescription** property and do not assign a default value, Axon assigns a blank value to all organization units.
statusID	Status of the Axon user in the IDP database. Set the value to `1`. **Note:** If you do not select the **statusID** property, or if you select the **statusID** property and do not assign a default value, Axon assigns the `Active` value to all the users.
profile	Profile of the user in Axon. Enter one of the following values: • Enter `4` for a WebUser profile. • Enter `5` for an Admin profile. • Enter `6` for a SuperAdmin profile. If you do not enter a value or you enter `0` for the field, Axon assigns the WebUser profile to the user. If you enter a value other than `0`, `4`, `5` or `6`, the user account is not created, and the user cannot log in to Axon.

Note: When you configure Axon to use SSO authentication, you can assign a default organization unit for all Axon users. If you choose to assign a default organization unit, the **orgunit**, **orgunitTitle**, and **orgunitDescription** properties are optional. To configure Axon to use SSO authentication, refer to "Configure Axon and Enter the SAML Metadata Values" on page 56.

6. In the IDP, provide access to the new Axon application to all Axon users and user groups.

The Axon installer assigns the following port numbers as the default value for the Apache server:

- If you use HTTP authentication, the installer assigns 9999 as the default port number for the Apache server.
- If you use HTTPS authentication, the installer assigns 9443 as the default port number for the Apache server.

You can enter a custom port number for the Apache server instead of choosing the default value.

Depending on whether you selected the default port number or entered a custom port number for the Apache server during Axon installation, you must enter the correct port number in the SSO domain URL and audience URL fields.

Generate a SAML Metadata for Axon in the IDP

After you have configured Axon in the IDP, you must obtain the SAML metadata. The SAML metadata is used when you configure Axon to use the IDP.

The SAML metadata contains the following properties:

- Identity Provider Endpoint
- SSO Entity ID
- X.509 Certificate

Make a note of the values of these properties. You must enter these values when you configure Axon.

Note: The SAML metadata is based on the configuration parameters that you have entered in the IDP. If you modify any of the parameters in the IDP, you must generate new SAML metadata values.

Configure Axon and Enter the SAML Metadata Values

You can configure Axon to use SSO by entering the SAML metadata values. You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **SAML Configuration**.
4. Click **Edit**.
5. Set the following properties for the IDP:

Property	Description
SSO URL	Enter `http(s)://<Axon_host>:<9999_or_9443_or_custom_port_number>/saml/login`. This URL redirects the user from the Axon login page to the SSO login page. **Note:** The default port number for HTTP authentication is 9999, and the default port number for HTTPS authentication is 9443.
Require SSO Authentication	Restrict guest users from viewing Axon objects. If you select the check box, guest users in Axon cannot view objects when they click **Search** on the toolbar. They must log in to Axon to browse and view Axon objects. If you clear the check box, guest users can view Axon objects when they click **Search** on the toolbar.
SSO Default Org Unit	Enter the reference number of the default organization unit that Axon assigns to all users. **Note:** If you enter a value for this property, the **orgunit**, **orgunitTitle**, and **orgunitDescription** properties in the IDP configuration are optional. To configure the IDP, refer to "Configure Axon Details for the New Application in the IDP" on page 54.
IDP Endpoint	Enter the value that you noted in the SAML assertion.
IDP Entity ID	Enter the value that you noted in the SAML assertion.
IDP X.509 Certificate	Enter the value that you noted in the SAML assertion. **Note:** Enter this value in a single line without any line breaks.

Property	Description
Create and Update Users	Enable if you want to allow IDP users to create and update users. If you do not enable this option, the following limitations apply. • New users are not created if they do not have access to Axon. • SAML user profile settings cannot be updated even if the users have access to Axon.

6. Click **Save**.

Clear the Axon Cache and Restart Services

After you have configured Axon in the identity management tool and configured the Axon SAML files, you must update the Axon cache with the new configuration values.

In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Configure Role Assignments

You can assign stakeholders to an object by selecting from a predefined list of users.

To configure role assignments, assign users to each role for a facet. The roles vary based on the selected facet. For example, if you select the Data Set facet, you can assign users to Data Steward and Data Owner roles. You must assign at least one user to a role.

You cannot create duplicate role assignments for a facet. For example, if you have assigned users to the Glossary Steward role for the Glossary facet, you cannot select the Glossary Steward role again for the Glossary facet to assign the users.

After you assign users to roles, you can view the list of facets, roles, and number of users. When you assign a stakeholder to an object, you have to select users with assigned roles from the predefined list. You can access all the assigned roles from the **My Account** page.

You can add, edit, or delete a role assignment from the **DG Operating Model** > **Role Assignment** page. To edit a role assignment, click the number of users and update the list of users. To delete a role assignment, select the role and click **Delete**.

After you enable the role assignment in the Admin panel and perform bulk upload, you can choose to add users configured in the role assignment as a stakeholder to the object. Use the **Add as a Stakeholder** option to add users as stakeholders to the object in the Role Assignment page. The stakeholder UI tab becomes available to the configured user.

You can view the change details for the users added or removed as a stakeholder in the Admin Activity Logs page. The users that are stakeholders appear with a green symbol prefixed in the Change details window.

Note: When you upgrade from a previous Axon version, the configured users are not added as stakeholders to the object.

When you create or edit an object and if a role assignment is present for the mentioned facet and role, enter values to get the users or directly show the configured user list from role assignment and select the

appropriate one. You can preview 200 users at a time from the drop down list of users for the stakeholder under the involved party.

The role assignment by default takes place when the following conditions are met during bulk upload of new items or during the object creation through API:

- Role assignment is present for the facet during bulk upload of an object.
- Bulk upload template does not explicitly mention a user.
- There is no role mapping present through Maker Checker or Maker Checker is disabled for the user.

Configuring a Role Assignment

To create a predefined list of stakeholders for a facet, assign Axon users to each role for a facet. Use this list to assign stakeholders to an object.

1. From the Axon toolbar, click **Admin Panel** under the user name.

   The **Admin Dashboard** appears with the Axon version details, status of Axon services, and Axon message queues.
2. On the left navigation pane, click **DG Operating Model** > **Role Assignment**.
3. To define a role assignment, click **Add**.
4. In the **Facet Role Selection** section, select a facet.
5. Select a role for the facet.
6. In the **Members** section, enter a user to add the user to the selected role.

   You must assign at least one user to a role. To add multiple users for a role, you can click the **+** sign.
7. Click **Save and Close**.

   You can view the role assignment that you added.

Configure Access Control

You can define access control for users that log in to Axon.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** dropdown, select **Environment**.
4. Click the **Edit** button, and then click the **Edit** icon.
5. In the **Provide Access Control List Configuration** window, define the Access Control List (ACL) method and role for Axon users.
6. In the **Provide Access Control List Configuration** window, click **Save** to save your changes.
7. In the **System Settings** page, click **Save** to save your settings.
8. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Change Default Administrator Password

Informatica recommends that you change the default administrator password after you install Axon.

1. From the Axon toolbar, click the **My Account** menu item under your user name.
2. In the **About Me** tab, click **Edit**.
3. In the **Password** and **Confirm** fields, enter your new administrator password.
4. Click **Save and Close**.

Unlock Locked Users

If a user enters incorrect login credentials 5 times in a row, Axon locks the user account, and the user cannot log in to Axon. You can unlock Axon users that have been locked out of their accounts.

You must have the SuperAdmin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Operational Management** category, click **Locked Users**.
3. In the **Locked Users** page, select the users that you want to unlock, and click **Unlock**.
4. Optionally, run the following command in the Linux environment to unlock a single user:

```
source <INSTALLATION_DIR>/scripts/exportEnv.sh
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console axon:unlocked --
email=<email_address>
```

View Licensed Users

You can view the count and list of active licensed users that have create or edit permissions in Axon. The active licensed users include SuperAdmin, Admin, and any other user that is assigned a role with create or edit permissions.

View the number of licensed users and the list of active users to verify that the licenses are effectively used in your organization. For example, if you buy 20 licenses and if you find that your organization uses only 15 licenses, you can assign the remaining licenses to additional users. If the licenses used are more than 20, then you can revoke some of the licenses.

Log in with the SuperAdmin profile to view the number of licensed users from the Admin Dashboard. To view the list of active licensed users, click **DG Operating Model** > **Licensed Users**. You can filter the results based on the type of user profile, first name, and last name.

If the **Axon Status** field for a user is "Inactive" or "Deleted," you cannot view them in the list of licensed users.

CHAPTER 4

Configuring Axon

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Configuration Overview

As the administrator, to set up Axon for use in your organization, you can configure your organization's email client with Axon to receive notification emails from Axon. You can also manage API tokens and modify the reindexing time.

Customize Browser Tab Title

You can customize the title that appears on the browser tab for all Axon pages to align with your organization's branding. The task requires SuperAdmin privileges.

Complete the following steps to customize the browser tab title:

1. From the Axon toolbar, click your user name and select **Admin Panel**.
2. In the navigation pane, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Environment**.
4. Click **Edit**.
5. In the **Browser Tab Title** field, enter the title that you want to display.

   Note: To revert to the default page-specific titles, clear the **Browser Tab Title** field.
6. Click **Save**.
7. On Linux, run the following paramsync script to clear the Axon cache and restart the required services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   Note: When you run the paramsync script, Axon restarts the HTTPD, Memcached, and email notification services, which may cause a brief service interruption.

After the script completes, custom title appears on the browser tab the next time you navigate to an Axon page.

Customize Email Templates

Axon offers a comprehensive set of default, customizable email templates specifically designed to facilitate the setup of various notifications.

Available Templates

The default pre-built templates support common use cases, including routine reports such as daily, weekly, or monthly reports, system notifications, and scheduled summaries. Each template serves as a solid foundation that you can easily tailor to your organization's unique requirements.

Customization

Customization options allow you to modify the email content, update logos, color schemes, and fonts and adjust layout and section ordering. Embed special tags to auto-populate data, such as recipient names, dates, or other relevant data—helping to make each email feel targeted and relevant to the recipient.

Configuration

Templates serve as base configurations that can be modified to meet specific organizational requirements while maintaining consistency across notifications.

Customize Email Templates

Customize existing email templates by modifying the content, design elements, and default placeholders, to create personalized email notifications.

You must have the SuperAdmin profile to perform the task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.

2. In the menu on the left, under the **Customize & Configure** category, click **Customize Email Templates**.

   The Customize Email Templates page displays.
3. In the **Email Template Name** list, select an email template.
4. Configure the following fields:

Field	Description
Email Template Name	Select an email template name to configure the email notification. For example, Daily Activity.
Language	Select the language for composing and sending the email.
Subject	Define the subject line for emails sent using the template.
Email Body	Define the email content using HTML and CSS. Supports default placeholders for personalized messaging. To modify the CSS style or HTML content, click the **Email Body Edit** button. A pop-up window appears with the following options: • **Email CSS Style** - Define custom CSS styles to control the email's visual presentation and format. • **HTML Content** - Edit the HTML structure to customize the email layout and design.

5. Click **Save**. You can also preview the current email content before saving.

   Note: You can choose to select Restore Default Settings to revert the template to its original state. A confirmation prompt appears to prevent accidental changes.

Configure Email Notifications

You can create a time-based job scheduler to receive emails from Axon. You must enter the details of the email system in the Admin Panel. You must also define the email schedule in the `axon.jobs` file.

You can schedule email notifications in the following time intervals:

- Daily
- Weekly
- Monthly

Configure Email Delivery Options

To receive email notifications, you must enter the details of the email server in Axon.

You must have the SuperAdmin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Email Options**.
4. Click **Edit**.

5. Configure the following properties:

Property	Description
Mailer Transport	Email protocol to configure your email delivery. Select a value from the list. For example, select **SMTP** to configure the SMTP account.
Mailer Host	Your email host address.
Mailer User Name	Your email user name.
Mailer Password	Your email password.
Mailer Port	Email server port.
Disable Email Delivery	Disable all email delivery to Axon users.
Disable Notification Emails for Tasks	Disable notification emails when Axon assigns tasks to users. **Note:** If you select the **Disable Email Delivery** option, Axon disables notification emails for tasks.
Disable Email Notifications for Data Quality Results	Disable email notifications to Axon users when a data quality score goes down for a data quality rule.
Mailer Encryption	The email encryption mode that you can configure based on the email protocol. You can select "`ssl`" or "`tls`." For example, if you use SendGrid, Office 365, or Gmail, you can configure mailer encryption to "`ssl`" or "`tls`" when you use the SMTP email protocol. **Note:** For Gmail accounts, it is recommended to use `ssl` as the mailer encryption mode. Leave this field blank if you do not want to use encryption.
Default Email Frequency	Set the default frequency of the notification emails sent to the users. **Note:** • The default value of the field is `Not receiving notification mails`. • The notification frequency set by the user in the **Activity Stream** tab of their **My Account** page will take precedence over the default value that you enter here. • If the user has not specified the notification frequency in the local profile, then Axon sends email notifications if you have configured a default email frequency.

6. Click **Save**.
7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Create a Daily Email Notification Schedule

Perform the following steps to create a daily email notification schedule:

1. Open the `axon.jobs` file available in the `<INSTALLATION_DIR>/axonhome/axon_scheduler` location.
2. In the `axon.jobs` file, configure the following properties:

Property	Description
name	Enter a name for the notification job.
cmd	Enter `<INSTALLATION_DIR>/axonhome/axon_scheduler/daily/dailyscript.sh`. The `dailyscript.sh` script runs the job according to the schedule that you define in the `axon.jobs` file.
time	Specify the daily email schedule. The six asterisks `* * * * * *` represent `<second> <minute> <hour> <date> <month> <day_of_the_week>`. For example, to receive email notifications at 5.00 p.m. everyday, enter the following value: `0 0 17 * * *`
onError	Enter `Stop`

3. Save your changes.
4. If you make changes to the `axon.jobs` file, you must reload the Jobber schedule settings. Run the following command to reload the settings:

```
<INSTALLATION_DIR>/axonhome/third-party-app/scripts/jobber reload
```

To send periodic email notifications to the Axon user, the Jobber utility runs the `dailyscript.sh` file that you created.

Create a Weekly Email Notification Schedule

Perform the following steps to create a weekly email notification schedule:

1. Open the `axon.jobs` file available in the `<INSTALLATION_DIR>/axonhome/axon_scheduler` location.
2. In the `axon.jobs` file, configure the following properties:

Property	Description
name	Enter a name for the notification job.
cmd	Enter `<INSTALLATION_DIR>/axonhome/axon_scheduler/weekly/weeklyscript.sh`. The `weeklyscript.sh` script runs the job according to the schedule that you define in the `axon.jobs` file.
time	Specify the weekly email schedule. The six asterisks `* * * * * *` represent `<second> <minute> <hour> <date> <month> <day_of_the_week>`. For example, to receive email notifications at 9.00 a.m. on the third day of each week, enter the following value: `0 0 9 * * 3`
onError	Enter `Stop`

3. Save your changes.

4. If you make changes to the `axon.jobs` file, you must reload the Jobber schedule settings. Run the following command to reload the settings:

   ```
   <INSTALLATION_DIR>/axonhome/third-party-app/scripts/jobber reload
   ```

To send periodic email notifications to the Axon user, the Jobber utility runs the `weeklyscript.sh` file that you created.

Create a Monthly Email Notification Schedule

Perform the following steps to create a monthly notification schedule:

1. Open the `axon.jobs` file available in the `<INSTALLATION_DIR>/axonhome/axon_scheduler` location.
2. In the `axon.jobs` file, configure the following properties:

Property	Description
name	Enter a name for the notification job.
cmd	Enter `<INSTALLATION_DIR>/axonhome/axon_scheduler/monthly/monthlyscript.sh`. The `monthlyscript.sh` script runs the job according to the schedule that you define in the `axon.jobs` file.
time	Specify the monthly email schedule. The six asterisks `* * * * * *` represent `<second> <minute> <hour> <date> <month> <day_of_the_week>`. For example, to receive email notifications at 2.30 p.m. on the seventeenth day of each month, enter the following value: `0 30 14 17 * *`
onError	Enter `Stop`

3. Save your changes.
4. If you make changes to the `axon.jobs` file, you must reload the Jobber schedule settings. Run the following command to reload the settings:

   ```
   <INSTALLATION_DIR>/axonhome/third-party-app/scripts/jobber reload
   ```

To send periodic email notifications to the Axon user, the Jobber utility runs the `monthlyscript.sh` file that you created.

Customize Workflow Notifications

For a change request, you can configure the actions on which Axon sends notifications to stakeholders of an object.

You must be a SuperAdmin user to customize workflow notifications for all the users in the organization.

1. In the Admin Panel, click **Customize & Configure** > **Application Settings**.
2. Click **Configure Notifications**.

   The **Configure Notifications** section opens.

3. On the **Configure Notifications** section, select the scenario in which a stakeholder receives notifications.

Option	Description
Enable notifications for change request activity.	You can configure Axon to not send a notification when a comment is added to a change request. If an object stakeholder is mentioned in the comment, they will receive a notification even if you disable notifications on comments.
Enable notifications for workflow activity.	You can configure Axon to not send a notification when a stakeholder completes a workflow step. If a workflow step requires an object stakeholder to take action, they will receive a notification even if you disable notifictations for workflow step completion.

4. Click **Save**.

Prerequisites for Modern Authentication

Verify the following prerequisites before you enable modern authentication.

Prerequisites for Azure modern authentication

- You must have the SuperAdmin profile to perform the enable modern authentication task.
- Enable SMTP authentication for the email account that sends or receives emails.
- The email account that sends or receives emails should have an active mailbox.
- Configure an email account for the application users.

Prerequisites for Gmail modern authentication

- You must have the SuperAdmin profile to perform the enable modern authentication task.
- Disable the two-factor authentication for the user account used for modern authentication.

Enable Modern Authentication for Email Options

Modern authentication combines the authentication and authorization methods between a client and a server with security measures, such as Multifactor Authentication and OAuth. You can enable modern authentication for Email Options.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Email Options**.
4. Click **Edit**.

5. Configure the following properties:

Property	Description
Modern Auth Provider	Select one of the following options as the modern authentication provider: - Gmail. - Azure. - None. **Important:** Axon installed on HTTPS protocol supports modern authentication. Axon on SUSE Linux Enterprise Server does not support modern authentication.
Client Secret	Required for Gmail and Azure authentication. Specify the client secret of the application. A client secret is known only to the application and the authorization server.
Tenant ID	Required for Azure authentication. Specify the tenant ID of the application. Tenant ID is a unique identifier or an alias for a tenant. For example, x0xxx10-00x0-0x01-0xxx-x0x0x01xx100.
Client or Application ID	Required for Gmail and Azure authentication. Specify the client or application ID. Client or application ID is a public identifier for applications.
Refresh Token	Required for Gmail and Azure authentication. Specify the refresh token to obtain additional access tokens. On the top right hand corner of the Admin Panel, you can click the **Refresh Token** button to generate new refresh tokens. The validity of a refresh token is as follows: - Gmail: 6 months - Azure: 3 months **Note:** A refresh token expires when the access or password changes in the user account that the modern authentication uses.

6. Click **Save**.

Modern Authentication Expiration

The modern authentication configuration expires in the following scenarios:

- A user changes the user account password or revokes access from the application.
- A user revokes the client secret or the client secret expires.
- The refresh token expires.

Configure Timeout Period for API Tokens

You can configure the timeout period for access tokens and refresh tokens. Use an access token to authenticate yourself when you make an API call. Use a refresh token to generate access tokens.

For information on using tokens to make API calls, see the *Authentication* topic in the *Axon Data Governance 7.2.1 REST-API Guide*.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **JWT Settings**.
4. Click **Edit**, and configure the following properties:

Property	Description
JWT Validity	Timeout period of the access token. Enter a value in seconds. For example, if you want to set the expiration date for 7 days, enter `604800`. The access token expires if you do not use it within the timeout period.
Refresh Token Validity	Timeout period of the refresh token. Enter a value in seconds. For example, if you want to set the expiration date for 30 days, enter `2592000`. The refresh token expires if you do not use it within the timeout period. **Note:** Each time you use a refresh token, the timeout period restarts from the usage date. For example, if you generate a token on June 1 with a timeout period of 60 days, the token will expire on July 30. If you use the token on June 3, the timeout period restarts, and the token will expire on August 2.

   Note:

   - If you change the timeout period, the tokens that have already been generated will continue to have their earlier timeout values.
   - It is recommended that the timeout period of the access token is less than the timeout period of the refresh token.

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Update Cache with Configuration Values

Some Axon configurations in the Admin Panel are saved to the Axon database, but these configuration values are not loaded to the Axon cache. To make the configuration changes available to the Axon users, you must update the Axon cache with the new configuration values.

You must update the Axon cache when you perform the following configurations:

- Display the logo of your organization in the Axon toolbar
- Change the colors of the Axon interface to reflect the theme and colors of your organization
- Configure email settings
- Configure regional settings
- Configure glossary types in the rollup view
- Configure saved searches for quick links
- Configure email delivery options
- Configure a SAML 2.0 application to work with Axon
- Configure Access Control
- Configure access to Enterprise Data Catalog
- Configure access to Data Quality
- Configure access to Data Privacy Management
- Configure access to the LDAP server
- Enable dashboard
- Specify prefix for the reference numbers of Axon objects
- Send diagnostic information to Informatica Global Customer Support

To clear the Axon cache and load the cache with the new configuration values, run the following command in the Linux environment:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Memcached and HTTPD services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Enable Data Migration

To migrate objects from Unison search, you must enable data migration in both source and target instances. A SuperAdmin user can perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. From the navigation pane, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Environment**.
4. Click **Edit**.
5. Select **Data Migration**.

6. In the **System Settings** page, click **Save** to save the settings.
7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Import Migrated Data

After you migrate data from a source Axon instance, you can import the migrated data into a target Axon instance.

Prerequisites
Verify the following prerequisites before you import the migrated data:

- You must be an Admin or a SuperAdmin user to perform this task.
- You must enable data migration from the Admin Panel.
- The mandatory custom fields and dropdown values in the target instance must be present in the source as well.
- If objects belong to segments in the source instance, the segments must be present in the target instance. You must have access to the segments in the target instance.
- When you migrate roles for a facet, ensure that the corresponding object stakeholders are present in the target instance and a role with the name exists in the target instance.

1. From the Axon toolbar, click your user name and click **Admin Panel**.
2. From the navigation pane, click **Operational Management** > **Import Migrated Data**.

   The **Import Migrated Data** page appears.
3. In the **Upload File** section, choose to upload a file with the `.zip` format.
4. Click **Import**.

   To check the import job status, go to the **Activity Stream** > **My Jobs** page of your account.

Send Diagnostic Information to Informatica Global Customer Support

You can send diagnostic information about the Axon services to Informatica using the Phone Home option. If you choose the phone home option, Axon periodically collects and sends the information to an Amazon S3 server that is hosted by Informatica. The Informatica Global Customer Support team can access this information to troubleshoot the Axon instance.

If you enable the phone home option, Axon collects and sends the following information to Informatica:

- Error message or log message for an Axon request
- Date and time of the request

- Axon component for the request
- User action for the request
- Module of the Axon facet
- Unique reference number for the request
- Data requested, such as a URL
- Number of the licensed users and active licensed users
- Axon facet and objects in the facet

Note:

- To send information to Informatica, the Axon instance must have a working Internet connection.
- Axon does not send sensitive or personally identifiable information to Informatica.

Enable Phone Home

The phone home option is disabled in the Axon installation by default. To send information about the Axon instance to Informatica, you must enable the phone home option in the Admin Panel.

You must have the Admin or SuperAdmin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Phone Home**.
4. Click **Edit**, and select the **Enable Phone Home** check box.
5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note:

1. You cannot use the phone home option if you install the PostgreSQL database on an external node. Edit the `phonehome_service.sh` file and remove the comment to enable the following parameter: `PHONE_HOME_ENABLED=true`
2. When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users use Axon.

Configure Scheduler to Send Information to Informatica

Perform the following steps to configure the scheduler that sends information about the Axon services to Informatica:

1. Open the `axon.jobs` file from the `<INSTALLATION_DIR>/axonhome/axon_scheduler` location.

2. In the `axon.jobs` file, configure the following properties:

Property	Description
name	`PhoneHomeService`
cmd	Enter `<INSTALLATION_DIR>/axonhome/third-party-app/scripts/phonehome_pushdata.sh`. In the `phonehome_pushdata.sh` section, you can define the schedule to send diagnostic information to Informatica Global Customer Support.
time	Specify the schedule to send the information to Informatica. The six asterisks `* * * * * *` represent `<second> <minute> <hour> <date> <month> <day_of_the_week>`. For example, to send the information at 8.00 p.m. everyday, enter the following value: `0 0 20 * * *`
onError	Enter `Stop`

3. Save your changes.
4. If you make changes to the `axon.jobs` file, you must reload the Jobber schedule settings. Run the following command to reload the settings:

```
<INSTALLATION_DIR>/axonhome/third-party-app/scripts/jobber reload
```

To send the information periodically, the Jobber utility runs the `script.sh` file.

Specify Support Email Address and Override Default Sender Identity

To contact the support team, Axon users can click the **?** icon and then select **Contact Support**. You can change the default email address for Axon support when users choose to contact the support team.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Email Options**.
4. Click **Edit**.
5. In the **Support Email** field, enter the default email address for Axon support.
6. Optionally, you can choose to update the sender information. For instance, to change the display name, edit the default **Sender Display Name** and enter the revised sender name.
7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.
8. Click **Save**.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save

your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Configure Timeout Limit Parameter

You can configure the timeout limit to increase or decrease the amount of time that can be allowed for reindexing services and precomputation process.

For reindexing services, enter the timeout value in seconds in the `etl_timeout` parameter. This value overrides the default value of 60 seconds.

For precomputation process, enter the timeout value in seconds in the `precomputation_status_refresh_timeout` parameter. This value overrides the default value.

Configure the `etl_timeout` and the `precomputation_status_refresh_timeout` parameters in the following file:

```
<INSTALLATION_DIR>/axonhome/axon/app/config/parameters.ymlc
```

Use the following syntax to configure the parameter: `etl_timeout: <value in seconds>` and `precomputation_status_refresh_timeout: <value in seconds>`

For example, to modify the timeout period to two hours, enter the following value: `etl_timeout: 7200`

Configure Connection Pools for Services in Axon

You can configure connection pools between the Unison and Segmentation services and between the PostgreSQL and OrientDB databases.

The following connections are values set up with the installer by default:

Unison:

For OrientDB: minimum 10 connections, maximum 300 connections

Configure the Unison values in this file: <Install_Dir>/axonhome/axon_unison_microservice/config/app.yml

Segmentation (Authorization):

For PostgreSQL: minimum 8 connections, maximum 32 connections

Configure the Segmentation values in this file: <Install_Dir>/axonhome/axon_authorization_microservice/config/app.yml

Lineage Consumer

For PostgreSQL: minimum 1 connections, maximum 1 connections

For OrientDB: minimum 1 connections, maximum 3 connections

Configure the Job Duration to Clear Pending Bulk Upload Jobs

Use the automated `clearPendingBUJobsScript.sh` scheduler to clear pending bulk upload jobs. The scheduler clears the pending jobs that exceed the job duration set in the Bulk Upload microservice.

The scheduler is available in the following location:
`<INSTALLATION_DIR>/axonhome/axon_scheduler/bulkupload/`

1. Open the `app.yml` file from the `<Install_Dir>/axonhome/axon_bulk_upload_microservice/service/` location.
2. Set the *clearBUJobsDuration* element value to the required job duration. By default, the job duration for a bulk upload job is 6 hours.
3. Save your changes.
4. Run the following command to restart the Bulk Upload microservice:
   `cmd- sh $AXONHOME/third-party-app/scripts/axon_bulk_upload_microservice_init restart`
5. Verify that the status of the pending bulk upload jobs is `undefined` in the **MyJobs** page.
   It takes at least 30 minutes to clear the pending bulk upload jobs.

Configure Log File Rotation and Deletion Mechanism

To address the increasing size of prod.log and dev.log files, and ensure efficient log management, implement a cron job for log rotation and deletion. The approach also extends to the prod_error.log and dev_error.log files, with an added focus on periodic deletion.

1. Introduce a new cron job running daily to manage log files. To do this, complete the following steps:
   a. Navigate to the axon.jobs file in the following Axon scheduler location:

      `<AXON Installation Directory>/axonhome/axon_scheduler/`

   b. Remove any comments related to the cron job for logs within the axon.jobs file.

      For example, locate the following commented section related to the cron job for logs:

      ```
      #- name: RotateLogs
      #  cmd: /data/nroot/Informatica/attrperf/axon7.3.5p23/scripts/axon_log_rotate.sh
      #  time: 0 0 4 * * *
      #  onError: Continue
      ```

      You can remove the comment # symbol against all the lines that relate to the cron job as follows:

      ```
      name: RotateLogs
       cmd: /data/nroot/Informatica/attrperf/axon7.3.5p23/scripts/axon_log_rotate.sh
       time: 0 0 4 * * *
      onError: Continue
      ```

   c. Set the specific time to run the cron job for logs.

      For instance, to set the time to 4:00AM in the cron job, set time value as follows:
      `0 0 4 * * *`
2. To set up the log rotation mechanism and to ensure separate dated files for easier management, complete the following steps:
   a. Rotate prod.log to a format `prod_<y-m-d>.log` (e.g., `prod_2024-05-14.log`).
   b. Rotate dev.log to a format `dev_<y-m-d>.log`.

3. Configure the job to check the number of log files in the `axon/var/logs/` directory. Identify logs older than the specified duration in the **Archive Log Retention Period** under **System Settings**, and delete any files exceeding this duration.

   Logs are both rotated and cleaned, maintaining efficient log storage and preventing unnecessary buildup.

   Note:

   - Ensure that the cron job is correctly scheduled and the required permissions are set for access to the log directory.
   - Regularly review and update the admin panel settings as necessary to adjust the retention period.

Modify Admin Panel Settings from the Linux Environment

One of the ways to change the Axon settings and parameters is to go to the **Customize & Configure** category in the Admin Panel. You can also modify some of the Admin Panel settings from the Linux environment.

You might want to modify the parameter values from the Linux environment under the following situations:

- You are the Axon Administrator in the Linux environment, but you do not have the SuperAdmin profile in Axon.
- Axon does not load, and you want to fix the issue by modifying some parameters that appear in the Admin Panel.

Identify Display Name, Internal Key, or Group for the Parameter

To modify an Admin Panel parameter from the Linux environment, you need to know the display name of the parameter, its internal key, or its group.

Identify display name

The display name is the name of the parameter as it appears in the Admin Panel. For example,**Mailer User Name** is the display name of a parameter in the **Email Options** group.

Identify internal key

The internal key is the name of the parameter as it is stored in the database. For example, the internal key for the **Mailer User Name** parameter in the **Email Options** group is `mailer_user`.

Identify groups from the Admin Panel

The settings in the Admin Panel are categorized into groups. To see the list of groups, perform the following tasks:

1. Log in to Axon with the SuperAdmin profile.
2. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
3. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
4. Click the **Group** list to see the groups for settings.

Identify groups from the Linux environment

Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

```
source <INSTALLATION_DIR>/scripts/exportEnv.sh
```

Go to the `<INSTALLATION_DIR>/axonhome/axon` directory and perform the following tasks to identify the groups.

- To see the list of groups that you can configure from the Linux environment, run the following command:

  ```
  <INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
  informatica:parameter:list -g all
  ```

- To see the list of parameter display names and internal keys for a group, run the following command:

  ```
  <INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
  informatica:parameter:list -g "<name of the group>"
  ```

  For example, to see the list of parameter display names and internal keys for LDAP configuration, run the following command:

  ```
  <INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
  informatica:parameter:list -g "LDAP Server"
  ```

Display the Parameter Value

You can display the value of an Admin Panel parameter in the Linux environment.. This value might be the default value assigned by the Axon installer, or it might be the value that you manually entered in Axon.

Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

```
source <INSTALLATION_DIR>/scripts/exportEnv.sh
```

Go to the `<INSTALLATION_DIR>/axonhome/axon` directory and perform the following tasks to display the parameter values.

Display value by internal key

To display the value of a parameter by entering the internal key, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:get -p "<parameter_internal_key>"
```

For example, the internal key for the **Mailer User Name** parameter in the **Email Options** group is `mailer_user`. To display the value of the **Mailer User Name** parameter, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:get -p "mailer_user"
```

Display value by display name

To display the value of a parameter by entering the display name, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:get -d "<parameter_display_name>"
```

For example, to display the value of the **Mailer User Name** parameter in the **Email Options** group, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:get -d "Mailer User Name"
```

Display all values in a group

To display the values of all parameters within a configuration group, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:list -g "<parameter_group_name>"
```

For example, to display the value of all parameters in the **Email Options** group, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:list -g "Email Options"
```

Reset Parameter Value

You can reset the value of a parameter to its default value. The default value of a parameter is the value assigned to it when you installed Axon.

Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

```
source <INSTALLATION_DIR>/scripts/exportEnv.sh
```

Go to the `<INSTALLATION_DIR>/axonhome/axon` directory and perform the following tasks to reset the parameter values.

Reset value by internal key

To reset the value of a parameter by entering the internal key, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:reset -p "<parameter_internal_key>"
```

For example, the internal key for the **Axon Agent Server Host** parameter in the **Data Quality** group is `agent_host`. To reset the value of the **Axon Agent Server Host** parameter, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:reset -p "agent_host"
```

Reset value by display name

To reset the value of a parameter by entering the display name, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:reset -d "<parameter_display_name>"
```

For example, to reset the value of the **Axon Agent Server Host** parameter in the **Data Quality** group, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:reset -d "Axon Agent Server Host"
```

Reset all values in a group

To reset the values of all parameters within a configuration group, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:reset -g "<parameter_group_name>"
```

For example, to reset the value of all parameters in the **Data Quality** group, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:reset -g "Data Quality"
```

Clear cache and restart services

After you reset the parameter values, run the following command to clear the Axon cache and restart the necessary services:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Set New Parameter Value

You can overwrite the existing value of a parameter and enter another value.

Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

```
source <INSTALLATION_DIR>/scripts/exportEnv.sh
```

Go to the `<INSTALLATION_DIR>/axonhome/axon` directory and perform the following tasks to set new parameter values.

Set value by internal key

To set the value of a parameter by entering the internal key, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:set -p "<parameter_internal_key>" --value=<new_value>
```

For example, the internal key for the **Enterprise Data Catalog Server Port** parameter in the **Enterprise Data Catalog** group is `eic_server_port`. To set the value of the **Enterprise Data Catalog Server Port** parameter to 9085, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
informatica:parameter:set -p "eic_server_port" --value=9085
```

Clear cache and restart services

After you set the new parameter values, run the following command to clear the Axon cache and restart the necessary services:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

CHAPTER 5

Feature Settings

This chapter includes the following topics:

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- Restrict Data View Access for Guest Users, 88
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Feature Settings Overview

As the administrator, you can customize Axon by enabling or disabling several of its features. You can configure custom fields, configure dropdown values, and customize static HTML pages. You can also manage dashboard ownership.

Enable Fuzzy Search

Enable fuzzy search to find matching results even when a user misspells, interchanges, or misses one of the keywords.

You must have the SuperAdmin profile to perform this task.

1. In the Admin Panel, click **Customize & Configure** > **Application Settings**.
2. Click **Search Settings**.

3. Select the **Enable fuzzy search** option.
4. Click **Save & Close**.

Configure Static Pages

Axon contains a number of static HTML pages. If you have an Admin or a SuperAdmin profile, you can customize the static pages.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, click **Static Page Editor** under **Meta-Model Administration**.
   - To create a new static page, click **New Static Page** from the **Actions** menu.
   - To edit an existing static page, click the title of the static page.
3. Configure the following static page properties:

Option	Description
Title	Specify a title for the static page. Axon does not display the title anywhere.
Link	Specify a link for the static page. Axon configures the URL of the page in the following format: `<Axon_host>:<port_number>/<link>`
Context	Specify a description for the static page. Axon does not display the context anywhere.
Content	Specify the content that is displayed in the static page. You can use HTML tags to format the content.
Homepage	Sets the page as the home page. **Note:** If the home page contains a link, you must apply the `<div class="hero-unit"> </div>` tag to the URL. For example, to add the www.yourorganization.com link for the "Home" text, enter the following tag: `<div class="hero-unit"> <a href = "https://www.yourorganization.com"> Home </a> </div>`

4. Click **Save and Close**.

Configure Custom Colors for Glossary Types

If you have a SuperAdmin profile, you can customize colors for the glossary types in a glossary facet.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **Customize Styles**.
3. Under the Custom CSS section, specify the color codes for the glossary types. You must specify the border color, background color, and the fill color.

Note: You must specify the color codes in hexadecimal format for the glossary type based on the ID. To view the ID number for the glossary type, click **Meta-Model Administration** > **Dropdown Configurations** > **Glossary Type**.

Use the following example to specify the color for the glossary type ID:

```
.glossary_type_id_4 {color: #ffffff !important;border-color: #A54BD2 !
important;background-color: #A54BD2 !important;}
```

4. Click **Save**.
5. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   For more information, see the "Clear the Axon Cache and Restart Services" on page 57 topic.

Configure Custom Fields

You can create custom fields that you want Axon to display when you create an object. You can use custom fields to capture specific additional data to meet your business requirements. For example, in your organization, all products might require a manufacturing lot number and an expiry date. You can create the **Lot Number** and **Expiry** fields for the Product facet. When you create a new Product object, Axon displays these fields in the interface so that you can enter the appropriate values.

When you search for Axon objects in Unison Search, you can choose to display the custom fields as columns in the search results. For more information about searching for Axon objects, refer the *Unison Search* topic in the *Search and View Objects* chapter of the *Axon Data Governance 7.2.1 User Guide*.

You can also enter several values for the custom fields using the bulk upload templates. For more information about uploading data to Axon using the bulk upload templates, refer the *Bulk Upload Templates Overview* topic in the *Bulk Upload Data* chapter of the *Axon Data Governance 7.2.1 Bulk Upload Reference Guide*.

Add Custom Fields

You can create custom fields to add additional fields to capture details according to your requirements.

You must have the Admin or Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under **Meta-Model Administration**, click **Custom Fields**.

   The **Configure Custom Fields** page appears.
3. In the **Facet** list, select the facet for which you want to create custom fields.

   The page displays the list of custom fields that you have created for the facet.
4. Click **Edit**, and then click the + icon to add a new field

   The **Add Custom Field** dialog box appears.

5. In the **Add Custom Field** dialog box, enter the following properties of the custom field:

Property	Description
Display Name	Name of the custom field that needs to appear in the Axon interface. Make sure that you enter a field name that does not match an existing field name in the facet. You can edit a display name later. **Note:** The following guidelines apply to custom field names: • When you first create a custom field, the Display Name you enter becomes its Technical Name, prefixed by cf. You cannot modify the Technical Name later. If you enter names with spaces in the Display Name, the Technical Name field stores the names without spaces. You need the Technical Name for API calls. • The field name must begin with a letter, number, or an underscore. • The following special characters are supported in the field name: `( ) [ ] { } - _ ?` • Do not use the following characters in an attribute name: `[ ]`
Mandatory	Determines if the Axon user must enter a value in the custom field. Mandatory fields appear in the **Definition** section of an Axon object, and non-mandatory fields appear in the **Optional Fields** section of an Axon object. **Note:** If you use SSO or retrieve users from the LDAP directory to create users in Axon, make sure that you do not create a mandatory field for the People or Org Unit facet.
Description	Custom field description that appears as a tooltip on hover over the tooltip icon.
Type	The type of value that the user can enter in the custom field. • Text: The user must enter a text string. • Number: The user must enter a numeric integer value. **Note:** The integer value cannot exceed 19 digits. • Decimal: The user must enter a numeric integer value or decimal value. **Note:** If the user enters a decimal value, the integer part of a decimal value cannot exceed 13 digits. • Date: The user must select a date. You cannot select the **Date** type for the **Data Attributes** facet. • Check Box: The field is a check box that the user can either select or clear. • Time: The user must select a time. You cannot select the **Time** type for the **Data Attributes** facet. • Percentage: The user must enter a numeric value. • Single Selection Dropdown. The user must select a value from a list of dropdown values. • Multiple Selection Dropdown. The user can select multiple values from a dropdown list.
Values	This property appears if you select the `Dropdown` type. Enter the list of values that must be available for the user. Enter each value in a new line. The user can select one of these values from the available dropdown list.
Default Value	Enter a value that must appear for the field by default. **Note:** If you configure a custom field as required, you must provide a default value.
Placeholder Text	Enter a descriptive text that instructs the user about the purpose of the custom field. The text that you enter in this field appears in grey color for the user.

6. Click **Save**.

View List of Custom Fields

To see the list of custom fields that you have created, perform the following steps:

You must have the Admin or Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under **Meta-Model Administration**, click **Custom Fields**.

   The **Configure Custom Fields** page appears.
3. In the **Facet** list, select the facet for which you want to create custom fields.

The page displays the list of custom fields that you created for the facet.

To edit a custom field, click the icon for a field and modify the field properties. You cannot modify a custom field type.

To delete a custom field, click the icon for the field.

Configure Dropdown Values

You can assign classifications when you create objects. If you have a SuperAdmin profile, you can create new classifications or modify the default classifications to suit the needs of your organization.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, click **Dropdown Configurations** under **Meta-Model Administration**.
3. In the **Dropdown Configurations** page, select a module from the list in the **Choice** property.

   Axon filters the list of choices to display the options available in the facet that you selected.

   Note: You must not update the values for the Axon Status module in the dropdown configuration list.
4. Click the module that you want to edit.

   Axon displays the configuration menu editor.
5. In the configuration page, click the **+** icon to add a new item to the dropdown menu.

   For example, if you selected Process Duration Type, you can add more values, such as seconds and minutes.
6. Configure the following properties:

Option	Description
PrimaryName	Specify a name that Axon will display in the dropdown menu.
Description	Specify a description that Axon will display in the dropdown menu.

7. If you want to delete a value that you added from the configuration page, click the **-** icon to delete the value.

   If the value is in use, Axon displays a warning message.
8. Click **Save and Close**.

Delete Relationship Type Values

If you select a relationship type from the list of dropdown values and delete a relationship type value that you added, the relationships between Axon objects that are linked using the relationship type are not applicable anymore. However, the Axon interface does not indicate that the relationships between the objects are broken.

Before you delete a relationship type value from the dropdown configuration, it is recommended that you ask the Axon users in your organization to edit the objects to remove the relationships.

To search for objects that are related to each other using a specific relationship type, use the **http(s)://<Axon_host>:<port>/unison/v2/facet/_search** API, and filter by the `relTypeName` parameter. For more information, see the *Retrieve Primary Objects and Their Related Objects* chapter in the *Axon Data Governance REST-API Guide*.

To delete the relationships after you find them, use the **http(s)://<Axon_host>:<port>/bulkupload/v1/relationships** API. For more information, see the *Create or Update a Single Object, and Create or Delete a Single Relationship* topic in the *Axon Data Governance REST-API Guide*.
After the users remove the relationships, you can delete the relationship type value from the dropdown configuration.

Delete Predefined Dropdown Values

You can delete the predefined values for dropdown configuration parameters. A SuperAdmin user can perform this task.

When you attempt to delete a predefined value that is in use, Axon validates the operation based on the following scenarios:

- For some dropdown configuration values, you can see a message to remove the values used in Axon objects before you delete the dropdown values. Before you delete such values, you need to remove the values for the objects from Axon.
- For some dropdown configuration values, you can remove them without removing their use in Axon objects. A confirmation message appears and you can delete the value after you confirm to proceed.

When you try to delete a predefined value that is not in use, Axon performs the following validations:

- If the value is present in the minimum range list for a dropdown configuration, you cannot delete the value because it is one of the required values.
- If the value is the only one that is remaining for a dropdown configuration, you cannot delete the value.

Deletion Exceptions

You cannot delete some of the default dropdown configuration values because of their impact and dependency across Axon. The following table lists the dropdown configuration parameters and the values that you cannot delete:

Dropdown Configuration Parameter	Values	Reason
Axon Status	**Active** and **Deleted**	Both values are used across all facets in Axon.
Data Set Lifecycle	**Draft**	During automated onboarding from Enterprise Data Catalog into Axon, the data sets are onboarded by default in the **Draft** stage of the data set lifecycle.

Dropdown Configuration Parameter	Values	Reason
Data Set Type	**Data Set**	During automated onboarding from Enterprise Data Catalog into Axon, the data sets are onboarded by default as the **Data Set** type.
Access Control Type	**Public** and **Non-Public**	Both values are used for the **Axon Viewing** field of an object.
Attribute Sourcing Type	All values	All values are required for lineage recommendations during automated onboarding from Enterprise Data Catalog into Axon.
Attribute Origination	**Enterprise Catalog**	During automated onboarding from Enterprise Data Catalog into Axon, the origin of onboarded attributes is **Enterprise Catalog**.
Regulation Status	**Active** and **Deleted**	Both values are used to indicate the status of Regulation objects in Axon.

Configure Glossary Rollup Settings

As an Admin user or as a SuperAdmin user, you can configure the glossary types that you want to see in the rollup view.

You can choose to configure the default glossary types that you want to see in the rollup view in the **Impact** and **Data** tabs of the Glossary object. The default glossary types include entity, term, and domain. You can enable or disable the glossary types to configure the default rollup view from the **Glossary Rollup Settings** link under **Customize & Configure** > **Application Settings**. For example, you can choose to view only entity type glossaries when you enable the rollup view in a Glossary object.

To save all the configuration changes, clear the Axon cache and load the cache with the new configuration values. In the Linux environment, run the following command:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Configure Saved Searches for Quick Links

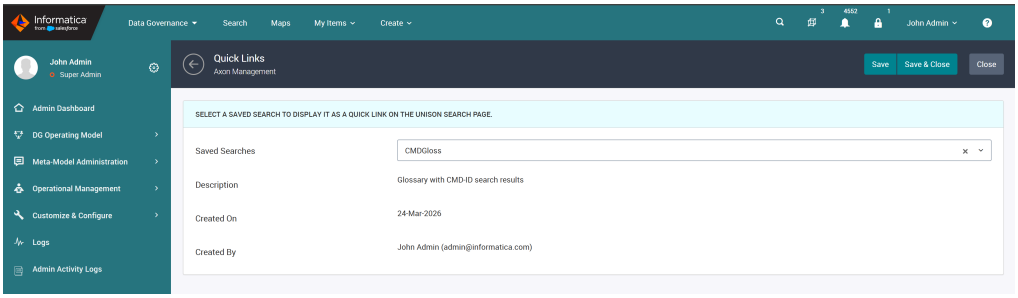
You can configure the saved searches to make them appear as a quick link button on the Unison search page. You can view the quick link button even if you do not log in to Axon.

You must have the Admin or Super Admin profile to perform this task.

Note: Informatica recommends that you create and save searches from Unison search instead of the Admin Panel. For more information, see the *Saving and Sharing Searches* topic in the *Search and View Objects* chapter of the *Axon Data Governance User Guide*

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **Application Settings**.
3. In the **Application Settings** page, click the **Quick Links** option.
4. On the **Quick Links** page, select a saved search that you want to display as a quick link button on the Unison search page.

   After you select a saved search, you can view the description of the search, the date the search was created, and the user who created the search.
   To delete a selected search, you click the **x** icon next to the name of the selected search.



5. Click **Save**.

After you save the changes, the name of the selected search appears as a quick link button on the Unison page.

Hide Empty Fields

A SuperAdmin user can hide fields with empty values from the view mode of the Summary page across all objects in Axon.

1. From the Axon toolbar, click the **Admin Panel** menu item located under the SuperAdmin user.
2. From the navigation pane, go to the **Customize & Configure** > **System Settings**.
3. From the **Group** list, select **Environment**.
4. Click **Edit**.
5. Select **Hide Fields**.
6. To save the settings, click **Save**.

   Axon hides the fields that have empty values from the view mode of the Summary page across all objects in Axon.

Hide Sensitive Personal Information

You can restrict the visibility of the **Email**, **Employee Type** and **LAN ID** information of objects listed under the **People** facet in Unison to only Axon users. Users who have not logged into Axon will not be able to view these fields or open the objects.

The feature is disabled by default. You need to be a SuperAdmin user to enable it. You can verify whether it is enabled or not using the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. From the navigation pane, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Environment**.
4. Review the status of the **Hide Personal Information** field to confirm whether the feature is enabled or not.

To modify the **Hide Personal Information** field, use the following steps:

1. Navigate to the following directory:
   `<INSTALLATION_DIR>/scripts/`
2. Locate and run the following script:
   - Hide sensitive people information:
     `sh hidePeople enable`
   - Show sensitive people information:
     `sh hidePeople disable`

Enable Dashboards

To view dashboards when you go to the Axon home page, you need to enable the option in the Admin Panel.

You must have the SuperAdmin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Dashboard**.
4. Click **Edit**, and select the **Enable My Dashboard** check box.
5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users use Axon.

Restrict Data View Access for Guest Users

You can configure Axon to restrict data views for users without a login. To enable restricted access, ensure you have SuperAdmin privileges.

- Select the **Admin Panel** menu item.
- From the navigation pane, under the **Customize & Configure** category, click **System Settings**.
- In the **Group** list, select **Environment**.
- Click **Edit**.
- Enable the **Restrict Access for Guest Users** option.
- Click **Save**.
- In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:
  `sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync`

  When you run the paramsync script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the System Settings pages. Additionally, perform this action during a maintenance period when very few users use Axon.

Restrict Non-Public Object Web User Access for Non-Stakeholders

You can configure Axon to restrict non-public objects for web users who are non-stakeholders. To enable restricted access, ensure you have SuperAdmin privileges.

- Select the **Admin Panel** menu item.
- From the navigation pane, under the **Customize & Configure** category, click **System Settings**.
- In the **Group** list, select **Environment**.
- Click **Edit**.
- Enable the **Restrict Non-Public Object Web User Access for Non-Stakeholders** option.
- Click **Save**.
- In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:
  `sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync`

  When you run the paramsync script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the System Settings pages. Additionally, perform this action during a maintenance period when very few users use Axon.

Transfer Ownership of Dashboards and Shared Searches

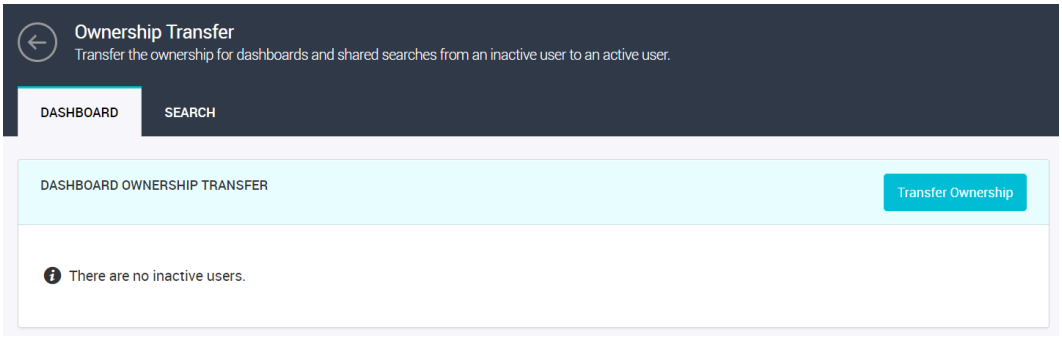
The SuperAdmin user can transfer the ownership of dashboards and shared searches from one Axon user to another. Before you transfer the ownership, set the dashboard or shared search owner's Axon status to a status other than active. Axon does not permit transfer of ownership between two active users.

Consider the following key guidelines when you manage content from deleted users:

- If any dashboard or saved searches are shared by deleted users, the users appear without hyperlinks on the Dashboard and Search Ownership Transfer page.
- The transfer ownership of the deleted users still functions correctly.
- The dashboards previously shared by deleted users appear on the main dashboard page or Home page without causing the page to stop responding.

Transfer Dashboard Ownership

The following image depicts the **Dashboard** tab in the **Ownership Transfer** page:



Before transferring the ownership to another active user, set the owner Axon status to deleted. To transfer the ownership of dashboards from one Axon user to another, perform the following steps:

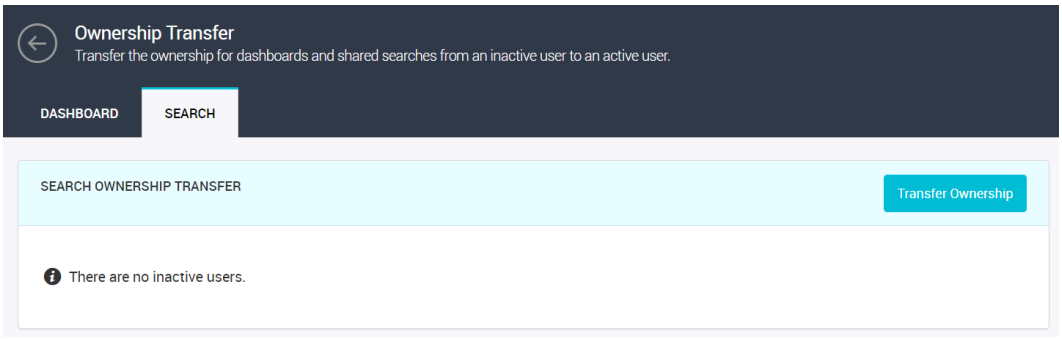
1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under **Operational Management**, click **Ownership Transfer**.

   The **Ownership Transfer** page appears.
3. In the **Dashboard** tab, select the dashboards that you want to transfer, and click **Transfer Ownership**.

   The **Transfer Dashboard Ownership** dialog box appears.
4. From the **New Owner** listing, select the user to whom the dashboard ownership is to be transferred, and click **OK** to complete the transfer.

Transfer Search Ownership

The following image depicts the **Search** tab in the **Ownership Transfer** page:



Before transferring the ownership to another active user, set the owner Axon status to deleted. To transfer the ownership of a shared searches from one Axon user to another, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under **Operational Management**, click **Ownership Transfer**.

   The **Ownership Transfer** page appears.
3. In the **Search** tab, select the shared searches that you want to transfer, and click **Transfer Ownership**.

   The **Transfer Search Ownership** dialogue box appears.
4. From the **New Owner** listing, select the user to whom the shared search ownership is to be transferred, and click **OK** to complete the transfer.

CHAPTER 6

Object Management

This chapter includes the following topics:

- Object Management Overview, 91
- Object Permissions, 91
- Configure Workflows and Change Requests, 93
- Manage Locks, 112
- Disable Export of Objects, 113
- Hide Non-Public Objects, 113
- Configure Periodic Review of Objects, 113
- Configure Threshold for Map Objects, 115
- Specify Prefix for Object Reference Numbers, 116

Object Management Overview

As the administrator, you can restrict object access to a limited set of users, prevent users from editing objects, and manage other permissions related to objects.

Object Permissions

Configure the actions that a role can perform by assigning permissions. When you assign permissions to a role, all users who are assigned the role inherit the permissions.

You can grant the following permissions:

New

A role with the **New** permission can create objects within the same facet as the object for which you have assigned the role. A user with only the **New** permission cannot edit objects.

Edit

A role that has the **Edit** permission on an object can edit and update the object properties. A user with only the **Edit** permission cannot create objects.

View

A user with the **View** permission can view objects. All Axon users have the **View** permission.

The following table lists the permissions for Axon users when they search for "Deleted" objects in Unison search:

User Profile	Deleted Objects Appear in Unison Search	Can Open Deleted Objects from Unison Search
User not logged into Axon	No	No
WebUser with no stakeholder role	No	No
WebUser with stakeholder role	No	No
Admin or SuperAdmin	Yes	Yes

The following table lists the permissions for Axon users when they search for "Non-Public" objects in Unison search:

User Profile	Non-Public Objects Appear in Unison Search	Can Open Non-Public Objects from Unison Search
User not logged in to Axon	No	No
WebUser with no stakeholder role	No if you choose to hide non-public objects from the Admin Panel. Otherwise, the non-public objects appear in Unison search for a WebUser with no stakeholder role.	No
WebUser with stakeholder role	Yes	Yes
Admin or SuperAdmin	Yes	Yes

Assign Permissions to Roles

Permissions define the level of access that users have on objects in a facet. You must define the scope of the permission by selecting the facet for which the role is valid. All users who are assigned the role inherit the permissions.

You must have the Admin profile to view role permissions. You must have the Super Admin profile to edit role permissions.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left of the page, click **Role Permissions** under **DG Operating Model**.
3. Click **Edit**.
4. Click the **+** icon against the role that you want to update.
5. Configure the following properties:

Option	Description
Module	From the list, select the facet for which the permissions is valid.
Role	From the list, select the role for which the permission is valid.

Option	Description
Permission	From the list, select the type of permission that you want to assign to the role.

6. Click **Save and Close**.

Configure Workflows and Change Requests

Configure the behaviour or workflows and change requests when users create or modify Axon objects.

Create a Default Workflow

You can configure Axon to use a default workflow when users create or modify objects. You can specify a default workflow for a facet, and Axon uses the default workflow whenever users create or modify objects for the facet.

You must have a SuperAdmin profile to create a default workflow.

For more information on workflows and change requests, see the *Workflows and Change Requests* chapter in the *Axon Data Governance User Guide*.

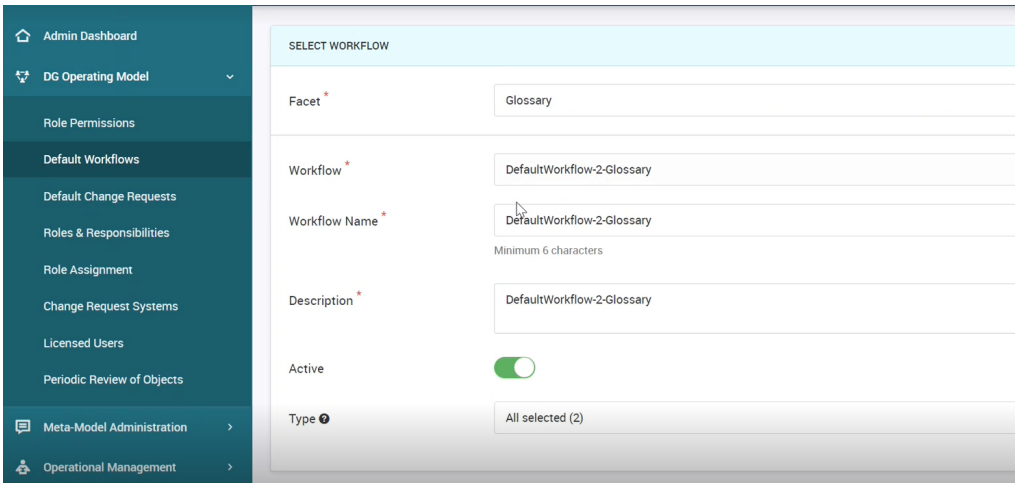
1. From the Axon toolbar, click **Admin Panel** under the user name.

   The **Admin Dashboard** appears with the Axon version details, status of Axon services, and Axon message queues.
2. On the left navigation pane, click **DG Operating Model** > **Default Workflows**.
3. Select the facet for which you want to configure a default workflow.
4. Choose to add a new workflow.
5. Enter a name for the workflow.

   The name must contain a minimum of six characters.
6. Provide a description for the workflow.

   Axon creates a canvas with many swimlanes based on the roles that are defined for the facet.
7. From the **Type** dropdown, select the required change request type.

   Note: Based on selected type, Axon allows user to start workflow automatically after raising change request.



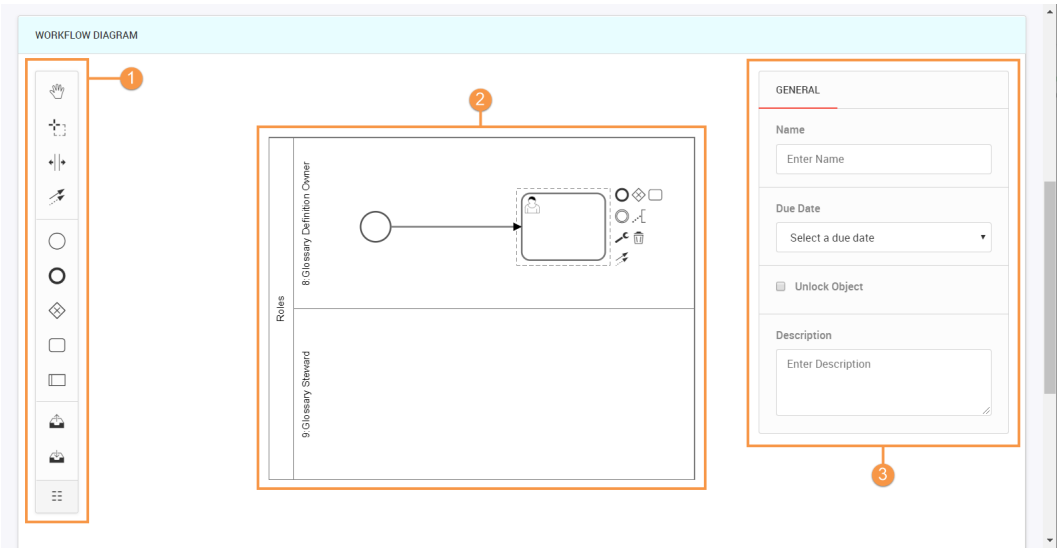
8. In **Workflow Diagram** section, create a workflow diagram on the canvas. Create swimlanes to assign tasks to the facet stakeholders or Requestor. Use the **Toolbar** to add objects and connect them. Use the **Properties** panel to specify additional properties for the objects.
9. In the workflow diagram, specify the required fields for the following items:
   - User Task. You need to enter a name and due date for a user task.
   - Exclusive Gateway. Specify short, unique labels for the lines that start from an exclusive gateway. The labels serve decision routing actions when you run the workflow. A gateway and the preceding user task must be close to one another in the same swimlane because they both are a logical unit and must be performed by the same user.
10. Click **Save**.

Workflow Designer

Axon implements a visual editor to create and run a workflow from a browser.

When you provide a name and description for a new workflow, Axon creates a canvas to create a workflow diagram. The Workflow Diagram section has three components: a toolbar on the left, a canvas in the center, and a properties panel on the right.

The following image shows the different sections of a workflow diagram:



1. Toolbar
2. Canvas
3. Properties

Toolbar

Use the toolbar to create a workflow diagram. The toolbar has three sections. The top section contains tools to select the items, resize the canvas, and connect the items. The middle section lists the building blocks of the workflow. You can select the items and drag to the canvas to create the workflow. The bottom section contains tools to show or hide the toolbar.

The following table describes the different tools that you can use to create a workflow diagram:

Tool Name	Description
Hand	Use the hand tool to select an item.
Lasso	Use the lasso tool to select a block on the swimlane.
Create/Remove Space	Use the create or remove tool to increase or decrease the canvas size.
Global Connect	Use the global connect tool to connect the items in a workflow.
Create StartEvent	Use the Create StartEvent tool to start a workflow event. You can decide the role that can start the event.
Create EndEvent	Use the Create EndEvent tool to end a workflow event. You can decide the role that can end the event.
Create ExclusiveGateway	Use the Create ExclusiveGateway tool to create decision boxes that can have two results. For example, a validate request can have two values: accept and reject. A gateway and the preceding user task must be close to one another in the same swimlane because they both are a logical unit and must be performed by the same user.

Tool Name	Description
Create UserTask	Use the Create UserTask tool to create tasks that a user can perform. You can enter a name, due date, and description for a user task. You can create user tasks such as validate request, capture impact assessment, confirm feasibility, and approve request.
Create Pool/Participant	Use the Create Pool or Participant tool to add a swimlane for a user role.
Upload BPMN	Use the Upload BPMN tool to upload a workflow diagram to the canvas.
Download BPMN	Use the Download BPMN tool to download a workflow diagram from the canvas.
Hide/Show	Use the Hide or Show tool to hide or show the toolbar.

Canvas

The canvas provides a visual representation of the workflow. When you create a custom or default workflow, Axon automatically creates an empty canvas with swimlanes for each role that is defined for a facet. For example, if a Glossary facet has four roles, Axon creates four swimlanes, one for each role.

Select a swimlane and drag a corner to resize the canvas vertically and horizontally. Select items from the toolbar, drag them to the canvas, and connect them. When you place an item on the canvas, you can click the item to view the toolbar options that are relevant to the object.

BPMN workflows use swimlanes to capture who is responsible for a set of tasks. The tasks that you specify in a swimlane becomes the responsibility of the role associated with the swimlane. If one or more roles do not contain any task in the workflow, you can leave the swimlanes empty or delete the swimlanes. If you want to create the swimlane again, you can use the tool bar to create a swimlane and you must specify the role name and the correct role ID.

Properties

In the **Properties** panel, you can specify additional properties for the items or tasks on the canvas. The properties that appear depend on the tool that you add to the workflow. All properties do not appear for all tools.

The following table describes the properties that you can configure:

Tool Name	Description
Name	Name of the workflow step. If you click a role in the workflow canvas, select the object stakeholder or requestor that participates in the workflow step.
Due Date	Date by which the workflow step must be completed.
Description	Description of the workflow step.
Axon Status	Status of the object during the workflow step as it appears in the **Axon Status** field.
Lifecycle	Lifecycle of the object during the workflow step as it appears in the **Lifecycle** field.

Tool Name	Description
Unlock Object	Select the check box to unlock the object at a specific step in the workflow. If you do not select the check box, the object remains locked.
Commit Changes	Select the check box to save the object with the changes and apply the values that you specified in the **Axon Status** and **Lifecycle** fields. This check box is typically selected when the workflow has reached the last step and a stakeholder has approved the changes to the object. Clear the check box to not save the object. This check box is typically cleared when the workflow has reached the last step and a stakeholder has rejected the changes to the object.

If you have a gateway in a workflow, you must specify the labels for the lines that start from the gateway. The labels need to be short and unique. The labels serves decision routing actions when you run the workflow.

Predefined Workflows

Axon has predefined default workflows for each facet. You can use these sample workflows to build your own workflows.

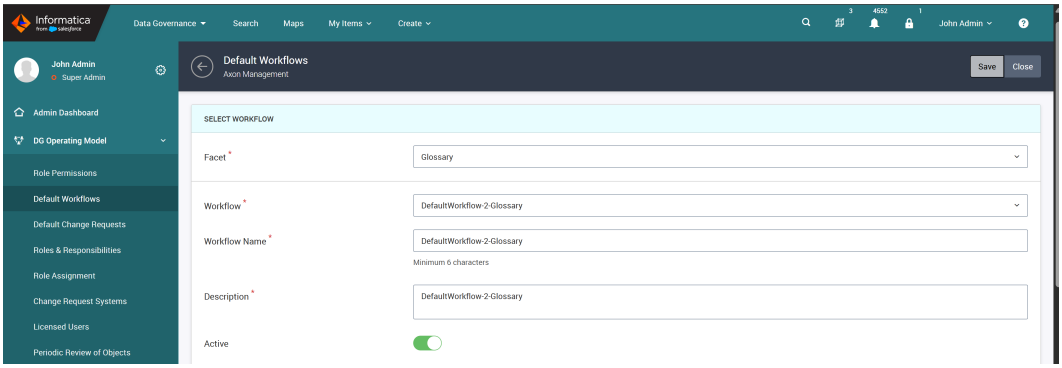
To view the predefined workflows from the Admin Panel, navigate to **DG Operating Model** > **Default Workflows** and select a facet. For each facet, you can view a sample predefined workflow. When you select the predefined workflow, Axon displays a canvas with the sample workflow for the facet.

The workflow diagram is spread across the swimlanes based on the roles that are defined for the facet. The swimlanes appear with a lane for each type of stakeholder role configured for a facet. For example, the Glossary facet has Glossary Steward and Glossary Definition Owner roles and the predefined workflow diagram is spread across both the lanes. The Business Area facet has Business Area Head role and the predefined workflow is present in a single lane. You can find an additional requestor swimlane in the diagram. A requestor can be any user in the organization, including users that are not stakeholders for a facet, who raises a change request.

The predefined workflow name is in the following format: `DefaultWorkflow-n-FacetName`. For example, if the Glossary facet has two stakeholder roles, the predefined workflow name is `DefaultWorkflow-2-Glossary`. If the Business Area facet has one stakeholder role, the predefined workflow name is `DefaultWorkflow-1-BusinessArea`.

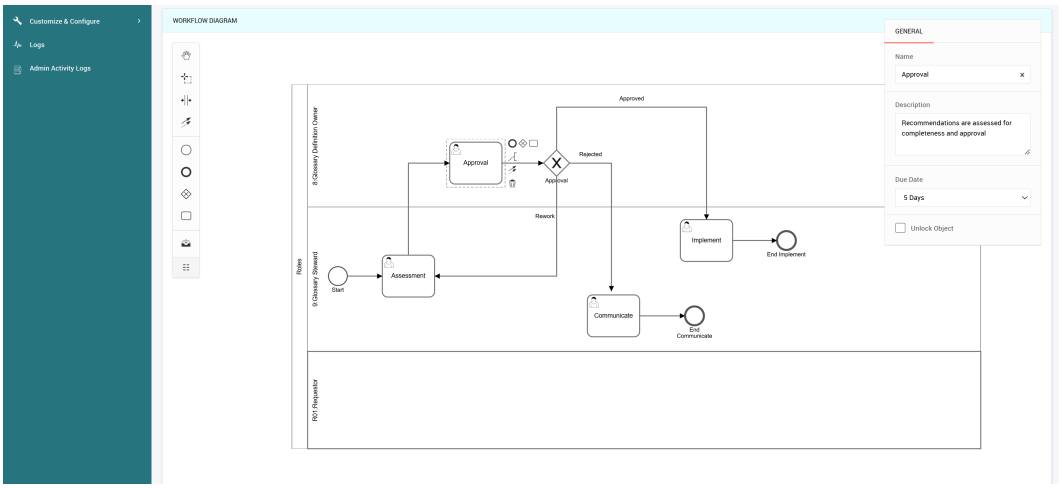
Consider that you select the predefined workflow for the Glossary facet. The rest of the fields are automatically populated.

The following image shows the predefined workflow selection for the Glossary facet:



You can see the predefined workflow tasks spread across two swimlanes in the Workflow Diagram section. The task starts and ends in the Glossary Steward swimlane. The Glossary Definition Owner swimlane includes a decision-making task.

The following image shows the workflow diagram for the Glossary facet:



1. Requestor
2. Glossary Steward
3. Glossary Definition Owner

In the Glossary workflow diagram, you can see that the steward creates a glossary term and sends the term to the definition owner for approval. If the glossary term is approved, the steward ends the workflow after implementation. If the glossary term requires rework, the steward reworks and sends to the definition owner again for approval. If the glossary term is rejected, the steward ends the workflow after communicating to other users. The Requestor swimlane is added by default in the sample workflow diagrams for all facets. You can make use of the Requestor swimlane by creating additional tasks for the requestor or delete the swimlane if you do not want.

You can view the sample predefined workflow for the following facets:

- Business Area
- Capability
- Glossary
- Data Set
- Client
- Committee
- Data Quality Rule
- Policy
- Process
- Product
- Project
- Regulation
- System

- System Interface

Workflow Diagram Example

The example shows a new Default Workflow being created. Apart from the different start point, the functionality to build the workflow is identical. The facet used in this example is Product. After you select the Product facet, you can see that one default workflow already exists. You can either select this one to edit, or create another.

1. From the Axon toolbar, click **Admin Panel** under the user name.

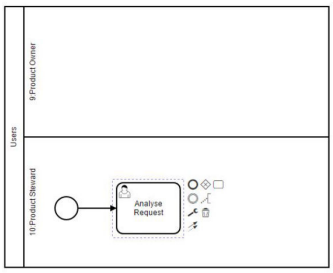
   The **Admin Dashboard** appears with the Axon version details, status of Axon services, and Axon message queues.
2. On the left navigation pane, click **DG Operating Model** > **Default Workflows**.
3. Select the **Product** facet for which you want to configure a workflow.
4. Choose to add a new workflow.
5. Enter a name for the workflow.

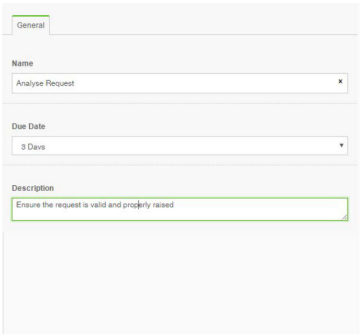
   The name must contain a minimum of six characters.
6. Provide a description for the workflow.

   Axon creates a canvas with many swimlanes based on the roles that are defined for the facet. Remember that the swimlanes appear with a lane for each type of stakeholder role configured for the facet, that is for the Product Steward, Product Owner role.
7. Click into one of the lanes, so that a dotted line appears, and use the bin icon on the menu to delete this lane. You can add it back in later if required.
8. From the toolbar, choose the **Start Event**, and place it in the Steward's swimlane. This means that the objects stewards receive the first request to action when a ticket is raised against this request.
9. Typically the first thing that anyone does is to validate that the request should have been routed to this workflow. If OK, then do something. If Not OK, inform the requestor and close the ticket.
10. The first step is to raise a User Action, and populate the Properties on the right.
    1. Click into the Start action.
    2. From the options cloud that appears, choose the User Action and drop it in the Steward's swimlane.

    A line automatically appears between the two concepts.

    The following image shows the start step and the first user task of the workflow:

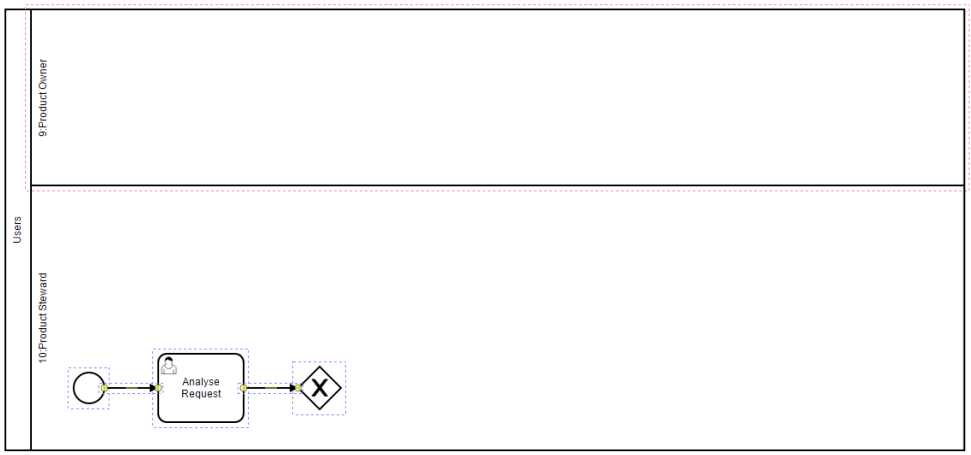




The User Action can be populated by completing the Properties section, as above.

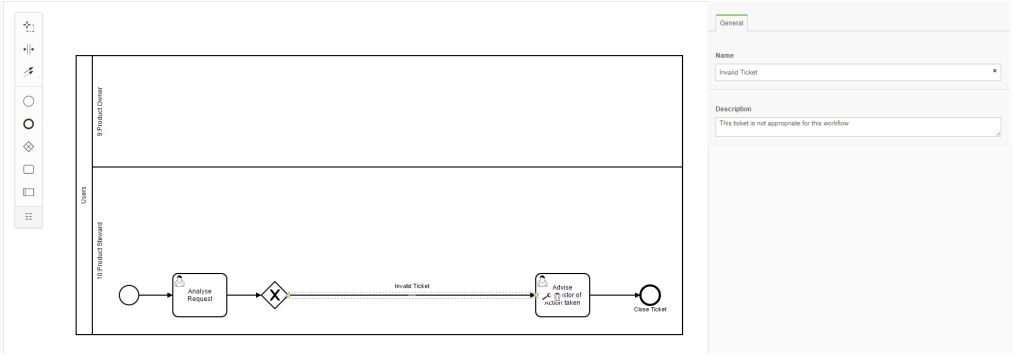
11. Next, you need a Gateway to show that there are two options - accept as valid request or reject.
    1. Click in the User Action and the same option cloud appears as before. This time select the diamond shaped concept, ensuring that it is placed in the Steward's swimlane.
    2. In the next picture the user has increased the horizontal width of all swimlanes, increased the height of the Steward's swimlane, and finally used the lasso tool on the Toolbar to grab and move the concepts you have created so far to the bottom of the swimlane.

    The following image shows the increased length of the swimlane:



12. Now, you can add the two possible paths. First, an invalid request needs a User Action to Inform the user and then a Close Action.
    1. Always remember to click the predecessor object, create the User Action, and close the workflow. Remember to populate the Properties for each object.
    2. Lines leaving a gateway MUST have a name added to them, because these will appear in the any active workflow as labels on the Accept/Reject buttons. Keep them short and clear.

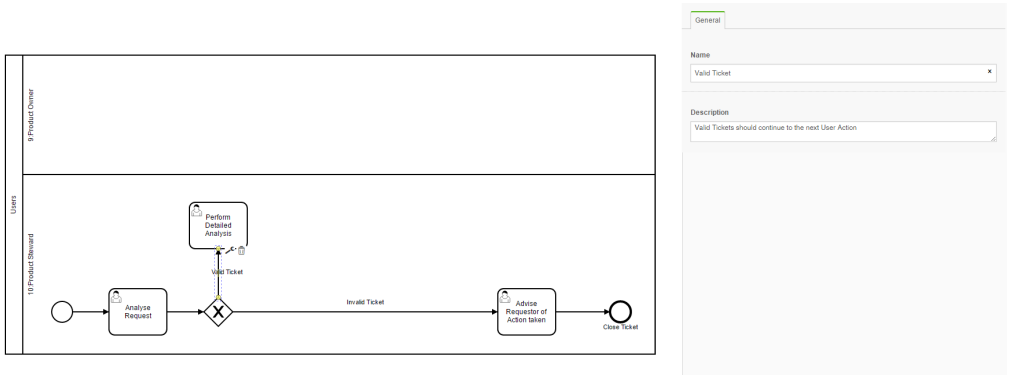
    The following image shows the workflow diagram with the user task:



13. If you want to leave the Gateway on the alternative (valid request) path, then you need to create the next User Task.

    This step is to 'Perform Detailed Analysis', and has been created in the picture below. As you can see, this is also performed by the Product Steward.

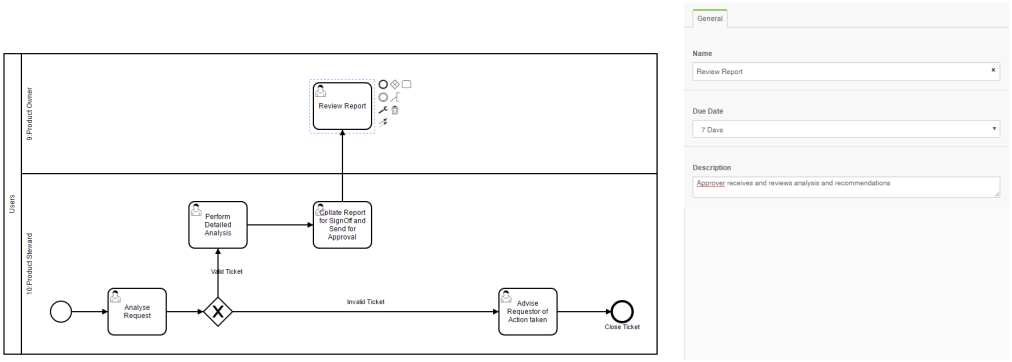
The following image shows the workflow diagram with the user task:



14. After some additional tasks by the Steward shown as an additional User Task (Collate Report) in the same swimlane, it is time to send to the Product Owner user for review and approval.

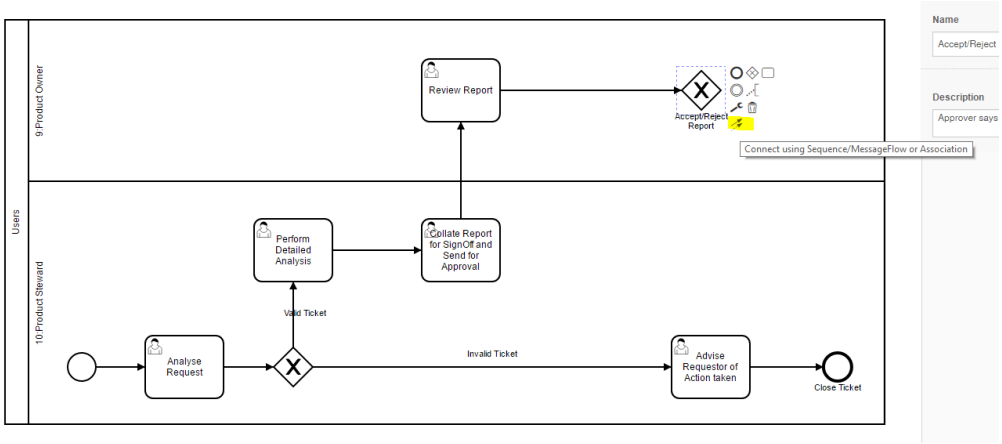
    This time, from the option cloud on the 'Collate Report' User Action, simply drag the User Action into the Owner's swimlane, and fill in the Properties.

    The following image shows the properties that you can fill in for the user task:



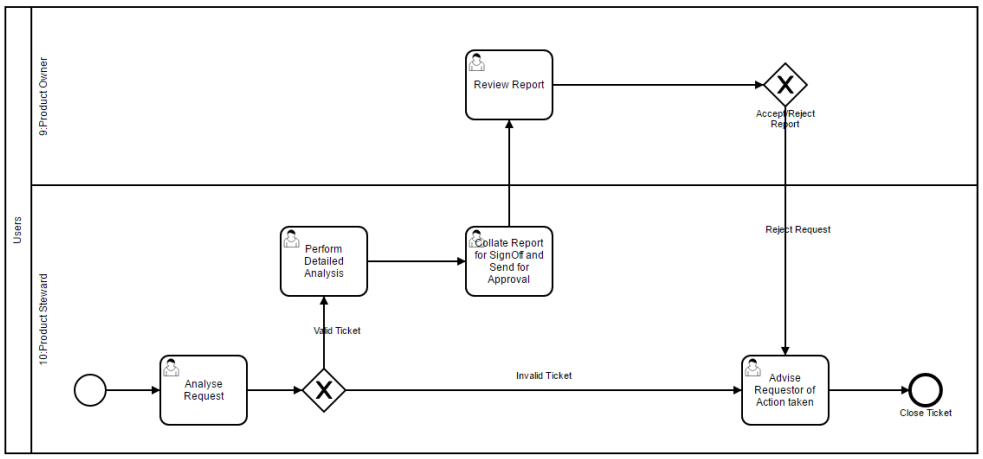
15. As the possible actions are to either accept or reject the findings in the report, you need a Gateway to show these two alternatives. If the Owner rejects the report, then you should connect this new Gateway to the User Action already created to 'Advise Requestor of Action taken', and then you close the workflow.

    The following image shows the alternatives for the gateway:



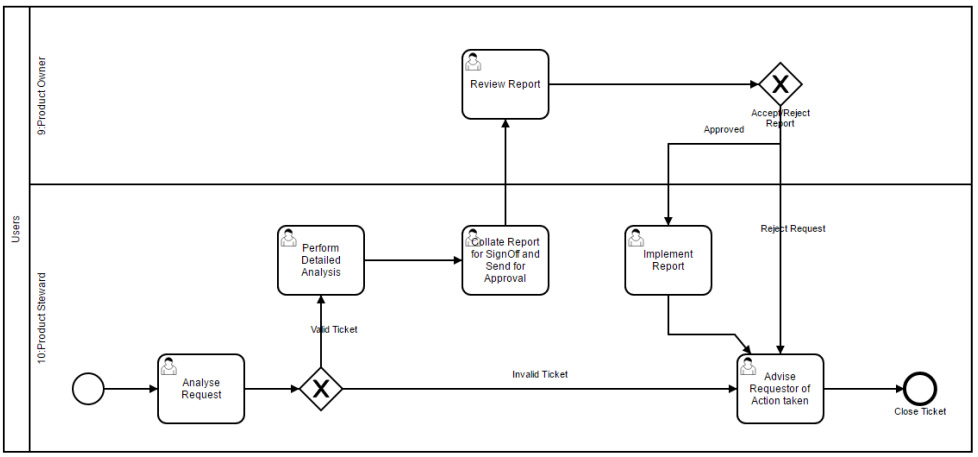
16. As this Gateway and the User Action both already appear on the page, you need to do something slightly different to connect them.
    1. Click on the Gateway, and select the lines icon, from the option cloud.
    2. Once clicked simply move your mouse toward the 'Advise Requestor…' User Action, and a line will be drawn between them.
    3. Remember to go back to this new line and name it. Click in the line between the two concepts and fill in the Properties. The line's name will later appear in the workflow step when the Owner has to accept/reject the step.

    The following image shows the workflow with the Reject path complete:



17. If the reviewed report is accepted, add something in to show what happens. The Steward is expected to implement the report, so create a User Action in the Steward's swimlane.

    The following image shows the action added to the steward role:



18. Save and close the workflow.

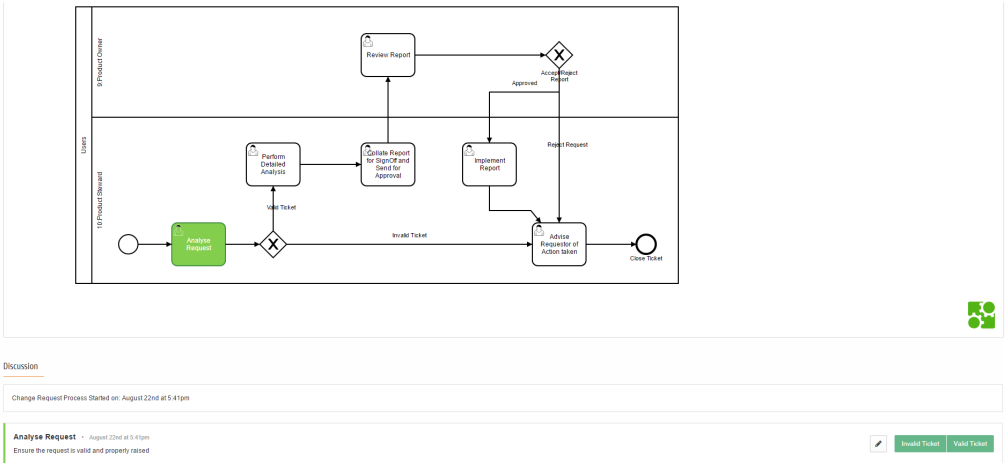
    You now have a completed workflow. All paths ultimately lead to the Close event. All lines leading FROM a Gateway have a name, which creates the approve/reject buttons in a live workflow.

19. When a Change Request is raised against the workflow, you can see the CR on the canvas.

You can see that:

1. The Active Step, i.e. the step awaiting action is Green. If orange it is alerting users that the Due By Date is less than a day away.
2. The description that was created for this workflow step (in the Properties section) appears as guidance on the action expected by the Steward in the Discussion section below the canvas..
3. There are two green buttons in the Discussion, which allow the Steward to accept/reject the step. They have the same names as the lines that leave the Gateway immediately after the active User Action.

The following image shows the workflow diagram for the change request:



Configure Behavior of Change Requests

You can configure the behavior of change requests when users create or modify Axon objects.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. On the navigation pane, click **Customize & Configure** > **System Settings**.
3. In the **Group** list, select **Change Requests**.
4. Click **Edit**.
5. Select **Auto-complete Change Requests** if you want Axon to automatically complete change requests that have reached the end of the workflow.
6. Select **Enable automatic deletion of change requests** if you want Axon to automatically delete change requests that is in complete or cancelled state. Also, ensure that the Axon objects associated with these change requests are deleted. This option is not checked by default.

   Note: When you upgrade to Axon version 7.3.2, if you want to remove the older change requests associated with the deleted Axon objects present in the earlier versions, run the following script after the upgrade:

   ```
   sh <INSTALLATION_DIR>/axonhome/axon_scheduler/changerequests/
   updateExistingChangeRequests.sh
   ```

   Enter a value in the **Days to automatically delete change requests** option to automatically delete the change requests:

   - Enter 0 to delete the change requests as soon as the associated Axon object is deleted.

- Enter a value greater than 0 to delete the change requests after the specified number of days of deletion of the Axon object.

Note: Do not enter a negative number.

7. Click **Save**.
8. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users use Axon.

Configure Default Values for Automatic Change Requests

You can configure Axon to start a change request automatically whenever users create or modify an object.

You must have the Admin or Super Admin profile to perform this task.

When users create or modify an object, they can save the object without review from other stakeholders. In the Admin Panel, you can configure Axon to automatically start a change request and mandatorily use a workflow whenever users create or modify an object. This type of mandatory approval ensures that the right stakeholders have the opportunity to review the object changes that users make.

For more information on workflows and change requests, see the *Workflows and Change Requests* chapter in the *Axon Data Governance User Guide*.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **DG Operating Model** category, click **Default Change Requests**.
3. In the **Facet** list, select the facet for which you want to create a change request.

   The option is available for the following facets:

   - Glossary
   - Data Set
   - Process
   - System
4. Select the **Enable Workflow Approval** option to use the workflow that you want to define for the facet object.

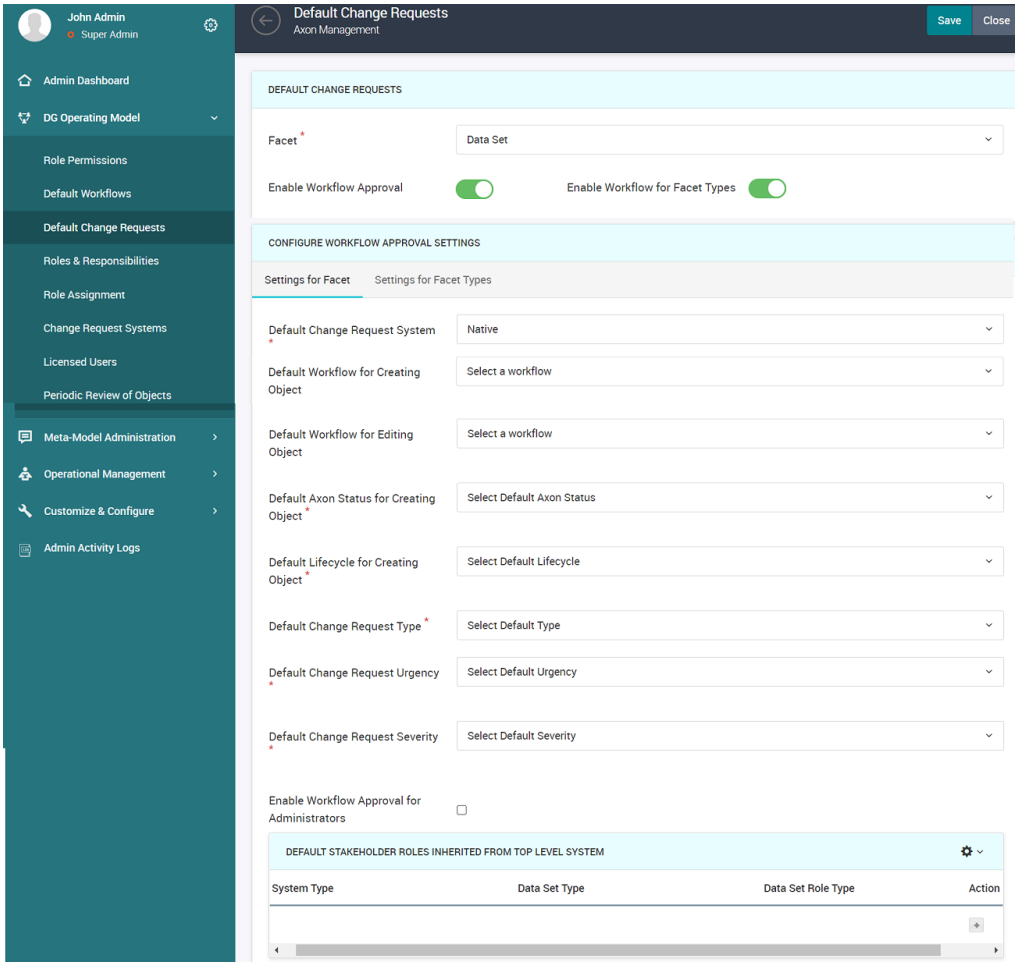
   The page displays the settings for the change request.
   If you do not select this option, you can create or modify the facet objects without using the workflow.
5. Select the **Enable Workflow for Facet Types** option to configure different workflow for different object types.
   For example, you can specify a workflow for change requests created for Glossary domain objects, and another workflow for change requests created for Glossary term objects.

   The page displays the settings for the change request for different facet types.

6. Configure the following properties for the change request for facet objects:

Property	Description
Default Change Request System	Default system that Axon must use for the change request. Select one of the following options: • Internal. Use the Axon change request system. • ServiceNow. Use the ServiceNow change request system. ServiceNow option appears when the external change request system is configured for the selected facet type. To configure Axon to connect to external change request system, see "Connect to External Change Request Systems" on page 108.
Default Workflow for Creating Object	Default workflow that Axon must use when you create an object. Select an active workflow that you have already created in Axon. To create or modify workflows, see "Create a Default Workflow" on page 93. **Note:** This field appears when you select `Internal` in the **Default Change Request** field.
Default Workflow for Editing Object	Default workflow that Axon must use when you modify an existing object. **Note:** This field appears when you select `Internal` in the **Default Change Request** field.
Additional Fields	Specify the additional fields that an external change request system requires to create a ticket. Enter the values in JSON format. When you specify the fields, the fields appear in the **External Fields** section of an Axon change request. For example, if the external system requires a change request to have a severity level and due date, you can configure `Severity` and `Due Date` as additional fields. **Note:** This field appears when you select `ServiceNow` in the **Default Change Request** field. For details on configuring the additional fields, see "Configure Additional Fields for External Change Request System" on page 110.
Default Axon Status for Creating Object	Default value for the **Axon Status** field when you create an object. **Note:** If you create an object through the bulk upload operation, Axon ignores the value you enter in the bulk upload file for the **Axon Status** field and uses the value that you specify here.
Default Lifecycle for Creating Object	Default value for the **Lifecycle** field when you create an object. **Note:** If you create an object through the bulk upload operation, make sure that in the bulk upload file you enter for the **Lifecycle** field the same value that you specify here.
Default Change Request Type	Default value for the **Type** field when you create an object.
Default Change Request Urgency	Default value for the **Urgency** field when you create an object.
Default Change Request Severity	Default value for the **Severity** field when you create an object.
Enable Workflow Approval for Administrators	Require workflow for changes made to objects by Admin and SuperAdmin users."

The following images shows an example for the **Default Change Requests** window:



g

7. In the **Default Stakeholder Roles to be Used From Top-Level Object** section, select the stakeholders that Axon must assign by default to the object that you want to create or modify.

   The section displays the list of stakeholders that you have configured for the facet. To add or remove stakeholders, see "Configure Roles" on page 40.

8. Configure the following properties for the change request for different object types:

Property	Description
Facet Type	Type of facet for which the change request and workflow setting is applicable. This option is available for the following facets: • Glossary • Data Set • Process • System

Property	Description
Default Change Request System	Default system that Axon must use for the change request. Select one of the following options: • Internal. Use the Axon change request system. Select **Inherited from Facet** option to use the workflow that Axon uses for change requests to the main facet object. • ServiceNow. Use the ServiceNow change request system. If the ServiceNow change request system is not configured for a particular facet type, the option will not appear in the dropdown list.
Default Workflow for Creating Object	Default workflow that Axon must use when you create an object of this facet type. You can change the default workflow configured for the specific object type. To create or modify workflows, see "Create a Default Workflow" on page 93. **Note:** For external change request system, default workflow for creating object is not applicable.
Default Workflow for Editing Object	Default workflow that Axon must use when you modify an existing object of this facet type. **Note:** For external change request system, default workflow for creating object is not applicable.
Default Change Request Type	Default value for the **Type** field when you create an object. Select **Inherited from Facet** option to use the workflow that Axon uses for change requests to the main facet object.

9. You can click **Restore Defaults** to reset all the changes made to the properties for the change request for the different object types.
10. Click **Save**.

Note:

- If you do not configure **Default Workflow for Creating Object** and **Default Workflow for Editing Object** properties, the fields will display the `Inherited from Facet` option.
- When a change request is in progress, and values are associated with the data set attribute, then a user cannot change the attribute name, attribute data type, or attribute length. To modify the data set attribute, unlink the values associated with the attributes and then modify the attributes.
- When a change request is enabled and no active change request is present, and values are associated with a data set attribute, then a user with the steward role cannot change the attribute name, attribute data type, or attribute length. To change the values, unlink the values and then change the attributes.
- When a WebUser modifies the name of an object, Axon starts a change request for the object. If the user bulk update the object during the change request process, the name of the object is changed to the value specified in the bulk upload template. The user can see the modified name of the object in the **View Original** or **View Changes** view of the object.

After you define a workflow for a change request and select the **Enable Workflow Approval** option, Axon uses the workflow when a user with the WebUser profile creates or modifies an object. For information on how to create Axon objects with a workflow, refer the *Creating an Object* topic in the *Axon Data Governance User Guide*

Connect to External Change Request Systems

You can configure Axon to connect to external change request systems that you use in your organization. When you create or modify an object, Axon can create a ticket in the change request system and include the object stakeholders in the workflow. For example, if you create a new Policy object in Axon, the change request system can create a new ticket to track the inputs and approvals for the policy. When a Policy stakeholder participates in the workflow for Policy objects, the change request system tracks the object creation process and notifies the Policy stakeholder.

You must have the Admin or Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **DG Operating Model** category, click **Change Request Systems**.

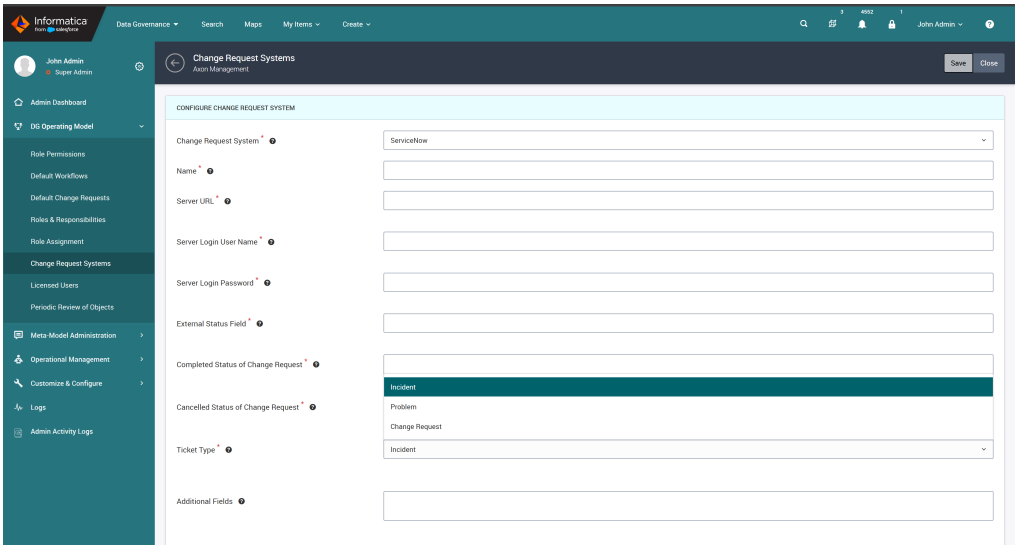
   The page displays the change request systems that you have configured. If you configure Axon to connect to several instances of an external change request system, the page displays all the instances.
3. Click on a change request system to configure its properties. Click **Create** to configure a new change request system for Axon to connect.

   The page displays the properties of the change request system.
4. Configure the following properties:

Property	Description
Change Request System	Select the change request system that you want to configure. Axon supports connection to ServiceNow and to Jira Cloud.
Name	Name of the change request system instance.
Server URL	URL of the change request system service in the following format: `http(s)://<host_name>` or `http(s)://<IP_address>`
Server Login User Name	User name to log in to the change request system
Server Login Password (ServiceNow only)	Password to log in to the change request system
Server Login Token (Jira Cloud only)	The token to log in to Jira Cloud. To get a server login token, log in to Jira Cloud, navigate to Account Settings > Security > Create
External Status Field	Enter the field name in the change requests system that indicates the ticket status. For example, if the change request system uses the "State" field to indicate the ticket status, enter `State` here.
Completed Status of Change Request	Enter the status in the **External Status Field** that corresponds to a completed change request. For example, if a completed status in the `State` field is indicated as "Resolved", enter `Resolved`.
Cancelled Status of Change Request	Enter the status in the change request system that corresponds to a cancelled change request. For example, enter `Cancelled` or `Revoked`.
Ticket Type	Select any one of the following ticket type based on your requirement: Incident: Track the status of an issue. Problem: Track the potential cause of one or more incidents. Change Request: Track the change request.

Property	Description
Project Key (Jira Cloud only)	Enter the key of the Jira Cloud project.
Issue Type (Jira Cloud only)	Enter the type of Jira Cloud issue. To get the Project Key, log in to Jira Cloud, navigate to Projects > View all Projects.
Additional Fields	Optional. Specify the additional fields that the change request system requires to create a ticket. Enter the values in JSON format. When you specify the fields, the fields appear in the **External Fields** section of an Axon change request. For example, if the ticketing system requires a change request to have a severity level and due date, you can configure `Severity` and `Due Date` as additional fields. For details on configuring the additional fields, see "Configure Additional Fields for External Change Request System" on page 110.

The following image shows an example of the Change Request System window:



5. Click **Save**.

Note:

- When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.
- If you connect Axon to Jira Cloud, Informatica recommends that do not create more than 100 issues at any given moment. If you do try to create more than 100 issues, the amount of time involved increases proportionally. Additionally, you cannot create more than 1000 issues.

Configure Additional Fields for External Change Request System

When you configure Axon to connect to an external change request system, the external system might require additional fields that do not appear by default in the Axon configuration page. Enter the details of the additional fields in JSON format so that Axon can create change requests in the external system.

Go to the Axon Admin Panel, and navigate to **DG Operating Model** > **Change Request Systems**. Select an external change request system, and configure the additional fields in the **Additional Fields** field.

Format

Add each additional field in the following format:

```
{
"key": "<field_name_in_external_system>",
"displayName": "<field_display_name_in_Axon",
"dataType": "<list_or_text_box>",
"defaultValue": "<default_value>",
"isMandatory": <required_or_not>,
"isHidden": <display_in_Axon_or_not>,
"isSensitive": <mask_field_or_not>
}
```

Enter square brackets `[` and `]` at the beginning and end of your entry. Separate each parameter by a a comma (,), and separate each field configuration by a comma (,).

Parameter Descriptions

The following table explains the parameters that you can enter for each field:

Parameter	Description
`key`	Name of the field in the external change request system. For example, if the field in the external system is "Urgency", enter the following parameter and value: `"key": "Urgency"`
`displayName`	Name of the field as you want it to appear on the Axon interface. For example, if you want the "Urgency" field in the external system to appear as **Importance** in Axon, enter the following parameter and value: `"displayName": "Importance"`

`dataType`	Specify whether the field must appear in Axon as a dropdown list or a blank text box.
	- To display the field in Axon as a dropdown list, enter the `list` value:
	`"dataType": "list"`
	- To display the field in Axon as a blank text box, leave the parameter as empty, or enter the `null` value:
	`"dataType": null`

`defaultValue`	Default values that you want to display for the field.
	- If you specify the field as a dropdown list in the `dataType` parameter, enter the dropdown values in a single line. Separate the values by a comma. For example, to specify the High, Medium, and Low values in the dropdown list, enter the following parameter and value:
	`"defaultValue": "High,Medium,Low"`
	- If you specify the field as a text box in the `dataType` parameter, enter the default text in the field. For example, to specify the "Accepted" default value in the text box, enter the following parameter and value:
	`"defaultValue": "Accepted"`

Parameter	Description

`isMandatory`	Specify whether the field is required or optional. For required fields, Axon appends an asterisk (*) to the field name.
	- To mark the field as required, enter the `true` value:
	`"isMandatory": true`
	- To mark the field as optional, enter the `false` value:
	`"dataType": false`

`isHidden`	Specify whether or not the field must appear in the Axon interface.
	- If the field is essential to create a ticket, and you do not want to display the field in the Axon interface, enter the `true` value:
	`"isHidden": true`
	- If the field is essential to create a ticket, and you want to display the field in the Axon interface, enter the `false` value:
	`"isHidden": false`

`isSensitive`	Specify whether you want to use asterisks to mask the values that the user enters in the field. For example, if the user enters an employee code, you might want to mask the entry.
	Note: To specify a field as sensitive, you must specify the field as a dropdown list in the `dataType` parameter.
	- To mask the user inputs, enter the `true` value:
	`"isSensitive": true`
	- To not mask the user inputs, enter the `false` value:
	`"isSensitive": false`

Example

You want to display the following fields from the external change request system in Axon:

Parameter	Field 1	Field 2	Field 3
"key"	The field in the external system is "Severity"	The field in the external system is "Description".	The field in the external system is "Category".
"displayName"	You want to display the field as **Importance** in Axon.	You want to display the field as **Description** in Axon.	You want to display the field as **Select the Issue Type** in Axon.
"dataType"	The field is a dropdown list.	The field is a text box.	The field is a dropdown list.
"defaultValue"	The dropdown values of the list are High, Medium, and Low.	The default entry in the text box is "Enter an optional comment."	The dropdown values of the list are Hardware, Software, and Other.
"isMandatory"	The field is mandatory.	The field is not mandatory.	The field is not mandatory.
"isHidden"	The field is not hidden in the Axon interface.	The field is not hidden in the Axon interface.	The field is not hidden in the Axon interface.
"isSensitive"	The field is not sensitive and does not need to be masked.	The field is not sensitive and does not need to be masked.	The field is not sensitive and does not need to be masked.

To configure the fields, enter the following values in the **Additional Fields** field:

```
[
{
"key": "Severity",
```

```
"displayName": "Importance",
"dataType": "list",
"defaultValue": "High,Medium,Low",
"isMandatory": true,
"isHidden": false,
"isSensitive": false
},
{
"key": "Description",
"displayName": "Description",
"dataType": null,
"defaultValue": "Enter an optional comment.",
"isMandatory": false,
"isHidden": false,
"isSensitive": false
},
{
"key": "Category",
"displayName": "Select the Issue Type",
"dataType": "list",
"defaultValue": "Hardware,Software,Other",
"isMandatory": false,
"isHidden": false,
"isSensitive": false
}
]
```


Manage Locks

Locks prevent users from editing an object. With an Admin profile, you can manage locked objects.

A content manager can assign one of the following types of locks:

Edit Lock

This lock prevents other users from editing an object that is currently being edited by a content manager. After a content manager finishes editing the object and clicks **Save and Close**, Axon removes the edit lock. You can manually remove an edit lock if Axon does not automatically remove the edit clock after the content manager finishes editing the object.

Permanent Lock

Content managers can assign a permanent lock to an object that they are editing. Permanent locks persist even after the object is closed. This prevents other users from editing the item until the content manager who assigned the lock unlocks the object.

Axon displays the name of the user who has assigned the permanent lock when you hover the cursor over the lock. As an administrator, you can unlock an object that has a permanent lock. Before you remove a permenant lock, you must contact the content manager who placed the lock to ensure that edits made to the object are not lost.

If the **Axon Status** of the user account of an object stakeholder is set to `Deleted` and if that object has an active change request, the SuperAdmin user can unlock the object.

Disable Export of Objects

You can disable export of People objects in Axon. The export of People objects is enabled by default. If you disable the export, users with the WebUser and Admin profiles will not see the export option when they display People objects in the Unison Search grid.

You must be a SuperAdmin user to configure facets for export.

For more information on exporting objects see the *Exporting Search Results* topic in the *Search and View Objects* chapter of the *Axon Data Governance 7.2.1 User Guide*.

1. In the Admin Panel, click **Customize & Configure** > **Application Settings**.
2. Click **Export Objects**.

   The export of People objects is enabled by default.
3. To disable export of People objects, clear the **People** check box.
4. Click **Save**.

Hide Non-Public Objects

You can hide objects with the **Axon Viewing** field set to "Non-Public" for users that are not object stakeholders. Axon hides the non-public objects from Unison search for WebUsers with no stakeholder role on any object of the facet type. A SuperAdmin user can perform this task. The object stakeholders, SuperAdmin, and Admin users can view the non-public objects.

1. In the Admin Panel, click **Customize & Configure** > **Application Settings**.
2. Click **Search Settings**.
3. Select **Hide non-public objects**.
4. Click **Save & Close**.

Configure Periodic Review of Objects

You can configure Axon to periodically initiate the review process to ensure that the objects are updated according to the data governance requirements of your organization.

You must have the Admin or Super Admin profile to perform this task.

In the Admin Panel, you can configure Axon to automatically start a periodic review of objects at the review start date. The action ensures that the stakeholders review and update objects regularly.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **DG Operating Model** category, click **Periodic Review of Objects**.
3. In the Facet list, click the facet for which you want to create a periodic review. You can choose one of the facet from the following list:
   - Data Set
   - Glossary

- Process
- System

4. Select the **Enable Periodic Review** option to configure periodic review for the selected facet.

   Note: The page displays the list of periodic reviews configured for the facet. You can click the name of the periodic review that you want to edit.

5. To delete a periodic review configuration, select the items form the list of periodic reviews.
6. Click **Delete**.
7. Click **Add** to configure a new periodic review of selected facet.

   The page displays the settings for review.

8. Configure the following properties in the **Definition** section:

Property	Description
Name	Name of the review configuration for the selected facet.
Description	Description of the review configuration.

9. Configure the following properties in the **Periodic Review date and Recurrence** section:

Property	Description
Periodic Review Start Date	The date from which the review must start. The default threshold period between change request and periodic review start date is 15 days.
Periodic Review Recurrence	Frequency of the periodic review. Click the edit icon to configure or edit the frequency of the periodic review. You can select the **Yearly**, **Monthly**, or **Do not recure** options. You can choose the corresponding settings for the selected recurrence option. Choose a frequency according to your requirement. For example, select the **Yearly** recurrence. You can configure one of the following options: • Start the recurrence one year after the periodic review start date. • Start the recurrence on first day of January after the periodic review start date. • Start the recurrence on first Monday of January after the periodic review start date.
Generate Change Request	Number of days before the object review date when Axon creates a change request. By default, the change request starts 15 days before the periodic review start date.

10. Configure the following properties in the **Filters** section:

Property	Description
Type	Select the type of the facet. The periodic review is created for the selected types for the facet. If you select more than two facet types, the field shows number of the selected types. Hover the mouse to view the details of the selected types.
Axon Status	Status of the object for which Axon starts a periodic review.
Lifecycle	Stage of the facet in the approval lifecycle.
Top Level Parent	Select the top level parent name for the selected facet. You can choose from 10K parent level filters.

11. Configure the following properties in the **Workflow and Change Request Configurations** section:

Property	Description
Default Change Request Title	Name for the automatic change request .
Default Change Request Summary	Summary for the automatic change request.
Default Change Request System	Default system to create change request. Select one of the following options: • Internal. Use the Axon change request system. • ServiceNow. Use the ServiceNow request system.
Default Workflow	Default workflow for the change request.
Default Change Request Type	Select the default reason that applies to the change request: • Request for Change • Request for Information
Default Change Request Urgency	Select the default urgency that applies to the change request.
Default Change Request Severity	Select the default severity that applies to the change request.
User Email	Active email address of the Axon user that creates the change request before Axon starts a periodic review.

12. Click **Save and Close**.
When you edit periodic review configuration or create another periodic review for the same facet, the filter configurations might overlap with the existing filter configurations. Due to the overlap, the facet objects may go through multiple periodic reviews, starting with the earliest review date. You can choose to save the new filter configuration with the overlap, or modify the filter configuration based on your requirement.

The page displays the list of periodic reviews configured for the facet objects.

Note: When you enable the periodic review for an object, the **Next Review Date** column and filter do not appear in the unison search page. However, the query API contains the next review date of the periodic review of the object.

Configure Threshold for Map Objects

When an Axon user generates a map that contains a large number of objects, the operation might take a long time and affect the system resources. You can configure the maximum number of System, Data Set, and Attribute objects that must appear in maps. If a user generates a map that exceeds the number of configured objects, Axon displays a notification that informs the user to narrow the filter criteria for maps.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.

3. In the **Group** list, select **Environment**.
4. Click **Edit**.
5. Configure the following threshold values for the map objects:

Property	Description
Map System Threshold	Maximum number of System objects in a map. The default value is 300.
Map Dataset Threshold	Maximum number of Data Set Objects in a map. The default value is 1000.
Map Attribute Threshold	Maximum number of Attribute objects in a map. The default value is 5000.

6. In the **System Settings** page, click **Save** to save your settings.
7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Specify Prefix for Object Reference Numbers

Each Axon object is stored in the database with a unique reference number. For each object type, you must specify the prefix for the reference number. This helps you quickly identify an object type when you search for the objects in the database. For example, if you prefix policy reference numbers with "POL-", you can identify the POL-2335446 object as a policy.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Reference Number Settings**.
4. Click the **Edit** button, and then click the **Edit** icon.
5. In the **Facets Reference Number** window, enter the prefixes for the Axon object types.
6. In the **Facets Reference Number** window, click **Save** to save your changes.
7. In the **System Settings** page, click **Save** to save your settings.
8. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

CHAPTER 7

Interface Customization

This chapter includes the following topics:

- Interface Customization Overview, 117
- Display Your Organization Logo, 117
- Configure Regional Settings, 118
- Configure Display Settings, 120
- Customize the Axon Interface Theme, 121

Interface Customization Overview

You can change the look and feel of the Axon Data Governance instance to match the specific needs of your organization.

Display Your Organization Logo

You can replace the logo in the Axon toolbar with the logo of your organization.

The available space for your organization logo is 143×44 pixels. Make sure that the logo image you upload conforms to these dimensions. You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **Change Logo**.
3. In the **Upload Logo** field, click **Browse** to select the image file that contains the logo of your organization.
4. Click **Save** to upload the file to Axon.
5. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Configure Regional Settings

You can configure Axon to display the interface so that it meets specific regional or locale requirements.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Localization**.
4. Click **Edit**, and configure the following properties:

Property	Description
Date	Format to display the date in the Axon interface. Enter a format that conforms to standard PHP convention. To view the supported formats, refer to "Supported Date Formats" on page 118
Date Time	Format to display the date and time in the Axon interface. Enter a format that conforms to standard PHP convention. To view the supported formats, refer to "Supported Date Formats" on page 118

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Supported Date Formats

When you configure regional settings, ensure that you enter the date and date time values in the proper format.

The following table lists the supported date formats with examples:

Format	Example
d-m-y	31-12-92
d-m-Y	31-12-1992
d-M-y	31-DEC-92
d-M-Y	31-DEC-1992
d-y-m	31-92-12
d-y-M	31-92-DEC

Format	Example
d-Y-m	31-1992-12
d-Y-M	31-1992-DEC
m-d-y	12-31-92
m-d-Y	12-31-1992
m-y-d	12-92-31
m-Y-d	12-1992-31
M-d-y	DEC-31-92
M-d-Y	DEC-31-1992
M-y-d	DEC-92-31
M-Y-d	DEC-1992-21
y-m-d	92-12-31
y-M-d	92-DEC-31
y-d-m	92-31-12
y-d-M	92-31-DEC
Y-m-d	1992-12-31
Y-M-d	1992-DEC-31
Y-d-m	1992-31-12
Y-d-M	1992-31-DEC

Use the same date formats when you enter the date in the **Date Time** field. To enter the time, use the `h:i:s` or `H:i:s` format, where `H` indicates the 24 hours time format and `h` indicates the 12 hours time format. You can enter `a` to indicate **AM** and `A` to indicate **PM**.

The following table lists some of the supported date and time formats with examples:

Format	Example
d-M-Y h:i:s	15-MAR-2021 01:35:22
d-m-y h:i:s	15-03-21 13:35:22
d-m-y h:i:s a	15-03-21 01:35:22 AM
d-M-y H:i:s A	15-MAR-21 13:35:22 PM

Configure Display Settings

You can configure the default facets and the order in which they must appear on the Unison search interface for all the users. For example, if users use only Glossary, System, and Data Set facets in an organization, the SuperAdmin user can configure these facets as default active facets. If users change and save layout for default active facets, Axon overrides the default settings.

You must be a SuperAdmin user to configure the default facets for all the users in the organization.

1. In the Admin Panel, click **Customize & Configure** > **Application Settings**.
2. Click **Display Settings**.

   The **Active Facets** section contains the list of facets that are currently visible as default facets in Axon. The **All Facets** section contains the remaining facets that you can choose to add to the default facets.
3. From the **All Facets** section, select a facet that you want to add to the default active facets.

   To choose multiple facets, hold the **Ctrl** key and select facets.
4. Click the **Move Right** arrow.

   The selected facet appears in the **Active Facets** section.
5. To remove facets from the **Active Facets** section, select a facet and click the **Move Left** arrow.
6. To move all the facets to the **Active Facets** section, click the **Move All to Right** arrow. To move all the facets to the **All Facets** section, click the **Move All to Left** arrow.

Customize Session Expiry Time and Password Reset Time Limit

The Axon instance displays a session timeout message based on the time duration that a SuperAdmin user had specified in the Admin panel. Only the user with the SuperAdmin role can modify the session expiry duration based on the requirement. The session expiry time determines the duration that a user can remain idle before the session gets terminated and the user must log in again. You can also specify the password reset time limit. The password reset time limit is the interval in minutes after which users can request password reset links.

You can specify the maximum limit for session expiry time in the `maxSessionExpireTime` variable in the `common.yml` file available in the following path:
`Axon/App/Config/common.yml`

. The maximum value that you can specify in `maxSessionExpireTime` variable is 1440 minutes.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Environment**.

4. Click **Edit**, and configure the following property:

Property	Description
Session Expire Time (in minutes)	Modify the session expiry timing by entering the required time in minutes. For example, if you want the Axon instance to display session timeout message after 5 hours, enter 300 in the **Session Expire Time** field. The default session timing in Axon is 180 minutes. **Note:** The valid session expiry time range is based on the value that you specify in `maxSessionExpireTime` variable in the `common.yml` file. The maximum value that you can specify in the `maxSessionExpireTime` variable is 1440 minutes. For example, if the `maxSessionExpireTime` variable value is 1000 minutes, the valid session expiry time range is between 1 and 1000 minutes. If you enter a value that exceeds the `maxSessionExpireTime` variable value, Axon displays an error message.
Session Alert Expiry (in minutes)	Set the time in minutes when the user gets notified before the session expires. For example, if you set 1 minute, the user gets the session expiration alert a minute before the session expiration time. Default is 2 minutes.
Password Reset Time Limit (in minutes)	Specify the interval in minutes after which a user can request a password reset link. Enter a positive integer value greater than zero. For example, if you set the limit to 10 minutes for a particular user, then the user can request a password reset link only after 10 minutes. Default is 15 minutes.

5. Click **Save**.
6. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

```
sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Customize the Axon Interface Theme

You can change the colors of the Axon interface to reflect the theme and colors of your organization.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **Customize Styles**.
3. In the **Upload Custom CSS** field, enter the stylesheet content for the theme and color patterns of your organization. To see the stylesheet values that you can enter for the main sections of the Axon interface, refer "Interface Elements and their CSS Values" on page 122. To reset the colors of an interface element to the default colors, delete the stylesheet values.
4. Click **Save**.

5. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

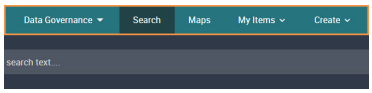
   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

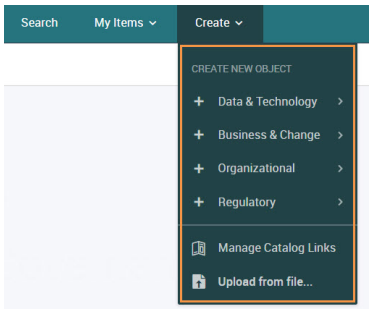
Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

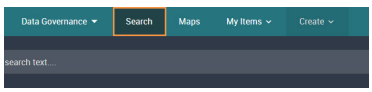
Interface Elements and their CSS Values

The following table describes the customization values for some of the sections of the Axon interface:

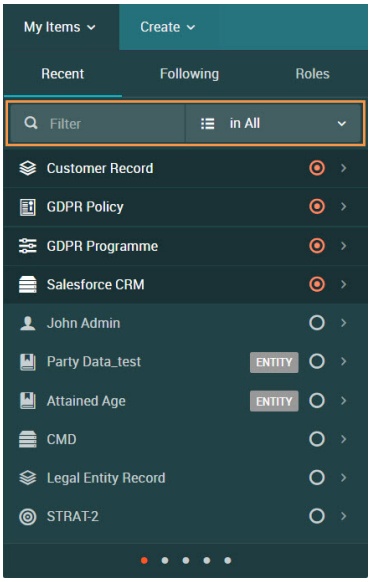
Interface Element	Stylesheet Value

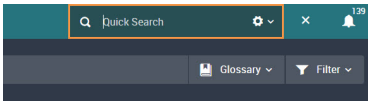
Toolbar—background color and border color
 
	`.navbar-inverse,.navbar-main .navbar-right > .navbar-nav`
	`{`
	`  background-color: <background_color_hex_code>;`
	`  border-color: <border_color_hex_code>;`
	`}`

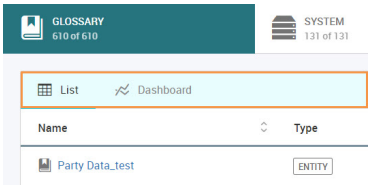
Dropdown lists in the toolbar—background color and border color
 
	`.dropdown-menu[class*=bg-], dropdown-toggle, .navbar-inverse .navbar-nav>.open>a`
	`{`
	`  background-color: <background_color_hex_code> !important;`
	`  border-color: <border_color_hex_code> !important;`
	`}`

Selection in the toolbar—background color and border color
 
	`.navbar-inverse .navbar-nav>.active>a`
	`{`
	`  background-color: <background_color_hex_code> !important;`
	`  border-color: <border_color_hex_code> !important;`
	`}`

Interface Element	Stylesheet Value

Filter input field and filter options in the **My Items** dropdown in the toolbar—background color and border color
 
	`.dropdown-filter>.row>div[class*=col-] .form-control, .dropdown-filter>.row>div[class*=col-] .select2-selection--single, .bg-lighter`
	`{`
	`background-color: <background_color_hex_code> !important;`
	`border-color: <background_color_hex_code> !important;`
	`}`

Quick Search input field—background color and border color
 
	`#quick-search-input, .quicksearch-options`
	`{`
	`background-color: <background_color_hex_code> !important;`
	`border-color: <border_color_hex_code> !important;`
	`}`

Section header for grids—background color
 
	`.panel-white>.panel-heading`
	`{`
	`background-color: <header_color_hex_code>;`
	`}`

CHAPTER 8

Database and Microservice Management

This chapter includes the following topics:

- Database and Microservice Management Overview, 124
- Back Up and Restore Database, 124
- Access Databases on External Nodes, 128
- PostgreSQL Database Service User Passwords, 129
- Working With Services, 131

Database and Microservice Management Overview

As the administrator, you can monitor the health of Axon Data Governance. You can verify if the application services and microservices are running. You can also back up the database. These are tasks that you must do on a regular basis to keep Axon running.

Back Up and Restore Database

Axon automatically backs up the PostgreSQL database according to the predefined schedule. You can also back up the Axon PostgreSQL database manually and restore it to another instance of the same Axon version. If you installed the PostgreSQL database on an external node, you can take a backup of the external database and restore it to another instance.

You might want to back up and restore the Axon database under the following circumstances:

- You want to have the same data for multiple Axon instances.
- You want to restore data that you backed up earlier.
- You want to move data from a test environment to a production environment.
- You want to move data

When Axon backs up the database automatically, the backup scheduler keeps the backups at the following location: `<INSTALLATION_DIR>/axonhome/db-backups/postgres_backups/`

Note: To perform a backup and restore of external databases, such as Microsoft Azure Database for PostgreSQL and Amazon RDS, you need to contact your database administrator.

Back Up Database

To back up the database, run the script file `axon_db_backup.sh` that contains the details of the backup. Perform the following tasks to back up the Axon database at the source location.

1. Go to the `<INSTALLATION_DIR>/scripts` directory and run the following command to back up the database:

   ```
   sh axon_db_backup.sh <BACKUP_TARGET_DIR>
   ```

2. If you have configured custom passwords for PostgreSQL database users during or after Axon installation, create a `.conf` file to store all the custom passwords and copy the file to a location that you can access later.

   Enter passwords in the `.conf` file in the same format as the `password.conf` file at the following location:

   ```
   <INSTALLATION_DIR>/config/password.conf
   ```

   Use the following required format:

   ```
   <configuration parameter>=<encrypted password>
   ```

   For example, enter the encrypted password of the Value List database user in the following format in the `.conf` file:

   ```
   VALUELIST_PWD=<encrypted VALUELIST_Password>
   ```

3. Go to the backup target directory to verify that the database backup file is created. The database backup is an SQL file with the timestamp: `<BACKUP_TARGET_DIR>/<backup_sql_file>`

   For example, your backup file might look like this: `<BACKUP_TARGET_DIR>/postgres_2018_06_24_025744.sql`

Restore Database

You can restore the Axon database to a target location. After restoring the database, restart Axon, reindex the database, and restore the Axon settings. If you have performed any file-level customization in the source instance, ensure that you make the same changes in the target instance.

1. Go to the `<INSTALLATION_DIR>/bin` target location where you have installed Axon and run the following command to shut down the Axon instance:

   ```
   sh shutdown.sh
   ```

2. Go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory, and run the following command to restart the PostgreSQL database service:

   ```
   sh postgresql restart
   ```

3. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   source <INSTALLATION_DIR>/scripts/exportEnv.sh
   ```

4. Run the following command to restore the database from the backup file:

   ```
   <INSTALLATION_DIR>/axonhome/third-party-app/postgresql/bin/psql -U
   <Axon_Administrator> -d postgres -h <INSTALLATION_DIR>/axonhome/third-party-app/
   config/postgres -f <full_path_to_backup_postgresql_file>
   ```

   For example, run the following command:

   ```
   $AXON_HOME/third-party-app/postgresql/bin/psql -U $AXON_USER -d postgres -h
   $AXON_HOME/third-party-app/config/postgres -f <BACKUP_TARGET_DIR>/
   postgres_2018_06_24_025744.sql
   ```

   You can ignore any user or role error message that might be displayed.

5. Reapply your Axon customization settings in the .ymlc or .yml files.
6. If you have not configured custom passwords for PostgreSQL database users, skip this step. If you have configured custom passwords for PostgreSQL database users, perform the following additional steps to restore them:
    1. If you have not configured the password for the Workflow service password, skip this step. If you have configured the password for the Workflow service, run the following command and update the unencrypted password manually:

        ```
        vi <INSTALLATION_DIR>/axonhome/camunda/server/apache-tomcat-9.0.37/conf/
        server.xml
        ```
    2. Ensure that the `.conf` file has all the custom passwords in the same format as the `<INSTALLATION_DIR>/config/password.conf` file in a location that you can access.
    3. To configure the Axon deployment with custom passwords, run the following commands:

        ```
        cd <INSTALLATION_DIR>/bin
        sh changeDBPassword.sh <Full_Path_to_Custom_Passwords_File>
        ```
7. Go to the `<INSTALLATION_DIR>/bin` location and run the following command to start Axon:

    ```
    sh startup.sh
    ```
8. Go to the `<INSTALLATION_DIR>/scripts` directory and run the following script to reindex the database:

    ```
    sh reIndex.sh
    ```
9. Check the Enterprise Data Catalog and Informatica Data Quality settings to make sure that they point to the target Axon instance.
10. Verify that all Axon services are running properly. Log in to Axon as an administrator, go to the **Admin Panel**, and view the **Admin Dashboard**. The **Status of Axon Services** section shows the services that are operational.

Back Up External PostgreSQL Databases

If the objects database is deployed on an external node, Axon automatically backs up the PostgreSQL database according to the predefined schedule. The backup scheduler keeps the backups at the following location: `<external_directory_PostgreSQL_path>/db_backup/postgres_backup`

If Axon is installed in HTTPS protocol, the Axon application communicates with the external database node using SSL authentication.

1. Log in to the external node, and go to the directory where the PostgreSQL database is deployed.
2. Run the following command to back up the database:

    ```
    sh axon_db_backup_external.sh
    ```
3. If you have configured custom passwords for PostgreSQL database users during or after Axon installation on the external database, create a `.conf` file to store all the custom passwords and copy the file to a location that you can access later.

    Enter the password in the `.conf` file in the same format as the `password.conf` file at the following location:

    ```
    <INSTALLATION_DIR>/config/password.conf
    ```

    Use the following required format:

    ```
    <configuration parameter>=<encrypted password>
    ```

    For example, enter the encrypted password of the Unison database user in the following format in the `.conf` file:

    ```
    UNISON_DB_PWD=<encrypted UNISON_DB_Password>
    ```

4. Go to the `<external_directory_PostgreSQL_path>/db_backup` directory to verify that the database backup file is created.

Restore External PostrgreSQL Databases

If the objects database is deployed on an external node, Axon automatically backs up the PostgreSQL database according to the predefined schedule. The backup scheduler keeps the backups at the following location: `<external_directory_PostgreSQL_path>/db_backup/postgres_backup`

1. On the Axon host, go to the `<INSTALLATION_DIR>/bin` destination location where you have installed Axon and run the following command to shut down the Axon instance:

   ```
   sh shutdown.sh
   ```

2. Log in to the external node, and go to the directory where the PostgreSQL database is deployed.
3. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   export LD_LIBRARY_PATH=<external_directory_PostgreSQL_path>/postgresql/
   lib:$LD_LIBRARY_PATH
   ```

4. Go to the `<external_directory_PostgreSQL_path>/postgresql/bin` directory and run the following command to restart the PostgreSQL database service:

   ```
   sh postgresql restart
   ```

5. Run the following command to restore the external database from the backup file:

   ```
   <external_directory_PostgreSQL_path>/postgresql/bin/psql -U <Axon_Administrator> -d
   postgres -h <external_directory_PostgreSQL_path>/postgresql/run/postgres -f
   <full_path_to_backup_postgresql_file>
   ```

   For example, run the following command:

   ```
   <INSTALLATION_DIR>/axonhome/third-party-app/postgresql/bin/psql -U nroot -d postgres
   -h <external_directory_PostgreSQL_path>/postgresql/run/postgres -f
   <BACKUP_DESTINATION_DIR>/postgres_2018_06_24_025744.sql
   ```

   You can ignore any user or role error message that might be displayed.

6. Reapply your Axon customization settings in the `.ymlc` or `.yml` files.
7. If you have not configured custom passwords for PostgreSQL database users on the external database, skip this step. If you have configured custom passwords for PostgreSQL database users on the external database, perform the following additional steps to restore them:
   1. If you have not configured the password for the Workflow service password, skip this step. If you have configured the password for the Workflow service, run the following command and update the unencrypted password manually:

      ```
      vi <INSTALLATION_DIR>/axonhome/camunda/server/apache-tomcat-9.0.37/conf/
      server.xml
      ```

   2. Ensure that the source `.conf` file has all the custom passwords in the same format as the `<INSTALLATION_DIR>/config/password.conf` file in a location that you can access.
   3. To configure the Axon deployment with custom passwords, run the following commands:

      ```
      cd <INSTALLATION_DIR>/bin
      sh changeDBPassword.sh <Full_Path_to_Custom_Passwords_File>
      ```

8. On the Axon host, go to the `<INSTALLATION_DIR>/bin` location and run the following command to start Axon:

   ```
   sh startup.sh
   ```

9. Go to the `<INSTALLATION_DIR>/scripts` directory and run the following script to reindex the database:

   ```
   sh reIndex.sh
   ```

10. Check the Enterprise Data Catalog and Informatica Data Quality settings to make sure that they point to the destination Axon instance.
11. Verify that all Axon services are running properly. Log in to Axon as an administrator, go to the **Admin Panel**, and view the **Admin Dashboard**. The **Status of Axon Services** section shows the services that are operational.

Backup and Restore Admin Activity History

When you migrate Axon from one instance to another, you must manually migrate the admin activity history. The admin activity history is stored in the Elasticsearch database.

1. Run the following commands in the Linux environment to take a snapshot of the admin activity history data:

```
TIMESTMP=`date +%m%d%Y%H%M%S`
SNAPSHOT_NAME="upgrade-snapshot-$TIMESTMP"
. <OLD_INSTALLATION_DIR>/scripts/elasticsearch_backup.sh $SNAPSHOT_NAME
```

2. Copy the snapshot files from the `<OLD_INSTALLATION_DIR>/scripts/axonhome/elasticsearch_backup` directory to the `<NEW_INSTALLATION_DIR>/axonhome/elasticsearch_backup/` directory.
3. Run the following commands to restore the admin activity history to the new Axon instance:

```
. <NEW_INSTALLATION_DIR>/scripts/elasticsearch_restore.sh $SNAPSHOT_NAME
rm -rf <NEW_INSTALLATION_DIR>/axonhome/elasticsearch_backup/*
sh <NEW_INSTALLATION_DIR>/axonhome/third-party-app/scripts/elasticsearch restart true
sh <NEW_INSTALLATION_DIR>/axonhome/third-party-app/scripts/logstash restart true
```

Access Databases on External Nodes

If you have deployed the PostgreSQL database on an external node, you must run a specific command to access the database console.

In the external node, run the following command to access the database console:

```
export LD_LIBRARY_PATH=<external_directory_PostgreSQL_path>/postgresql/
lib:$LD_LIBRARY_PATH
```

Run the following command to log in to the PostgreSQL database:

```
<external_directory_PostgreSQL_path>/postgresql/bin/psql -U <Axon_Administrator> -d
postgres -h <external_directory_PostgreSQL_path>/postgresql/run/postgres
```

If you want to log in as a database user on the external PostgreSQL database, run the following command:

```
<external_directory_PostgreSQL_path>/postgresql/bin/psql -U <DB_User> -d <Database> -h
127.0.0.1
```

PostgreSQL Database Service User Passwords

Your database administrator might modify a database user password according to the security policies of your organization. After your database administrator modifies the password, you must update the password in Axon configuration to connect to the database.

Perform the following task to modify and encrypt a database user password, and update the password in the Axon configuration:

1. Back up the PostgreSQL database.
   See "Back Up and Restore Database" on page 124 for more information.
2. Modify the database user password.
3. Encrypt the modified database user password.
4. Save the encrypted password in a `.conf` file according to your requirement.
5. Update the Axon configuration with the modified user password.

The following table lists the databases for which the database administrator can modify the password:

Service	Database	Database User	Configuration Parameter
Axon Core	axon	axon	DB_PWD
Bulk Upload	axon_bulkupload	bulkupload	BULKUPLOAD_DB_PWD
Unison	unison	axon_unison	UNISON_DB_PWD
Change Request	axon_changerequest	changerequest	CHANGEREQUEST_DB_PWD
Workflow Service	camunda	camunda	CAMUNDA_PWD
Value List	axon_data	axon_data	VALUELIST_PWD
Data Marketplace	marketplace	marketplace	MARKETPLACE_PWD
Document Upload	axon_docupload	docupload	DOCUPLOAD_DB_PWD
Notification	axon_notification	notification	NOTIFICATION_DB_PWD
Segmentation	segmentation	segmentation	SEGMENTATION_PWD

Update PostgreSQL Database Service User Password

You can update the PostgreSQL database service user password in the Axon configuration. You must enter database password to access PostgreSQL database.

1. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   source<INSTALLATION_DIR>/scripts/exportEnv.sh
   ```

2. Run the following command to log in the database for which you want to modify the user password:

   ```
   <INSTALLATION_DIR>/axonhome/third-party-app/postgresql/bin/psql -U <DB_User> -d
   <Database> -h 127.0.0.1
   ```

Enter the existing password to access the PostgreSQL database service. If you do not know the existing password, contact Informatica Global Customer Support for assistance.

For example, to access the Bulk Upload database, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/postgresql/bin/psql -U bulkupload -d
axon_bulkupload -h 127.0.0.1
```

If you have deployed the database on an external node, see "Access Databases on External Nodes" on page 128 to access the database console and log in to the database as an administrator or database user, and then continue to change the password.

3. To modify the PostgreSQL database service password, run the following commands and enter the new password of the database:

```
axon=> \password
Enter new password:
Enter it again:
axon=> \q
```

For example, to modify the password of the Bulk Upload database, run the following command and enter the new password:

```
axon_bulkupload=> \password
Enter new password:
Enter it again:
axon_bulkupload=> \q
```

4. To encrypt the modified password, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php <INSTALLATION_DIR>/axonhome/
axon/bin/console axon:encrypt <password>
```

For example, to encrypt the modified password of the Bulk Upload service, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php/console axon:encrypt
<BULKUPLOAD_DB_Password>
```

5. Enter the encrypted password in the `.conf` file.

Note: Enter the password in the `.conf` file in a specific format. For reference, see the sample `passwordNew.conf` file at the following location:

```
<INSTALLATION_DIR>/config/passwordNew.conf
```

Use the following format to specify the encrypted password:

```
<configuration parameter>=<encrypted password>
```

For example, enter the encrypted password of the Bulk Upload database user in the following format in the `.conf` file:

```
BULKUPLOAD_DB_PWD=<encrypted BULKUPLOAD_DB_Password>
```

6. To update PostgreSQL database service with the modified user password, go to the `<INSTALLATION_DIR>/bin/` directory, and run the following command:

```
sh changeDBPassword.sh <path to your .conf file>
```

To maintain data security, make sure that only authorized users can modify the PostgreSQL database service user passwords.

Note: The Workflow service cannot read an encrypted password. Make sure to manually update the Workflow service database user password in plain text in the following file:

```
<INSTALLATION_DIR>/axonhome/camunda/server/apache-tomcat-9.0.37/conf/server.xml
```

After you update the user password, restart the Workflow service. To stop the running Workflow service and start it again, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh camunda restart
```

Working With Services

Axon uses several application services and microservices to provide specific functionality that enable you to implement data governance. These services are independent modular services that work together to enhance Axon capabilities.

The services and microservices are located in the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment.

Axon uses the following services:

Service	Service Name	Service Script
Authorization	`authorization`	`axon_authorization_microservice_init`
Bulk Upload	`axon_bulk_upload_microservice`	`axon_bulk_upload_microservice_init`
Change Request	`axon_changerequest_microservice`	`axon_changerequest_microservice`
Daemon Tools	`axon_daemontools`	`daemontools`
Data Marketplace	`axon_marketplace_microservice`	`axon_marketplace_microservice_init`
Document Upload	`axon_document_upload_microservice`	`axon_document_upload_microservice`
Email Notification	`jobber`	`jobber`
Event Monitor	`enable_eventmonitor`	`enable_eventmonitor.sh`
Graph Database and Search	`orientdb`	`orientdb`
HTTPD	`httpd`	`httpd`
Log Search	`elasticsearch`	`elasticsearch`
Manage Logs	`Logstash`	`logstash`
Memcached	`memcached`	`memcached`
Messaging	`rabbitmq-server`	`rabbitmq`
Notification	`axon_notification_microservice`	`axon_notification_microservice`
OdbConsumer	`OdbConsumer`	`OdbConsumer`
PostgreSQL	`postgresql`	`postgresql`
Unison Search	`axon_unison_microservice`	`axon_unison_microservice_init`

Service	Service Name	Service Script
Value List	axon_value_list_microservice	axon_valuelist_microservice_init
Workflow	camunda	camunda

Note: After you enable the Event Monitor service, you cannot disable it.

Check the Status of a Service

To check the status of a specific service, go to the <INSTALLATION_DIR>/axonhome/third-party-app/scripts directory in the Linux environment, and run the following command:

```
sh <service_script> status
```

For example, to check the status of the Email Notification service, run the following command:

```
sh jobber status
```

Check the Status of All Services

To check the status of all Axon services, go to the <INSTALLATION_DIR>/axonhome/third-party-app/scripts directory, and run the following command:

```
sh axonStatus
```

Start a Service

To start a service that is shut down, go to the <INSTALLATION_DIR>/axonhome/third-party-app/scripts directory in the Linux environment, and run the following command:

```
sh <service_script> start
```

For example, to start the Document Upload service, run the following command:

```
sh axon_document_upload_microservice start
```

Stop a Service

To stop a running service, go to the <INSTALLATION_DIR>/axonhome/third-party-app/scripts directory in the Linux environment, and run the following command:

```
sh <service_script> stop
```

For example, to stop the Unison Search service, run the following command:

```
sh axon_unison_microservice_init stop
```

Restart a Service

To stop a running service and start it again, go to the <INSTALLATION_DIR>/axonhome/third-party-app/scripts directory in the Linux environment, and run the following command:

```
sh <service_script> restart
```

For example, to restart the Value List service, run the following command:

```
sh axon_value_list_microservice_init restart
```

CHAPTER 9

Monitor Axon

This chapter includes the following topics:

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- Configure Audit History Period, 134
- Log Object History, 134
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- Monitor Data Inconsistency Issues, 139
- View Usage Statistics, 140
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Monitoring Overview

As the administrator, you can manage logs, view scheduled jobs and view usage statistics to monitor user activity in Axon.

View System Information

Administrators can view system information from the Admin Panel. Axon displays static information and real time information. Use the system information to troubleshoot Axon before you contact Informatica Global Customer Support.

Axon displays the following static information in the Admin Panel:

- Axon version
- Axon build
- Axon core
- Build date

Axon displays the following real time data in the Admin Panel:

- Status of the Axon services
- Axon message queue

Axon represents the status of Axon services using colors, and reports the health of the application services. A green-colored dot next to a service means that the service is healthy. A red-colored dot indicates that the service has shut down.

You can also check the status of specific services from the Linux environment. To check the status of a service, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh <service_script> status
```

For example, to check the status of the Email Notification service, run the following command:

```
sh jobber status
```

For the complete list of services used by Axon, refer the *Working with Services* topic in the *Monitoring Axon* chapter of the *Axon Data Governance 7.2.1 Administrator Guide*.

Configure Audit History Period

You can configure the period of audit history that Axon displays for an object. For example, if you set the history to 12 months, Axon displays changes made to the object in the last 12 months. You can typically see the audit history in the **History** tab of an object.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Environment**.
4. Click **Edit**.
5. In the **Default History Duration** field, enter a value in months. The default setting is 6 months.
6. In the **System Settings** page, click **Save** to save the settings.
7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Log Object History

You can configure Axon to track and store the audit history of an object retrieved using a REST API.

You must have the Super Admin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.

3. In the **Group** list, select **Environment**.
4. Click **Edit**.
5. Enable the **Log Object History** option.
6. In the **System Settings** page, click **Save** to save the changes.

Enable or Disable Viewing of Logs

When you installed Axon, you might have chosen not to enable viewing of the Axon logs. You can enable the viewing of logs after installation. Similarly, if you enabled the viewing of logs during Axon installation, you can disable this option.

To enable the viewing of logs, go to the `<INSTALLATION_DIR>/bin` location and run the following command:

```
sh enableLogging.sh
```

When you enable the viewing of logs, users with the SuperAdmin profile can view the Axon logs in the **Logs** section of the Admin Panel.

Note: If you choose to enable viewing of logs, make sure that the Linux system has 28 GB of memory. For more information on the requirements to run Axon, see the *System Requirements* topic in the *Axon Data Governance Installation Guide*.

To disable the viewing of logs, go to the `<INSTALLATION_DIR>/bin` location and run the following command:

```
sh disableLogging.sh
```

When you disable the viewing of logs, the **Logs** option does not appear in the Admin Panel.

View Logs

When you create objects, modify existing objects, or perform tasks that use the application services and microservices. Axon records those actions in log files. You can view user name, user email, and session timeout details in the log files. When you apply onboarding rules to a system or enable automated onboarding from a system, Axon captures system-level audit logs.

You can view the content of the log files and filter the information based on specific criteria. Axon searches all the log files for the criteria that you specify. In the search result, you can see snippets from the log files that match your criteria. You can then export the search results as a CSV file.

Axon Log Levels

Each Axon component records actions for specific log levels. These log levels are set as default in your Axon installation.

The following table lists the Axon components and their default log levels:

Axon Component	Default Log Level
Axon Core	ERROR
Bulk Upload service	INFO
Change Request service	ERROR
Document Upload service	ERROR
Lineage Consumer process	FINEST
Notification service	ERROR
Unison Search service	INFO
Value List service	INFO

Contact your system administrator to modify the log levels for the Axon components.

View System Log File Content

You need to select an Axon component to view the log file content that applies to the component. Perform the following steps to search and display the content of the log files.

You must have the SuperAdmin profile to perform this task.

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, click **Logs**.
3. Click the gear icon to specify the search parameters.
4. Enter the following parameters to filter the search results:

Option	Description
Component	Select the component for which you want to display the content of the log file. You can choose one of the following options: • All components • Axon Core component • Bulk Upload service • Change Request service • Document Upload service • Lineage Consumer process • Notification service • Unison search service • Value List service

Option	Description
Correlation ID	If you know the unique reference number for a specific request made by a service, enter the reference number. For example, if you uploaded a document, and the Document Upload service displays the `SDF21354654564` correlation ID, enter `SDF21354654564` here.
Message	Enter any specific string value within the log files that you want to search.
Date From	Enter the date and time if you want to search for log files that were created after a specific date and time.
Date To	Enter the date and time if you want to search for log files that were created before a specific date and time.
User Name	Enter the user name if you want to display the logs for a specific Axon user.
Log Level	Select the log level for which you want to display the content of the log file. You can choose from the following log levels: • All log levels • ALERT: Messages that require you to monitor an application or operation. You might have to take steps to prevent the application or operation from failing. The messages in this level also include internal server errors. This log level applies only to the Axon Core component. • DEBUG: Messages that are helpful to debug an operation or application that might have failed. • ERROR: Messages that indicate that an operation has failed. • INFO: Messages that indicate the status of an application or operation. • TRACE: Messages that are helpful to trace a specific code or function. • WARN: Messages that indicate that an application or operation is not running as expected. • FATAL: Messages that indicate that an application or operation has shut down. This log level applies only to the Axon Core component. • CRITICAL: Messages that indicate that an application or operation faces a severe error. This log level applies only to the Axon Core component. • CONFIG: Messages for static configurations. This log level applies only to the Lineage Consumer process. • FINE: Messages that provide simple tracing information. This log level applies only to the Lineage Consumer process. • FINEST: Messages that provide detailed tracing information. This log level applies only to the Lineage Consumer process. • SEVERE: Messages that indicate that an application or operations has encountered a serious failure. This log level applies only to the Lineage Consumer process.

Note: The search parameters are optional. You can enter one or more parameters to search within the log files.

5. Click **Search**.

   Axon displays snippets from the log files that match the criteria. You can scroll down the page to fetch and display more records. Axon displays a maximum of 10000 records.

6. Click the gear icon and then select **Export to CSV** to export the search results as a CSV file. The CSV file displays all the columns for the search results. You can export a maximum of 100000 records.

Download Logs

You can download application logs to troubleshoot the errors that occur in Axon.

Application logs store internal errors that occur while you work with the Axon interface. For example, you might not be able to edit an object even if you have permissions.

A user with the Admin or SuperAdmin profile can view the application logs. To download application logs, click **Download Logs** from the **Operational Management** menu on the Admin Panel page. You can open or save the `application.log` file to view the logs.

View Admin Activity Logs

Axon generates activity logs for an Admin or SuperAdmin user and displays the logs on the **Admin Activity Logs** page. The activities include all the actions that an Admin or SuperAdmin user performs on the Admin Panel. You can view the logs when an Admin or SuperAdmin user creates, edits, and deleted data onboarding rules from the Admin Panel. You can also view the logs when a SuperAdmin user deletes a single object from the object page or bulk deletes objects from the Unison search. You must be an Admin or SuperAdmin user to view the activity logs. An Admin user can view all the logs only for the changes that an Admin user performs. An SuperAdmin user can view all the logs for the changes that Admin and SuperAdmin users perform.

The activity logs display the following information:

Id

The serial number for the activity log.

Setting

Settings that were changed by the Admin or SuperAdmin user. Examples include Role Permissions, Default Workflows, and Dropdown Configuration.

Component

The component within a setting that the Admin or SuperAdmin user has changed. For example, in the System Settings setting, Dashboard is a component.

User

The name and email address of the Admin or SuperAdmin user who made the changes.

Change Type

The type of change that an Admin or SuperAdmin user has made. The values include Create, Update, Delete, and Other Actions. The Create type includes actions, such as creating roles and adding a default workflow. The Update type includes actions, such as updating display settings and changing default change request settings. The Delete type includes actions, such as deleting objects and role permissions. The Other Actions type includes downloading logs, running glossary rules engine, and running the LDAP synchronization.

Timestamp

The date and time at which an Admin or SuperAdmin user made the changes. The timestamp is displayed in the Coordinated Universal Time (UTC) format.

Details

The details of the change made by an Admin or SuperAdmin user. For example, if an Admin or SuperAdmin user changes a user permission from "View" to "Edit," you can view the previous and changed values when you click the **Details** link.

To view a specific activity log, you can search for a keyword or apply filters in the search area. For example, if you want to see the updates made for the dropdown configuration, choose **Dropdown Configuration** from the list of configuration, select the **Update** change type, and click **Search**.

You can export the logs to a CSV format. For example, if you want to view a consolidated report of all activities within a time period, choose the **From Date** and **To Date** values, and click **Search**. After the logs appear, click the **Settings** icon. Click **Export to CSV** to export and download the logs to a CSV file. In the CSV file, you cannot view the previous values for workflow diagrams, styles, and logos.

Monitor Data Inconsistency Issues

Axon notifies you by email if the data in the database is inconsistent with the data that you might have entered while creating, modifying or deleting an asset.

When you create, modify or delete an asset, the record is created in the objects database. The objects database is indexed, and the same record is created in the graph database so that the object appears in Unison Search. However, the database index might not happen properly and the data may not get recorded correctly to the graph database. Here we have an inconsistency between the data you entered and the data as it appears in Unison Search.

When a data inconsistency occurs, Axon sends a notification email.

To see the objects that are inconsistent, open the log file at the following location: `<INSTALLATION_DIR>/axonhome/axon/app/Resources/etl/logs/indexing_failure.log`

Email Addresses for Notification

To specify the addresses where Axon should send the notification email, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
3. In the **Group** list, select **Environment**.
4. Click **Edit**.
5. In the **Email ID for Search Index Failures** field, enter the email addresses separated by commas.
6. Click **Save**.
7. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

   When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Fix the Data Inconsistency

Axon automatically fixes the inconsistency by running a partial reindex of the objects database. This partial indexing happens at 12:00 A.M.

You can choose to run a partial reindex job manually to fix the inconsistency. For more information on partial reindex of the database, see "Troubleshooting Search" on page 159.

View Usage Statistics

You can access reports on Axon usage statistics, such as resolved change requests, top change request creators, and user login activity. For example, a data governance officer can monitor Axon performance by viewing the number of users logged in to Axon and the number of change requests that are resolved every day. You can use external visualization applications, such as Tableau, Power BI, and ClickView, to view the usage reports.

After you install or upgrade Axon, you need to create materialized views to access the Axon usage statistics. After you enable materialized views, a scheduled job is triggered every night to collect the updated data.

You can view the following types of usage statistics:

- Number of resolved change requests and the duration of resolved change requests.
- Number of change requests created and edited for an object over a period.
- Status and count of change requests.
- Number of created, edited, or deleted objects for a facet over a period of time.
- Creators of top 10 change requests.
- Number of created and deleted users.
- Number of automatic and manual change requests.
- Login activity of users.

If you enable the Event Monitor service, you can also view the following types of usage statistics:

- Number of searches in Unison and query API for a facet over a period of time.
- Frequently used keywords to search for objects in Unison over a period of time.
- Most searched for objects in Unison over a period of time.
- Users who conducted the most searches in Unison over a period of time.
- Number of quick searches over a period of time.
- Most viewed object in a facet over a period of time.
- Number of change requests created by a user over a period of time.

Note:

- The date values are in the UTC time zone.

- The duration over which the above mentioned usage statistics are captured for the last 180 days. After 180 days, the data is stored in a CSV file in the following directory:

  `<INSTALLATION_DIR>/axonhome/db-archives`

  Axon uses a scheduled job created by the Jobber utility to archive this data. This job is executed every week at 6 AM UTC on Saturday.

  If you configure the PostgreSQL database on external database instances such as Amazon RDS and Microsoft Azure Database, the AWS administrator or the Azure administrator is responsible for creating a backup for the data that exceeds 180 days.

Create Materialized Views for Usage Statistics

To view usage statistics, you must create materialized views of the database. Statsuser is a predefined user that can access materialized views. To create the statsuser profile, you must provide a password. With the statsuser credentials, you can access the usage statistics reports. After you install Axon, you can run the `matviewenable.sh` script to create materialized views. A user with the SuperAdmin or Admin profile can perform these tasks.

1. In the Linux environment, navigate to the following directory:

   `<INSTALLATION_DIR>/scripts`
2. To create materialized views, run the `sh matviewenable.sh` command.
3. Enter `1`.

   The materialized views are created.
4. To create the statsuser profile, enter `1` and type a password. Confirm the password.

   The statsuser is created.
5. Run the `sh matviewenable.sh` command again. Choose one of the following options:

Option	Description
Refresh	Refresh the instance of materialized views.
Drop and Recreate	Discard the materialized views and create a fresh instance of materialized views again.

6. If you want to create the statuser profile before you enable materialized views or if you did not choose to create the statsuser earlier, run the `matview_user_creation.sh` command and enter the password to create the statsuser.
7. To disable materialized views, run the `sh matviewdisable.sh` command.

Resolved Change Requests

You can view the average number of change requests that are resolved every day in the `cr_resolution_stats` table.

The following table lists the fields and description of the `cr_resolution_stats` table:

Field	Datatype	Description
rsltn_date	Date	Date on which the change request is resolved.
avg_rsltn_dur	String	Average resolution time for a change request.
rsltn_cnt	Number	Number of change requests resolved per day.
min_rsltn_dur	String	Minimum resolution time for a change request.
max_rsltn_dur	String	Maximum resolution time for a change request.

The following table shows a sample data that you can view in the `cr_resolution_stats` table:

rsltn_date	rsltn_cnt	avg_rsltn_dur	min_rsltn_dur	max_rsltn_dur
2/25/2021	1	00:00:12.259	00:00:12.259	00:00:12.259
2/24/2021	1	00:19:58.916	00:19:58.916	00:19:58.916
2/23/2021	2	00:08:45.223	00:07:31.142	00:09:59.304
2/22/2021	5	00:31:48.4034	00:02:48.556	00:46:33.246
2/21/2021	1	00:10:22.066	00:10:22.066	00:10:22.066

Created and Edited Change Requests for Facets

You can view the daily created and edited change requests for each facet in the `cr_facet_stats` table.

The following table lists the fields and description of the `cr_facet_stats` table:

Field	Datatype	Description
date	Date	Date on which the change requests that are created or edited.
obj_cnt	Number	Number of objects that contain change requests.
cr_create_cnt	Number	Number of automatic change requests that are created when you create an object.
cr_edit_cnt	Number	Number of automatic change requests that are created when you edit an object.

Field	Datatype	Description
cr_mnl_cnt	Number	Number of manual change requests.
facet_nm	String	Name of the facet.

The following table shows a sample data that you can view in the `cr_facet_stats` table:

date	obj_cnt	cr_create_cnt	cr_edit_cnt	cr_mnl_cnt	facet_nm
2/24/2021	1	2	1	2	policy
2/23/2021	1	0	2	1	project
2/23/2021	1	3	0	1	process
2/11/2021	1	1	1	1	dataset

Change Request Status

You can view the count of change request statuses in the `cr_status_stats` table.

The following table lists the fields and description of the `cr_status_stats` table:

Field	Datatype	Description
date	Date	Date on which the change requests are created, updated, or deleted.
create_cnt	Number	Number of created change requests.
rsltn_cnt	Number	Number of resolved change requests.
in_prgrs_cnt	Number	Number of change requests that are in progress.
rjctd_cnt	Number	Number of rejected change requests.

The following table shows a sample data that you can view in the `cr_status_stats` table:

date	create_cnt	rsltn_cnt	in_prgrs_cnt	rjctd_cnt
1/8/2021	2	1	1	0
1/9/2021	3	0	3	1
1/10/2021	6	2	4	3
1/11/2021	23	1	24	2
1/12/2021	1	0	16	9

Facet Objects

You can view the number of created, updated, and deleted objects for a facet in the `obj_facet_stats` table.

The following table lists the fields and description of the `obj_facet_stats` table:

Field	Datatype	Description
date	Date	Date on which the objects are created, updated, or deleted.
create_cnt	Number	Number of created objects for a facet.
upd_cnt	Number	Number of updated objects for a facet. This count is updated only when the **Last Updated** value changes in the **Summary** tab of an object.
del_cnt	Number	Number of deleted objects for a facet. If a delete operation does not apply to a facet or an object is permanently deleted from a database, the "NaN" value appears.
facet_nm	String	Name of the facet.

The following table shows a sample data that you can view in the `obj_facet_stats` table:

date	create_cnt	upd_cnt	del_cnt	facet_nm
2/18/2021	0	1	0	people
2/24/2021	2	1	NaN	changerequest
2/24/2021	1	0	NaN	role
2/20/2021	2	0	0	dataset
2/19/2021	3	1	1	glossary

Top 10 Change Request Creators

You can view the top 10 users that created the maximum number of change requests in the `cr_top_creators` table. You can see a maximum of 100 rows in the `cr_top_creators` table depending on the number of users that created maximum change requests. For example, consider that 10 users created 10 change requests each and five users created five change requests each. The users that created 10 change requests are

ranked as "1", and the users that created five change requests are ranked "2". You can see that 15 rows are populated in the `cr_top_creators` table.

The following table lists the fields and description of the `cr_top_creators` table:

Field	Datatype	Description
name	String	Name of the user that created the change requests.
email	String	Email address of the user that created the change requests.
role	String	Profile of the user that created the change requests. Axon has WebUser, Admin, and SuperAdmin user profiles.
cr_cnt	Number	Number of change requests that the user created.
rank	Number	Rank of the change requests.

The following table shows a sample data that you can view in the `cr_top_creators` table:

name	email	role	cr_cnt	rank
WebUser1	wu1@infa.com	WebUser	1	3
John Admin	jadmin@informatica.com	SuperAdmin	3	2
InfAdmin	infadmin@infa.com	Admin	4	1

Created and Deleted Users

You can view the number of created and deleted users in the `usr_sync_stats` table.

The following table lists the fields and description of the `usr_sync_stats` table:

Field	Datatype	Description
date	Date	Date on which the users are created or deleted.
usr_create_cnt	Number	Number of users that are created.
usr_del_cnt	Number	Number of users that are deleted.

The following table shows a sample data that you can view in the `usr_sync_stats` table:

date	usr_create_cnt	usr_del_cnt
1/11/2021	0	1
1/10/2021	2	0

date	usr_create_cnt	usr_del_cnt
1/9/2021	1	0
1/8/2021	2	1

Manual and Automatic Change Requests

You can view the number of manual and automatic change requests in the `cr_initiator_stats` table.

The following table lists the fields and description of the `cr_initiator_stats` table:

Field	Datatype	Description
date	Date	Date on which the change requests are created.
usr_initiated_cnt	Number	Number of manual change requests.
sys_initiated_cnt	Number	Number of automatic change requests.

The following table shows a sample data that you can view in the `cr_initiator_stats` table:

date	usr_initiated_cnt	sys_initiated_cnt
2/26/2021	1	0
2/25/2021	0	10
2/24/2021	2	5
2/23/2021	3	0

User Login Activity

You can view the login activity for Axon users in the `usr_login_stats` table.

The following table lists the fields and description of the `usr_login_stats` table:

Field	Datatype	Description
date	Date	Latest date on which the user logged in.
login_cnt	Number	Number of times the user logged in.
name	String	Name of the user that logged in.
role	String	Profile of the user that logged in. Axon has WebUser, Admin, and SuperAdmin user profiles.
email	String	Email address of the user.

The following table shows a sample data that you can view in the `usr_login_stats` table:

date	login_cnt	name	email	role
3/4/2021	1	John Admin	admin@informatica.com	SuperAdmin
3/2/2021	53	John Admin	admin@informatica.com	SuperAdmin
3/2/2021	2	Perf First Name 1 Perf Last Name 1	p1@informatica.com	WebUser
3/1/2021	60	John Admin	admin@informatica.com	SuperAdmin

Note: If you upgrade to Axon 7.3.1, the `usr_login_stats` table does not display any data.

Number of Searches in Unison For a Facet

In the `searches_for_<facet name>_in_180d` and the `guest_searches_for_<facet name>_in_180d` tables, you can view the number of searches in Unison for a facet. Specify the name of facet the in the `<facet name>` field to retrieve data that is relevant to the facet. For example, to retrieve the number of searches in Unison for the System facet, see the `searches_for_system_in_180d` and the `guest_searches_for_system_in_180d` tables.

The following table lists the fields and descriptions of the `searches_for_<facet name>_in_180d` table:

Field	Datatype	Description
searchterm	String	The search term used to initiate the search in Unison.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which a search was conducted.

The following table lists the fields and descriptions of the `guest_ searches_for_<facet name>_in_180d` table:

Field	Datatype	Description
searchterm	String	The search term used to initiate the search in Unison.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which a search was conducted.

Note:

- When performing a search in Unison, if the page is refreshed before the search is completed, Axon does not store the search term used to initiate the search. It stores only the latest search term for which the search was completed.

- If you conduct a search in non-default facets as a guest user, and perform a successive search without clearing the previous search after you log in, Axon stores the search performed after you log in as two identical entries.
- When you search for an object in Unison, if you apply filters, Axon stores only the latest search term that was used to perform a search that did not involve the use of filters.
- If you use a search term that is enclosed in double quotation marks, Axon stores it as a separate search term instead of updating the search term entry that does not contain double quotation marks.
- The **Regulatory** facet and the **Geography** facet are not available to guest users. Therefore you cannot create materialized views for data pertaining to guest users for these facets.

Frequently Used Keywords to Search for Objects in Unison

In the `searches_in_180d` and the `guest_searches_in_180d` table, you can view the frequently used keywords to search for objects in Unison.

The following table lists the fields and descriptions of the `searches_in_180d` and `guest_searches_in_180d` table:

Field	Datatype	Description
searchterm	String	The search term used to initiate the search in Unison.
date	Date	Date on which search is made.
fullname	Varchar	Fullname of the user.
role	Varchar	Role of the user.

Note:

- When performing a search in Unison, if the page is refreshed before the search is completed, Axon does not store the search term used to initiate the search. It stores only the latest search term for which the search was completed.
- If you conduct a search in non-default facets as a guest user, and perform a successive search without clearing the previous search after you log in, Axon stores the search performed after you log in as two identical entries.
- When you search for an object in Unison, if you apply filters, Axon stores only the latest search term that was used to perform a search that did not involve the use of filters.
- If you use a search term that is enclosed in double quotation marks, Axon stores it as a separate search term instead of updating the search term entry that does not contain double quotation marks.
- The **Regulatory** facet and the **Geography** facet are not available to guest users. Therefore you cannot create materialized views for data pertaining to guest users for these facets.

Most Searched For Objects in Unison

In the `top_25_unison_searches_in_180d` and the `guest_top_25_unison_searches_in_180d` tables, you can view the most searched for objects in Unison.

The following table lists the fields and descriptions of the `top_25_unison_searches_in_180d` table:

Field	Datatype	Description
facet_name	String	The name of the facet.
searchterm	String	The search term used to initiate the search in Unison.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which a search was conducted.

The following table lists the fields and descriptions of the `guest_top_25_unison_searches_in_180d` table:

Field	Datatype	Description
facet_name	String	The name of the facet.
searchterm	String	The search term used to initiate the search in Unison.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which a search was conducted.

Note:

- When performing a search in Unison, if the page is refreshed before the search is completed, Axon does not store the search term used to initiate the search. It stores only the latest search term for which the search was completed.
- If you conduct a search in non-default facets as a guest user, and perform a successive search without clearing the previous search after you log in, Axon stores the search performed after you log in as two identical entries.
- When you search for an object in Unison, if you apply filters, Axon stores only the latest search term that was used to perform a search that did not involve the use of filters.
- If you use a search term that is enclosed in double quotation marks, Axon stores it as a separate search term instead of updating the search term entry that does not contain double quotation marks.
- The **Regulatory** facet and the **Geography** facet are not available to guest users. Therefore you cannot create materialized views for data pertaining to guest users for these facets.

Users Who Conducted the Most Searches

In the `users_who_conducted_the_most_searches_in_180d` table, you can view the Axon users who conducted the most searches.

The following table lists the fields and descriptions of the `users_who_conducted_the_most_searches_in_180d` table:

Field	Datatype	Description
username	String	The name of the Axon user.
hit_count	Number	The number of searches conducted with a search term.

Number of Quick Searches

In the `quick_searches_in_180d` and the `guest_quick_searches_in_180d` tables, you can view the number of quick searches. This also includes searches that you initiate using the Axon Quick Look browser extension.

The following table lists the fields and descriptions of the `quick_searches_in_180d` table:

Field	Datatype	Description
searchterm	String	The search term used to initiate the quick search.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which a search was conducted.

The following table lists the fields and descriptions of the `guest_quick_searches_in_180d` table:

Field	Datatype	Description
searchterm	String	The search term used to initiate the quick search.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which a search was conducted.

Note: These do not include searches conducted using the Axon Quick Look desktop application.

Most Viewed Object in a Facet

In the `most_viewed_objects_in_<facet name>_in_180d` and the `guest_ most_viewed_objects_in_<facet name>_in_180d` tables, you can view the most viewed object in a facet. Specify the name of the facet in the `<facet name>` field to retrieve data that is relevant to the facet. For example, to retrieve the most viewed

object in the System facet, see the `most_viewed_objects_in_<facet name>_in_180d` and the `guest_ most_viewed_objects_in_<facet name>_in_180d` tables.

The following table lists the fields and descriptions of the `most_viewed_objects_in_<facet name>_in_180d` table:

Field	Datatype	Description
facet_name	String	The name of the facet.
object_name	String	The name of the facet object that is accessed.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which the object was viewed.

The following table lists the fields and descriptions of the `guest_ most_viewed_objects_in_<facet name>_in_180d` table:

Field	Datatype	Description
facet_name	String	The name of the facet.
object_name	String	The name of the facet object that is accessed.
hit_count	Number	The number of searches conducted with a search term.
date	Date	The latest date on which the object was viewed.

Note: The **Regulatory** facet and the **Geography** facet are not available to guest users. Therefore you cannot create materialized views for data pertaining to guest users for these facets.

Number of Change Requests Created by a User

In the `change_requests_created_by_user_in_180d` table, you can view the number of change requests created by Axon users.

The following table lists the fields and descriptions of the `change_requests_created_by_user_in_180d` table:

Field	Datatype	Description
date	Date	Latest date on which the change request was created.
hit_count	Number	Number of change requests created by the user.
name	String	Name of the Axon user.

Monitor Scheduled Jobs

You can monitor the scheduled jobs created by the Jobber utility. You can view the details of scheduled jobs and the error messages for failed jobs.

See all scheduled jobs

To see the list of all jobs scheduled by Jobber, run the following command:

```
<INSTALLATION_DIR>/axonhome/third-party-app/jobber/bin/jobber -
sock=<INSTALLATION_DIR>/axonhome/third-party-app/config/jobber/jobber_daemon.sock -
user=<Axon_Administrator_non-root_user> -jobfile=<INSTALLATION_DIR>/axonhome/
axon_scheduler/axon.jobs list
```

See email notification logs

The `<user_name>` file contains the error messages, if any, for the email notifications. To view the error messages, run the following command:

```
vi /var/spool/mail/<user_name>
```

CHAPTER 10

Troubleshooting

This chapter includes the following topics:

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Troubleshooting Overview

If you encounter errors while using Axon Data Governance, try the following solutions before you contact Informatica Global Customer Support.

Axon Does Not Load in the Browser

You can access Axon by typing the fully qualified host name in the address bar of a web browser. In the address field, type the fully qualified host name for the Axon login page in the following format: `http(s)://<host_name>:<port_number>/`

If Axon does not load in the browser, perform the following tasks.

Verify the Firewall Settings

You must enable the firewall of the machine where you installed Axon to allow requests for the Axon port.

Verify that the HTTPD Service is Running

To verify that the HTTPD service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh httpd status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Unison Search Service is Running

To verify that the Unison Search service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh axon_unison_microservice_init status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify Log Files

Verify the following log files:

- `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/logs/<hostname>-error_log` for HTTP authentication
- `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/logs/<hostanme>-ssl_error_log` for HTTPS authentication
- `<INSTALLATION_DIR>/axonhome/axon/var/logs/prod.log`

Verify Your Axon License Validity

Check if your Axon license has expired. If your license expires, users cannot log in to Axon. Perform the following tasks if your Axon license expires:

1. Contact Informatica Global Customer Support to obtain a new license key.
2. In the Linux environment, replace the expired license key file with the new license key file in the following directory:

   `<INSTALLATION_DIR>/axonhome/axon/`

   Note: Make sure that the new license key file and expired license key file share the same name.
3. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

   ```
   sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
   ```

When you run the `paramsync` script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Verify Axon Settings and Parameters

Check if the Axon settings and **Admin Panel** parameters are correct. You can modify some of the **Admin Panel** settings from the Linux environment. You can run specific commands to display the current parameter values, reset the parameters to their default values, or enter new parameter values.

For more information, refer to "Modify Admin Panel Settings from the Linux Environment" on page 75.

Troubleshooting Login Issues

Cannot Log In to Axon

If your Axon license expires, users cannot log in to Axon. Logged in users automatically log out of Axon at the time of license expiry.

Perform the following tasks if your Axon license expires:

Verify that the PostgreSQL Database Service is Running

To verify that the PostgreSQL database service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh postgresql status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Note: Use the same commands to verify that the database services is running for the PostgreSQL database deployed on a different node and external nodes, such Amazon RDS and Microsoft Azure Database for PostgreSQL.

Replace Your Axon License Key

To replace your Axon license key, perform the following steps:

1. Contact Informatica Global Customer Support to obtain a new license key.
2. In the Linux environment, replace the expired license key file with the new license key file in the following directory:

   `<INSTALLATION_DIR>/axonhome/axon/`

   Note: Make sure that the new license key file and expired license key file share the same name.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Cannot Log In to Axon Using SSO

When you enable Single Sign-On (SSO), users can log in to Axon using login credentials that are common to other applications in your organization. Axon uses SSO to authenticate users based on the Identity Provider (IDP) credentials.

If users cannot log in to Axon using SSO, perform the following tasks:

Verify the SSO Settings

Make sure that you have configured the SSO settings correctly. For SSO configurations, refer to "Configure Single Sign-On" on page 53.

1. In the IDP, make sure that Axon is configured as a SAML 2.0 application.
2. In the IDP, make sure that Axon details and the properties for the SAML attributes are entered correctly.
3. If you cannot access the SSO settings in Axon, disable SSO from the Linux environment.
   a. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

      ```
      source <INSTALLATION_DIR>/scripts/exportEnv.sh
      ```

   b. Go to the `<INSTALLATION_DIR>/axonhome/axon` directory, and run the following command to disable SSO:

      ```
      <INSTALLATION_DIR>/axonhome/third-party-app/php/bin/php bin/console
      informatica:sso-disable
      ```

4. In Axon, make sure that the values of the SAML metadata are entered correctly.

Verify the SSO Settings in the SAML Response File

After Axon users enter their SSO credentials, the IDP sends a SAML response to Axon. Open the response file and verify that the following properties display the correct values for the Axon users:

Property	Value
orgunit	Organization unit of the Axon user
statusID	Status of the Axon user in the IDP database. The value must be `1`.
firstName	First name of the Axon user
lastName	Last name of the Axon user
orgunitDescription	Organization unit description for the Axon user
orgunitTitle	Title of the Axon user in the organization unit
email	Email address of the Axon user
profile	Profile of the user in Axon. The value must be `4`, `5` or `6`.

For example, the values for the Axon user John Wilson might look like this:

```
<saml2:AttributeStatement
    xmlns:saml2="urn:oasis:names:tc:SAML:2.0:assertion">
    <saml2:Attribute
        Name="orgunit"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">FIN
        </saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute
        Name="status"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">1
        </saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute
        Name="firstName"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">John
        </saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute
        Name="lastName"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">Wilson
        </saml2:AttributeValue>
    </saml2:Attribute>

    <saml2:Attribute
        Name="orgunitDescription"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">Finance
    </saml2:Attribute>
    <saml2:Attribute
        Name="orgunitTitle"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">Accountant
    </saml2:Attribute>
    <saml2:Attribute
        Name="email"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:basic">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">jwilson@yourorganization.com
        </saml2:AttributeValue>
    </saml2:Attribute>
    <saml2:Attribute
        Name="profile"
        NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified">
        <saml2:AttributeValue
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:type="xs:string">6
        </saml2:AttributeValue>
    </saml2:Attribute>
</saml2:AttributeStatement>
```

Verify that the X.509 certificate values are correctly retrieved from the IDP. For example, the X.509 certificate value might look like this:



```
<ds:X509Data>
<ds:X509Certificate>MIIDpDCCAoygAwIBAgIGAVyqDY5AMA0GCSqGSIb3DQEBCwUAMIGSMQswCQYDVQQGEwJVUzETMBEG
A1UECAwKQ2FsaWZvcm5pYTEWMBQGA1UEBwwNU2FuIEZyYW5jaXNjbzENMAsGA1UECgwET2t0YTEU
MBIGA1UECwwLU1NPUHJvdmlkZXIxEzARBgNVBAMMCmRldi01OTEzMzExHDAaBgkqhkiG9w0BCQEW
DWluZm9Ab2t0YS5jb20wHhcNMTcwNjE1MDQzOTM0WhcNMjcwNjE1MDQ0MDMzWjCBkjELMAkGA1UE
BhMCVVMxEzARBgNVBAgMCkNhbGlmb3JuaWExFjAUBgNVBAcMDVNhbiBGcmFuY2lzY28xDTALBgNV
BAoMBE9rdGExFDASBgNVBAsMC1NTT1Byb3ZpZGVyMRMwEQYDVQQDDApkZXYtNTkxMzMxMRwwGgYJ
KoZIhvcNAQkBFg1pbmZvQG9rdGEuY29tMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA
p6W0v5qQRU9yFEdafg70eyQiiWU6orQlAeMluBJw+HRyFXk43b9J2A7CXVx8l+goM+bDxtQ7A6eu
htBlMc0Yjsyj/0JB6xUohr0SqD0sxhEAnfYmwu8cOvDASucZQC8H6+FKPBWoAHwMJb/qj6HuB5eU
N8qa3mHCX5przBKUVaiwT2nPoTLHtj7zUUkdyHOr4UWVTxNHxMIwrFx5sT7pC5Dml6C3mw1SVFED
jNteF/V1QHY7cxEW+qGnOdARD1FhuHk/BGOSJtNF1CMz9/v60bim+rk/lq8fzJXO5Tj/1SnD1jS2
xp0O0wXU5EBKRS5G0XhRUav3sKV0mXXXRettpwIDAQABMA0GCSqGSIb3DQEBCwUAA4IBAQCS/GJB
0JBiHR+rGLmctWMFFAxSscVTwl+kFcm+3Na9a8a0cofvdyKAVbeMANdqQ9hPpKo35jwSNkeFouUc
GC443XZ8T+N3BkUjr6tvuDaIN2IDbY0MyMrnGEAgjEEc/C8sLv6aaTdNg0ijievZXJpI/Ffmj6XS
gbphRf7L4ZBGcooTfKwbH3IH0grqSLaFfbmUZ4tr6qT1NwHr3vNm8WUTnBxIBYb0ZyG0CYBcfUf1
yO7+Lzx75nP4vi+omOGN2K4s6dKG5LnU7mkBdx/Wry2lbvq1hXNRB1jMW0r+xuDiu2WwaIrpimfD
Coqevi4fVNDe9uZaLZiuo/3vxC52irCV</ds:X509Certificate>
</ds:X509Data>
```

If you have configured the properties in the IDP correctly, the values in the SAML response file must match the values you enter in the IDP.

Cannot Open, Create, or Save Objects

Depending on the user profile, Axon users can view existing Axon objects, create new objects, or save changes to existing objects.

You might encounter the following issues when you work with Axon objects:

- You cannot create new Axon objects.
- You cannot open existing Axon objects.
- You cannot save new Axon objects.

Perform the following task to troubleshoot this issue:

Verify that the PostgreSQL Database Service is Running

To verify that the PostgreSQL database service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh postgresql status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Check the Status of Axon Services and Available Memory

Go to the Admin Panel to view the status of Axon services. Axon represents the status of Axon services using colors and reports the health of the application services. A green-colored dot next to a service means that the service is healthy. A red-colored dot indicates that the service has shut down.

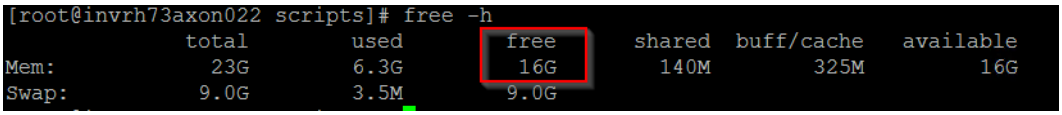
You can also check the status of Axon services from the Linux environment. Go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory, and run the following command:

```
sh axonStatus
```

If all the services are shut down, run the following command to see the available memory in the Linux system:

```
free h
```

The value in the `Mem` row and `free` column indicates the available memory in the system.



If the system has less than 16 GB memory available, contact the IT department of your organization to clear the system memory.

After you verify that the Linux system has 16 GB or more available memory, go to the `<INSTALLATION_DIR>/bin` location and run the following command to start Axon:

```
startup.sh
```

Troubleshooting Search

When you use the Unison Search feature, Axon searches for the object that you select and displays all linked objects in all the available facets. You can also use the Quick Search feature to search for a particular object across all facets.

You might encounter the following issues with Unison Search or Quick Search:

- Objects that exist in Axon do not appear in Quick Search results.
- Items that are newly created in Axon do not appear in Quick Search or Unison Search results.
- Axon displays a message that the Unison Search service is not available.

Perform the following tasks to fix search issues:

Verify that the Graph Database Service is Running

To verify that the Graph Database service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh orientdb status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

If the Axon maps are displayed incorrectly after you restart the Graph Database service, perform the following steps in the Linux environment to reindex the database:

1. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   source <INSTALLATION_DIR>/scripts/exportEnv.sh
   ```

2. Go to the `<INSTALLATION_DIR>/scripts` directory, and run the following script to reindex the database:

   ```
   sh reIndex.sh
   ```

Reindex the Search Database with Full Index

Reindex the search database from the Linux environment.

1. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   source <INSTALLATION_DIR>/scripts/exportEnv.sh
   ```

2. Go to the `<INSTALLATION_DIR>/scripts` directory, and run the following script to reindex the search database:

   ```
   sh reIndex.sh
   ```

3. In case of failure, the index job stops with an error and the Unison microservices shuts down. You can then choose to resume from the point of failure or restart the reindex with the following command: `sh reIndex.sh`
4. Choose to run the reindex with one of the following options:
   1. Resume the previous unsuccessful index operation.
   2. Restart the index operation.
5. You can review the logs in the following directory:
   `<INSTALLATION_DIR>/axonhome/axon/app/Resources/etl/logs/reIndex.log`

Reindex the Search Database with Partial Index

Use the partial index to index the search database from the specified date.

1. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   source <INSTALLATION_DIR>/scripts/exportEnv.sh
   ```

2. Go to the `<INSTALLATION_DIR>/scripts` directory, and run the following script to run the partial index of the search database:

   ```
   sh partialIndex.sh
   ```

3. Select one of the following options to choose a start date to run the partial index:
   1. Date of the last successful index.
   2. Custom start date to index objects. Ensure that you enter the date in the format, `YYYY-MM-DD` and it is less than or equal to the current date.
4. Verify the selected date to proceed. Date specified to run the partial index is: `<date>` Do you wish to continue?
   1. Yes
   2. No
5. In case of failure, the index job stops with an error and the Unison microservices shuts down. You can then choose to resume from the point of failure or restart the partial index with the following command: `sh partialIndex.sh`
6. Choose to run the partial index with one of the following options:
   1. Resume the previous unsuccessful index operation.
   2. Restart the index operation.
7. You can review the logs in the following directory:
   `<INSTALLATION_DIR>/axonhome/axon/app/Resources/etl/logs/partialIndex.log`

Verify that the Messaging Service is Running

To verify that the Messaging service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Message Broker**.

4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

```
sh rabbitmq status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Daemon Tools Service is Running

To verify that the Daemon Tools service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh daemontools status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the OdbConsumer Service is Running

To verify that the OdbConsumer service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh OdbConsumer status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Unison Search Service is Running

To verify that the Unison Search service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh axon_unison_microservice_init status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify the Axon URL

Effective in version 5.3, the Axon installer assigns the following port numbers as the default value for the Apache server:

- If you use HTTP authentication, the installer assigns 9999 as the default port number for the Apache server.
- If you use HTTPS authentication, the installer assigns 9443 as the default port number for the Apache server.

Depending on whether you selected the default port number or entered a custom port number for the Apache server during installation, you must enter the correct port number in the browser address field. Make sure that users access Axon by typing the fully qualified host name and port number in the address bar of the web browser. The Axon URL must be in the following format: `http(s)://<Axon_host>:<9999_or_9443_or_custom_port_number>`

Troubleshooting Bulk Upload

You can use the bulk upload option to create or edit a large number of objects, relationships, and roles in Axon.

You might encounter the following issues with bulk upload:

- You get a connection error or access error during bulk upload.
- You get an error in the **Map Columns** step of bulk upload.
- The bulk upload task fails during the **Upload** step.
- Axon does not display items that have been successfully uploaded using bulk upload.

Perform the following tasks to fix bulk upload issues:

Modify the hosts File

In the local machine of the Axon user, map the IP address and the domain name address of the Axon host. To identify the Axon host domain name address, perform one of the following tasks:

- Open the `parameters.yml` file to see the domain name address for the Axon host.
- Run the `hostname -f` command to see the domain name address for the Axon host.

After you have identified the Axon host domain address, modify the `hosts` file in the local machine of the Axon user.

1. Open the `hosts` file available in the following location:

   `C:\Windows\System32\drivers\etc`

2. At the end of the `hosts` file, add the IP address and domain name address for the Axon host.

   For example, if the `parameters.yml` file contains `bulk_upload_host: AXONL70VM01.informatica.com`, add the entry `192.0.2.5 AXONL70VM01.informatica.com` to the hosts file.

To bulk upload data, Axon users must enter this host name instead of the IP address in the browser address bar.

Verify that the Bulk Upload Service is Running

To verify that the Bulk Upload service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Bulk Upload Service**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

   ```
   sh axon_bulk_upload_microservice_init status
   ```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Messaging Service is Running

To verify that the Messaging service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.

3. In the **Status of Axon Services** section, see **Message Broker**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

```
sh rabbitmq status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Troubleshooting Document Upload

You can upload supporting files that provide relevant information about the Axon objects that you work on. You can view these files in the **Documents** section of the **Summary** tab for the object.

You might encounter the following issues with document uploads:

- When you open an object, the attached documents do not appear.
- You cannot attach new documents to an object.

Perform the following task to fix document upload issues:

Verify that the Document Upload Service is Running

To verify that the Document Upload service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Document Upload**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

```
sh axon_document_upload_microservice status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Troubleshooting Change Requests

Change requests provide a controlled and audited change approval process. You can create a change request to facilitate communication and discussion between the object stakeholders and the wider Axon community.

You might encounter the following issues with change requests:

- You cannot create a change request.
- You cannot open a change request.
- You cannot see your change requests when you go to the **Change Request** tab of the **My Account** user menu.

Perform the following task to fix change request issues:

Verify that the Change Request Service is Running

To verify that the Change Request service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Change Request Service**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

```
sh axon_changerequest_microservice status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Troubleshooting Workflows

Workflows are predetermined steps that a change request must follow to ensure that the right people have the opportunity to provide input, challenge, and approve matters related to the objects that they change.

You might encounter the following issues with workflows:

- Workflows do not appear when you create a change request.
- New workflows do not appear for your account.
- You cannot create a new workflow.

Perform the following tasks to fix workflow issues:

Verify that the Workflow Service is Running

To verify that the Workflow service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Workflow**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

```
sh camunda status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Messaging Service is Running

To verify that the Messaging service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Message Broker**.

4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

   ```
   sh rabbitmq status
   ```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Daemon Tools Service is Running

To verify that the Daemon Tools service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh daemontools status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Troubleshooting Maps

Maps provide a graphical representation of the relationships between Axon objects across different facets.

You might encounter the following issues with maps:

- Maps do not appear in the **Interfaces** tab for Axon objects.
- Relationships between objects do not reflect in maps.

Perform the following tasks to fix notification issues:

Verify that the Graph Database Service is Running

To verify that the Graph Database service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh orientdb status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

If the Axon maps are displayed incorrectly after you restart the Graph Database service, perform the following steps in the Linux environment to reindex the database:

1. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   source <INSTALLATION_DIR>/scripts/exportEnv.sh
   ```

2. Go to the `<INSTALLATION_DIR>/scripts` directory, and run the following script to reindex the database:

   ```
   sh reIndex.sh
   ```

Verify that the OdbConsumer Service is Running

To verify that the OdbConsumer service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh OdbConsumer status
```

If the service is shut down, go to the <INSTALLATION_DIR>/bin directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that Data Set and Attribute Reference Numbers Have Prefixes

To verify that the data sets and attributes have prefixes for the reference numbers, perform the following steps:

1. Log in to Axon with the SuperAdmin profile.
2. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
3. In the menu on the left, under the **Customize & Configure** category, click **System Settings**.
4. In the **Group** list, select **Reference Number Settings**.
5. In the **Facets Reference Number** window, make sure that the CatalogueItem and CatItemComponent parameters have a prefix value. The CatalogueItem parameter represents data set prefix, and the CatItemComponent parameter represents attribute prefix.

   ```
   "CatalogueItem": {
       "RefNumber": "DS-"
   },
   "CatItemComponent": {
       "RefNumber": "AT-"
   }
   ```

6. If the CatalogueItem and CatItemComponent parameters do not have a prefix value, click the **Edit** icon.
7. In the **Facets Reference Number** window, enter a prefix value for the CatalogueItem and CatItemComponent parameters.

   ```
   "CatalogueItem": {
       "RefNumber": "<Data_Set_prefix>"
   },
   "CatItemComponent": {
       "RefNumber": "<attribute_prefix>"
   }
   ```

8. In the **Facets Reference Number** window, click **Save** to save your changes.
9. In the **System Settings** page, click **Save** to save your settings.
10. In the Linux environment, run the following command to clear the Axon cache and restart the necessary services:

    ```
    sh <INSTALLATION_DIR>/axonhome/third-party-app/scripts/paramsync
    ```

    When you run the paramsync script, Axon restarts the HTTPD, Memcached, and email notification services.

Note: When you clear the cache and restart the Axon services, the Axon web interface might be disrupted for some users that are logged into Axon. Informatica recommends that you update the cache after you save your changes in all the **System Settings** pages. Additionally, perform this action during a maintenance period when very few users are using Axon.

Troubleshooting Notifications

The notification center displays information that apply to your stakeholder role, change requests that you might be required to act upon, and status of your bulk upload tasks.

You might encounter the following issues with notifications:

- You cannot see your existing notifications.
- New notifications for recent actions do not appear for your account.
- Notifications do not appear for bulk upload tasks.

Perform the following tasks to fix notification issues:

Verify that the Notification Service is Running

To verify that the Notification service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Notification**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

```
sh axon_notification_microservice status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Run Bulk Upload Notification Script

To display the notifications for bulk upload tasks, perform the following steps:

1. Make sure that no bulk upload tasks are in progress.
2. Go to the `<INSTALLATION_DIRECTORY>/scripts` directory and run the `deploy-bulkupload.sh` file.

Axon now displays notifications for new bulk upload tasks.

Troubleshooting Data Migration

When you choose to import the migrated data, Axon performs a bulk upload operation to import the objects and relationships.

If an error occurs, all subsequent jobs might fail irrespective of whether it is dependent on the failed import job.

To rectify the error, perform the following steps:

1. Download the error report from
2. Unzip file and rectify error for that corresponding template.
3. Incase few of the jobs have already passed remove those entries from Manifest.json to avoid duplicates.
4. Zip it back and upload.

Troubleshooting Value Lists

You can upload a large number of values for the data set attributes by using the predefined templates. Axon writes these values directly to the database using the Value List service.

You might encounter the following issues with value lists:

- Axon displays an error when you edit a data set attribute.
- Axon displays an error when you upload a template that contains value list entries.

Perform the following task to fix value list issues:

Verify that the Value List Service is Running

To verify that the Value List service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Value List Service**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

   ```
   sh axon_valuelist_microservice_init status
   ```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the Bulk Upload Service is Running

To verify that the Bulk Upload service is running, perform the following steps:

1. From the Axon toolbar, click the **Admin Panel** menu item under your user name.
2. In the Super Admin menu on the left of the page, click **Admin Dashboard**.
3. In the **Status of Axon Services** section, see **Bulk Upload Service**.
4. Optionally, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command to check the status of the service:

   ```
   sh axon_bulk_upload_microservice_init status
   ```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Troubleshooting Logs

Logs provide a record of your actions when you create objects, modify existing objects, or perform tasks that use the application services and microservices.

You might encounter the following issues with logs:

- The **Logs** page in the Admin Panel does not display the logs for your actions.
- The **Logs** page does not display the components or the log levels correctly.

Restart the Logging Service

To fix the issues, go to the `<INSTALLATION_DIR>/bin` location and run the following command to restart the logging service:

```
sh enableLogging.sh
```

Data Quality Dashboards Do Not Display Data

The Data Quality dashboard displays a visual representation about the data quality rules, data stewardship, processes, and data privacy metrics.

You might find that the **Dashboard** tab for data quality rules appears blank and does not display scores. Perform the following task to fix dashboard issues:

Verify that the Graph Database Service is Running

To verify that the Graph Database service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh orientdb status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

If the Axon maps are displayed incorrectly after you restart the Graph Database service, perform the following steps in the Linux environment to reindex the database:

1. Run the following command to define the Axon environment variable so that the Linux system can access the Axon resources at the correct location:

   ```
   source <INSTALLATION_DIR>/scripts/exportEnv.sh
   ```

2. Go to the `<INSTALLATION_DIR>/scripts` directory, and run the following script to reindex the database:

   ```
   sh reIndex.sh
   ```

Cannot Start the HTTPD Service because SSL is Password-Protected

If your SSL certificate is protected by a password, the HTTPD service cannot start. You are prompted for the password each time the HTTPD service is started or restarted.

To avoid entering the password each time the HTTPD service starts, perform the following steps:

1. Create a new .conf file to store your SSL password.
   For example, you can create a `password.conf` file in the `<INSTALLATION_DIR>/axonhome/third-party-app/httpd/` directory.

2. In the SSL password file you created, add the following lines to store your SSL password:

   ```
   #!/bin/sh
   echo "<your_SSL_password>"
   ```

For example, add the following lines to the <INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf file:

```
#!/bin/sh
echo "Qwpt123%"
```

3. Run the following command to provide the execute permission to the SSL password file:

   ```
   chmod +x <path_to_file>
   ```

   For the sample file <INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf created in Step 1, run the following command:

   ```
   chmod +x <INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf
   ```

4. Modify the SSL certificate file at <INSTALLATION_DIR>/axonhome/third-party-app/httpd/conf.d/ssl.conf to point applications to refer to the SSL password file you created. Change the SSLPassPhraseDialog builtin parameter to SSLPassPhraseDialog exec:<path_to_your_SSL_password_file>.
   For the sample file <INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf created in Step 1, open the ssl.conf file and change the SSLPassPhraseDialog builtin parameter to SSLPassPhraseDialog exec:/<INSTALLATION_DIR>/axonhome/third-party-app/httpd/password.conf.

   After you make the changes, the Axon HTTPD service retrieves the SSL password from the password.conf file instead of the ssl.conf file.

Cannot Share Diagnostic Information with Informatica Global Customer Support

You can send diagnostic information about the Axon services to Informatica using the Phone Home option. If you have enabled the Phone Home option but Axon does not send the diagnostic information as scheduled, perform the following tasks to identify and troubleshoot the issue:

Verify that the Data Collection Service is Running

To verify that the data collection service is running, go to the <INSTALLATION_DIR>/axonhome/third-party-app/scripts directory in the Linux environment, and run the following command:

```
sh phonehome_collectdata.sh
```

If the console displays a license error, the HTTPD service might be shut down. If the console displays a psql error, the PostgreSQL database might be shut down.

Verify that the HTTPD Service is Running

To verify that the HTTPD service is running, go to the <INSTALLATION_DIR>/axonhome/third-party-app/scripts directory in the Linux environment, and run the following command:

```
sh httpd status
```

If the service is shut down, go to the <INSTALLATION_DIR>/bin directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

Verify that the PostgreSQL Database Service is Running

To verify that the PostgreSQL database service is running, go to the `<INSTALLATION_DIR>/axonhome/third-party-app/scripts` directory in the Linux environment, and run the following command:

```
sh postgresql status
```

If the service is shut down, go to the `<INSTALLATION_DIR>/bin` directory, and run the following command to restart the Axon services:

```
sh startup.sh
```

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